



Hardware Installation Guide for Cisco 8010 Series Large and Medium Density Fixed Routers

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CHAPTER 1

Safety Warnings

This handout lists the safety warnings necessary for handling this chassis. Before you install or service the chassis, review these safety warnings to avoid injuring yourself or damaging the equipment.

For a complete list of translated safety warnings, see the [Regulatory Compliance and Safety Information — Cisco 8000 Series Routers](#).

The safety warnings are grouped under the following sections:

- [Standard warning statements, on page 1](#)
- [Safety guidelines for personal safety and equipment protection, on page 3](#)
- [Safety Precautions for Module Installation and Removal, on page 3](#)
- [Safety with electricity, on page 4](#)
- [Cautions and regulatory compliance statements for NEBS, on page 6](#)
- [Power supply considerations, on page 8](#)
- [Preventing ESD damage, on page 9](#)

Standard warning statements



Warning

Statement 1071—Warning Definition

IMPORTANT SAFETY INSTRUCTIONS

Before you work on any equipment, be aware of the hazards involved with electrical circuitry and be familiar with standard practices for preventing accidents. Read the installation instructions before using, installing, or connecting the system to the power source. Use the statement number at the beginning of each warning statement to locate its translation in the translated safety warnings for this device.

SAVE THESE INSTRUCTIONS





Note **Statement 407**—Japanese Safety Instruction

You are strongly advised to read the safety instruction before using the product.

<https://www.cisco.com/web/JP/techdoc/pldoc/pldoc.html>

When installing the product, use the provided or designated connection cables/power cables/AC adapters.

〈製品使用における安全上の注意〉

www.cisco.com/web/JP/techdoc/index.html

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Warning **Statement 1003**—DC Power Disconnection

To reduce risk of electric shock or personal injury, disconnect DC power before removing or replacing components or performing upgrades.



Warning **Statement 1006**—Chassis Warning for Rack-Mounting and Servicing

To prevent bodily injury when mounting or servicing this unit in a rack, you must take special precautions to ensure that the system remains stable. The following guidelines are provided to ensure your safety:

- This unit should be mounted at the bottom of the rack if it is the only unit in the rack.
 - When mounting this unit in a partially filled rack, load the rack from the bottom to the top with the heaviest component at the bottom of the rack.
 - If the rack is provided with stabilizing devices, install the stabilizers before mounting or servicing the unit in the rack.
-



Warning **Statement 1017**—Restricted Area

This unit is intended for installation in restricted access areas. Only skilled, instructed, or qualified personnel can access a restricted access area.



Warning **Statement 1022**—Disconnect Device

To reduce the risk of electric shock and fire, a readily accessible disconnect device must be incorporated in the fixed wiring.

**Warning****Statement 1024—Ground Conductor**

This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

**Warning****Statement 1049—Rack Installation**

To reduce the risk of bodily injury, mount the chassis on a rack that is permanently affixed to the building.

**Warning****Statement 9001—Product Disposal**

Ultimate disposal of this product should be handled according to all national laws and regulations.

Safety guidelines for personal safety and equipment protection

The following guidelines ensure your safety and protect the equipment. This list does not include all the potentially hazardous situations. Therefore, you must be alert.

- Before moving the system, always disconnect all power cords and interface cables.
- Never assume that power is disconnected from a circuit; always check.
- Before and after installation, keep the chassis area clean and dust free.
- Keep tools and assembly components away from walk areas where you or others can trip over them.
- Do not work alone if potentially hazardous conditions exist.
- Do not perform any action that creates a potential hazard to people or makes the equipment unsafe.
- Do not wear loose clothing that may get caught in the chassis.
- When working under conditions that may be hazardous to your eyes, wear safety glasses.

Safety Precautions for Module Installation and Removal

Be sure to observe the following safety precautions when you work on the chassis.

**Warning****Statement 1006**—Chassis Warning for Rack-Mounting and Servicing

To prevent bodily injury when mounting or servicing this unit in a rack, you must take special precautions to ensure that the system remains stable. The following guidelines are provided to ensure your safety:

- This unit should be mounted at the bottom of the rack if it is the only unit in the rack.
- When mounting this unit in a partially filled rack, load the rack from the bottom to the top with the heaviest component at the bottom of the rack.
- If the rack is provided with stabilizing devices, install the stabilizers before mounting or servicing the unit in the rack.

**Warning****Statement 1008**—Class 1 Laser Product

This product is a Class 1 laser product.

**Note****Statement 1089**—Instructed and Skilled Person Definitions

An instructed person is someone who has been instructed and trained by a skilled person and takes the necessary precautions when working with equipment.

A skilled person or qualified personnel is someone who has training or experience in the equipment technology and understands potential hazards when working with equipment.

**Warning****Statement 1090**—Installation by Skilled Person

Only a skilled person should be allowed to install, replace, or service this equipment. See statement 1089 for the definition of a skilled person.

Safety with electricity

**Warning****Statement 1003**—DC Power Disconnection

To reduce risk of electric shock or personal injury, disconnect DC power before removing or replacing components or performing upgrades.

**Warning****Statement 1022**—Disconnect Device

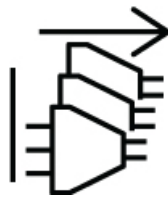
To reduce the risk of electric shock and fire, a readily accessible disconnect device must be incorporated in the fixed wiring.

**Warning** **Statement 1024—Ground Conductor**

This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

**Warning** **Statement 1028—More Than One Power Supply**

This unit might have more than one power supply connection. To reduce risk of electric shock, remove all connections to de-energize the unit.

**Warning** **Statement 1046—Installing or Replacing the Unit**

To reduce risk of electric shock, when installing or replacing the unit, the ground connection must always be made first and disconnected last.

If your unit has modules, secure them with the provided screws.

**Warning** **Statement 2004—Grounded Equipment**

This equipment is intended to be grounded to comply with emission and immunity requirements. Ensure that the switch functional ground lug is connected to earth ground during normal use.

**Warning** **Statement 1073—No User-Serviceable Parts**

There are no serviceable parts inside. To avoid risk of electric shock, do not open.

**Warning** **Statement 1074—Comply with Local and National Electrical Codes**

To reduce risk of electric shock or fire, installation of the equipment must comply with local and national electrical codes.

When working on equipment that is powered by electricity, follow these guidelines:

- Locate the room's emergency power-off switch. If an electrical accident occurs, you know where to quickly turn off the power.

- Before starting work on the system, turn off the DC main circuit breaker and disconnect the power terminal block cable.
- Disconnect all power when:
 - Working on or near power supplies
 - Installing or removing a device chassis or network processor module
 - Performing most hardware upgrades
- Never install equipment that appears damaged.
- Carefully examine your work area for possible hazards, such as moist floors, ungrounded power extension cables, and missing safety grounds.
- Never assume that power is disconnected from a circuit; always check.
- Never perform any action that creates a potential hazard to people or makes the equipment unsafe.
- If an electrical accident occurs and you are uninjured:
 - Use caution to avoid injuring yourself.
 - Turn off power to the device.
 - Seek medical attention, if necessary.

Use the following guidelines when working with any equipment that is disconnected from a power source, but connected to telephone wiring or network cabling:

- When installing or modifying telephone lines, use caution.
- Never install telephone jacks in wet locations unless the jack is designed to handle such locations.
- Never install telephone wiring during a lightning storm.

Cautions and regulatory compliance statements for NEBS

The NEBS-GR-1089-CORE regulatory compliance statements and requirements are discussed in this section.



Note **Statement 7001**—ESD Mitigation

This equipment may be ESD sensitive. Always use an ESD ankle or wrist strap before handling equipment. Connect the equipment end of the ESD strap to an unfinished surface of the equipment chassis or to the ESD jack on the equipment if provided.



Warning **Statement 7003**—Shielded Cable Requirements for Intrabuilding Lightning Surge

The intrabuilding port(s) of the equipment or subassembly, which is the management Ethernet port, must use shielded intrabuilding cabling/wiring that is grounded at both ends.

**Warning****Statement 7005—Intrabuilding Lightning Surge and AC Power Fault**

The intrabuilding port(s) of the equipment or subassembly, which is the management Ethernet port, must not be metalically connected to interfaces that connect to the OSP or its wiring. These interfaces are designed for use as intrabuilding interfaces only (Type 2 or Type 4 ports as described in GR-1089-CORE) and require isolation from the exposed OSP cabling. The addition of Primary Protectors is not sufficient protection in order to connect these interfaces metalically to OSP wiring.

**Warning****Statement 7011—Surge Protection Device Requirements for GR-1089 Antenna Ports**

Protect equipment antenna ports, that are classified as Type 6 according to GR-1089-CORE, with lightning surge protectors that are rated at a minimum of 600 V peak surge of 1.2/50 uS duration.

Connecting a Cable to the GNSS Antenna Interface

- GNSS modules have built-in ESD protections on all pins, including the RF-input pin. However, additional surge protection is required if an outdoor antenna is being connected. The Lightning Protector must be able to provide a low clamping voltage (less than 600V).
- A lightning protection must be mounted at the place where the antenna cable enters the building. The primary lightning protection must be capable of conducting all potentially dangerous electrical energy to PE (Protective Earth).
- Surge arrestors should support DC-pass and suitable for the GPS frequency range as per the constellation used with low attenuation.

**Warning****Statement 7012—Equipment Interfacing with AC Power Ports**

Connect this equipment to AC mains that are provided with a surge protective device (SPD) at the service equipment that complies with NFPA 70, the National Electrical Code (NEC).

**Note****Statement 7013—Equipment Grounding Systems—Common Bonding Network (CBN)**

This equipment is suitable for installations using the CBN.

**Note****Statement 7016—Battery Return Conductor**

The battery return conductor of this equipment shall be treated as (DC-I).

**Note** **Statement 7018**—System Recover Time

The equipment is designed to boot up in less than 30 minutes provided the neighboring devices are fully operational.

**Note** **Statement 8015**—Installation Location Network Telecommunications Facilities

This equipment is suitable for installation in network telecommunications facilities.

**Note** **Statement 8016**—Installation Location Where the National Electric Code (NEC) Applies

This equipment is suitable for installation in locations where the NEC applies.

Power supply considerations

Check the power at your site to ensure that you are receiving clean power (free of spikes and noise). If necessary, install a power conditioner.

Power Connection Guidelines

This section provides guidelines for connecting the device power supplies to the site power source.

**Warning** **Statement 1024**—Ground Conductor

This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

Guidelines for DC-Powered Systems

Basic guidelines for DC-powered systems include the following:

- Each chassis power supply has its own dedicated input power source. The source must comply with the safety extra-low voltage (SELV) requirements in the UL 60950-1, UL 62368-1, CSA 60950-1, CSA 62368-1, EN 60950-1, EN 62368-1, IEC 60950-1, and IEC 62368-1 standards.
- Protect the circuit by a dedicated two-pole DC circuit breaker. Ensure that the circuit breaker is sized according to the power supply input rating and local or national code requirements.
- The circuit breaker is considered as the disconnect device and is easily accessible.
- The system ground is the power supply and chassis ground.
- Use the grounding lug to attach a wrist strap for ESD protection during servicing.

- Do not connect the DC return wire to the system frame or to the system-grounding equipment.
- Ensure that the DC return is grounded at the source side.
- Ensure that each power feed of the equipment is connected to different power sources.

Guidelines for AC-Powered Systems

Basic guidelines for AC-powered systems include the following:

- Each chassis power supply has its own dedicated branch circuit.
- Ensure that the circuit breaker is sized according to the power supply input rating and local or national code requirements.
- The AC power receptacles that are used to plug in the chassis must be the grounding type. The grounding conductors that connect to the receptacles must connect to protective earth ground at the service equipment.

Prevent Power Loss

Use the following guidelines to prevent power loss to the device:

- To prevent input power loss, ensure that the maximum load on each circuit supplying the power is within the current ratings of the wiring and breakers.
- In some systems, you can use an UPS to protect against power failures at your site. Avoid UPS types that use ferroresonant technology. These UPS types can become unstable with systems such as the device, which can have substantial current-draw fluctuations due to bursty data traffic patterns.

Determining power requirements is useful for planning the power distribution system to support the device.

Preventing ESD damage

Electrostatic Discharge (ESD) can damage equipment and impair electrical circuitry. ESD may occur when electronic printed circuit cards are improperly handled and can cause complete or intermittent failures. When removing and replacing modules, always follow these ESD prevention procedures:

- Ensure that the device chassis is electrically connected to earth ground.
- Wear an ESD-preventive wrist strap, ensuring that it makes good skin contact. To channel unwanted ESD voltages safely to ground, connect the clip to an unpainted surface of the chassis frame. To guard against ESD damage and shocks, the wrist strap and cord must operate effectively.
- If no wrist strap is available, ground yourself by touching a metal part of the chassis.
- When installing a component, use any available ejector levers or captive installation screws to properly seat the bus connectors in the backplane or midplane. These devices prevent accidental removal, provide proper grounding for the system, and help to ensure that bus connectors are properly seated.
- When removing a component, use available ejector levers or captive installation screws, if any, to release the bus connectors from the backplane or midplane.
- Handle components by only their handles or edges; do not touch the printed circuit boards or connectors.

- Place a removed component board side up on an antistatic surface or in a static-shielding container. If you plan to return the component to the factory, immediately place it in a static-shielding container.
- Avoid contact between the printed circuit boards and clothing. The wrist strap only protects components from ESD voltages on the body; ESD voltages on clothing can still cause damage.
- Never attempt to remove the printed circuit board from the metal carrier.

For the safety of your equipment, periodically check the resistance value of the antistatic wrist strap. Maintain the value between 1 and 10 MOhm.



CHAPTER 2

Cisco 8010 Large and Medium Density Routers Overview

The Cisco 8010 Series Large and Medium Density Fixed Routers, featuring the Cisco Silicon One A100 ASIC, deliver comprehensive Class C timing support and end-to-end encryption. Designed for seamless network expansion and scalability, these routers are built on the robust Cisco IOS XR operating system.

The Cisco 8010 Series Large and Medium Density Fixed Routers are temperature-hardened, fixed port, one rack unit form-factor routers. The primary use cases of the router are:

- CSG (Cell Site Gateway),
- L3 Pre-aggregation,
- L2 Pre-aggregation,
- L3 WAN, and
- OLT Aggregation

For more information about the router features and benefits, see the [Cisco 8010 Large Density Routers Data Sheet](#).

Router Variants

This table provides the variants for the Cisco 8010 Series Large and Medium Density Fixed Routers.

Table 1: Router variants and applications

Variant name	Classification	Deployment Mode
Cisco 8011-4G24Y4H-I	Large Density	Central office, outdoor
Cisco 8011-32Y8L2H2FH	Large Density	Central office
Cisco 8011-12G12X4Y-A	Medium Density	Central office, outdoor
Cisco 8011-12G12X4Y-D	Medium Density	Central office, outdoor



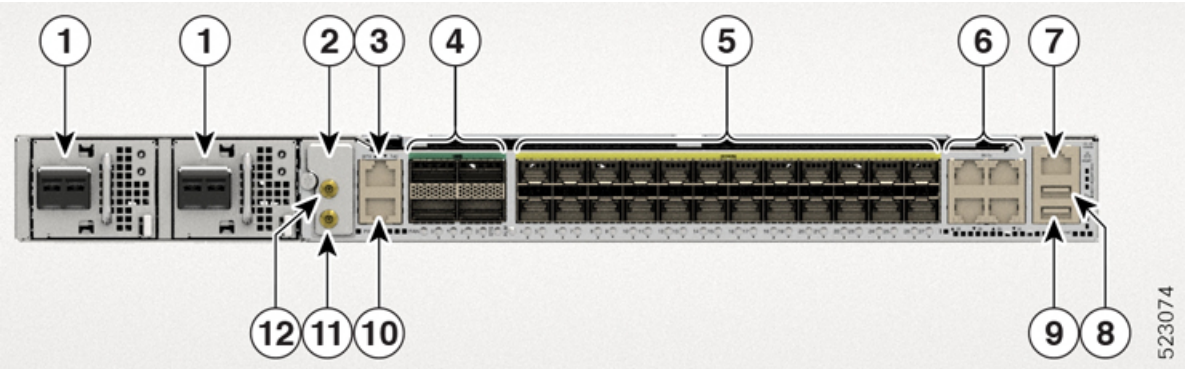
Note While installing in outdoor environment, we recommend you to have an IP65/IP66 sealed cabinet with heat exchanger complying with Telcordia GR487.

- [Hardware Features of Cisco 8000 Series Large and Medium Density Routers](#), on page 12
- [Network Interfaces](#), on page 15
- [Interface Naming](#), on page 17
- [Network Timing Interfaces](#), on page 19
- [External Alarm Inputs](#), on page 27
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- [RS232 Console](#), on page 27
- [Online Insertion and Removal](#), on page 28
- [Supported Transceiver Modules](#), on page 28

Hardware Features of Cisco 8000 Series Large and Medium Density Routers

Cisco 8011-4G24Y4H-I Router

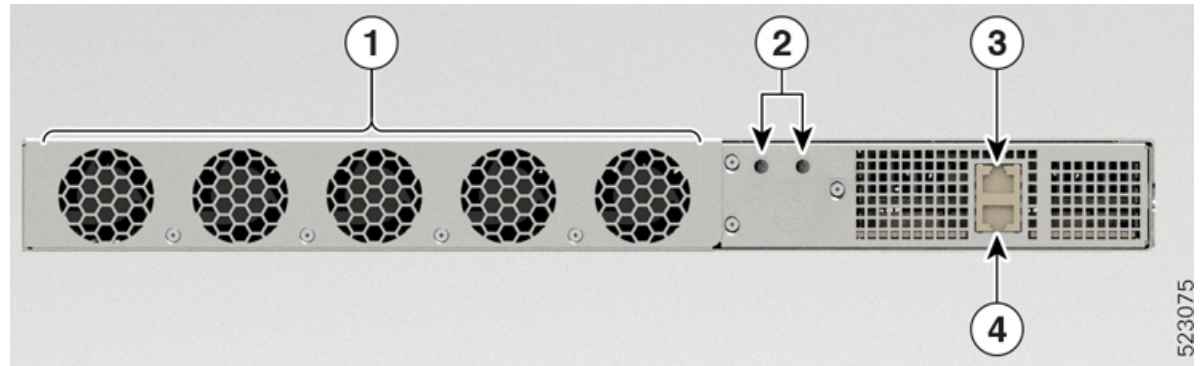
Figure 1: Cisco 8011-4G24Y4H-I Router Port and LED details on Front View



1	DC or AC PSU Power Module (PM0 and PM1)	2	• 8000-TIC (shown): • Timing Interfaces Card: 10MHz, 1PPS ports • 8000-TIC-GNSS (optional): • PRTC-B GNSS Receiver Card: 10MHz, 1PPS, Antenna ports
3	BITS port	4	100G QSFP ports

5	1/10/25G SFP ports	6	Copper ports
7	1G Ethernet Management port	8	USB memory port
9	USB Console port	10	1PPS/Time of Day (ToD) port
11	1PPS port	12	10MHz port

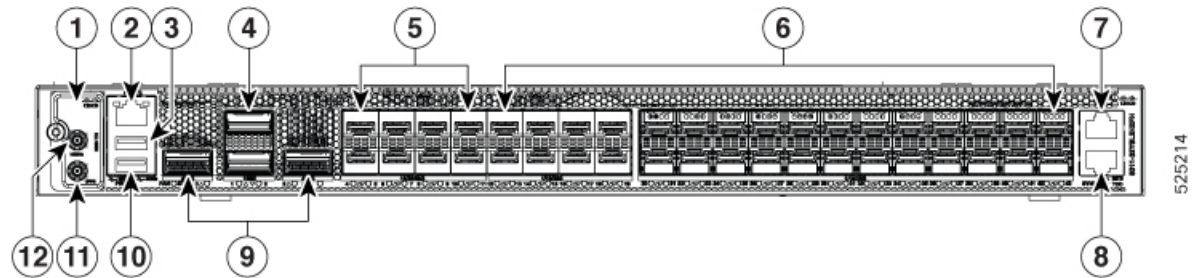
Figure 2: Cisco 8011-4G24Y4H-I Router Fan and Alarm Port details on Rear View



1	Fixed Fan Modules	2	Grounding Lug Holes
3	Alarm port	4	RS232 Console port

Cisco 8011-32Y8L2H2FH Router

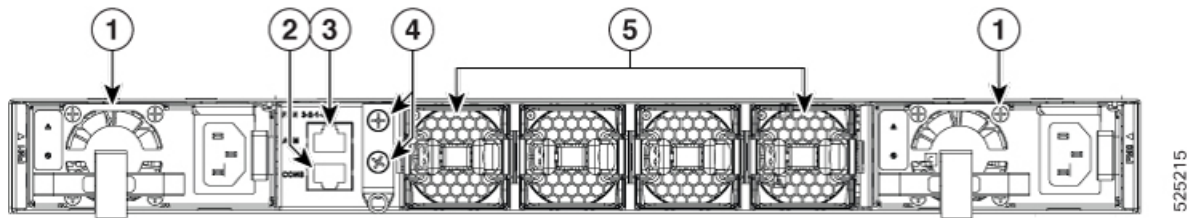
Figure 3: Cisco 8011-32Y8L2H2FH Router Port and LED details on front view



1	8000-TIC (shown): - Timing Interfaces Card: 10MHz, 1PPS ports 8000-TIC-GNSS (optional): - PRTC-B GNSS Receiver Card: 10MHz, 1PPS, Antenna ports	2	10/100/1000M Ethernet Management port
3	USB memory port	4	100G QSFP ports

5	10/25/50G SFP ports	6	1/10/25G SFP ports
7	BITS port	8	1PPS/Time of Day (ToD) port
9	400G QSFP ports	10	USB console port
11	1PPS port	12	10MHz port

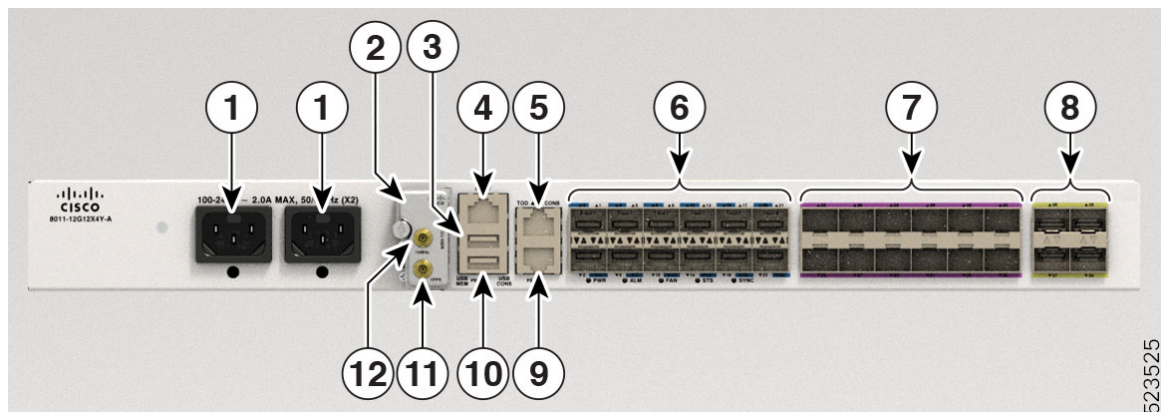
Figure 4: Cisco 8011-32Y8L2H2FH router fan and alarm port details on rear view



1	DC or AC PSU power module (PM0 and PM1)	2	RS232 console port
3	Alarm port	4	Grounding lug holes
5	Fan modules (0 to 3)		

Cisco 8011-12G12X4Y-A Router

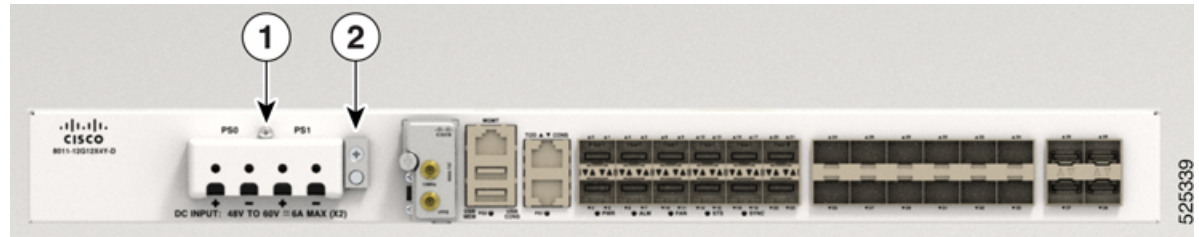
Figure 5: Cisco 8011-12G12X4Y-A Router Components on Front View



1	AC PSU PS0/PS1 PM	2	8000-TIC (by default) or 8000-TIC-GNSS (optional)
3	USB Memory port	4	Management port
5	1PPS/Time of Day (ToD) port	6	1G CSFP SFP ports
7	1/10G SFP+ ports	8	1/10/25G SFP28 ports
9	RS232 Console port	10	USB Console port

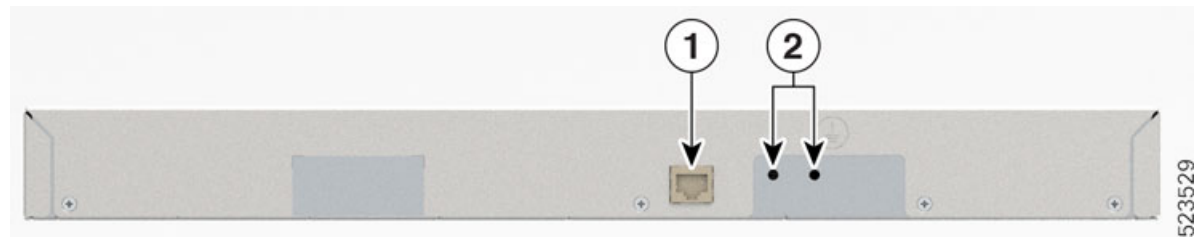
11	1PPS	12	10MHz
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Figure 6: Cisco 8011-12G12X4Y-D Router Components on Front View



1	DC PSU PS0/PS1 PM	2	Surge pin
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Figure 7: Cisco 8011-12G12X4Y Router Components on Rear View



1	Alarm port	2	Lug Holes
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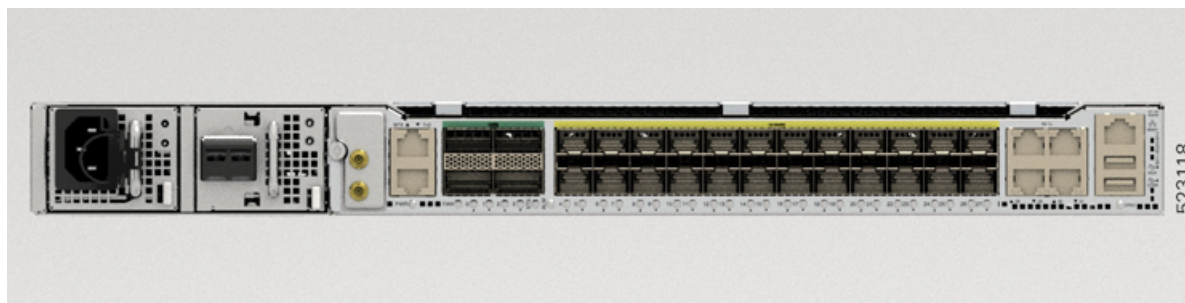
For temperature and physical specifications, see the [Cisco 8010 Series Routers Data Sheet](#).

Network Interfaces

Cisco 8011-4G24Y4H-I Router

The Cisco 8011-4G24Y4H-I router supports the following network interfaces:

- 4x1G Copper ports
- 24x1/10/25G SFP ports
- 4x100G QSFP ports

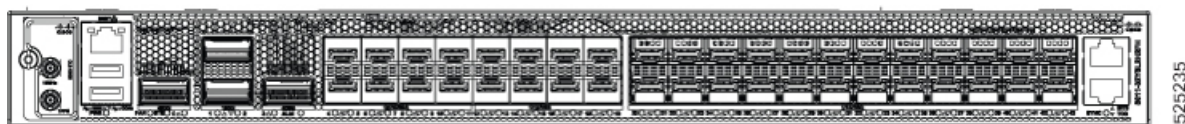
Figure 8: Cisco 8011-4G24Y4H-I Router

Note The ports are color coded in the chassis for ease of identification; for example, the 100G ports are in green, and the 1/10/25G ports are in yellow.

Cisco 8011-32Y8L2H2FH Router

The Cisco 8011-32Y8L2H2FH router supports these network interfaces:

- 2x400G QSFP56-DD ports
- 2x100G QSFP28 ports
- 8x10/25/50G SFP ports
- 32x1/10/25G SFP ports

Figure 9: Cisco 8011-32Y8L2H2FH Router

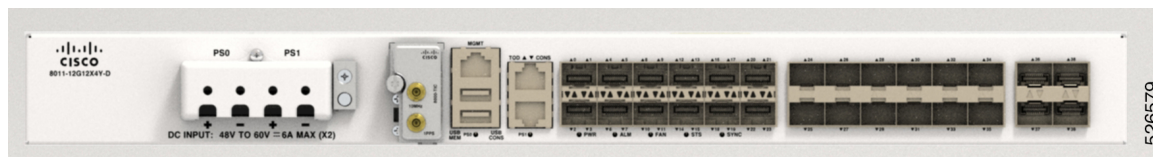
Note The ports are color coded in the chassis for ease of identification; for example, the 400G ports are in blue, the 100G ports are in green, the 10/25/50G ports are in orange, and the 1/10/25G ports are in yellow.

Cisco 8011-12G12X4Y Router

The Cisco 8011-12G12X4Y-A router and Cisco 8011-12G12X4Y-D router support the following network interfaces:

- 12x1G CSFP ports
- 12x1G SFP ports
- 12 x1/10G SFP+ ports
- 4x1/10/25G SFP28 ports

Figure 10: Cisco 8011-12G12X4Y Router

**Note**

The ports are color coded in the chassis for ease of identification; for example, the 1G CSFP ports are in blue, the 1/10G SFP+ ports are in pink, and the 1/10/25G SFP28 ports are in yellow.

Interface Naming

Cisco 8011-4G24Y4H-I Router

This table shows the interface naming of the Cisco 8011-4G24Y4H-I router:

Table 2: Port numbers

100G QSFP ports	1/10/25G SFP ports	1G Copper ports (10/100/1000M)
0 to 3	4 to 27	28 to 31

The *interface-path-id* is *rack/slot/module/port*. The slash between values is required as part of the notation.

- **HundredGigE**—0/0/0/0 to 0/0/0/3
- **TwentyFiveGigE**—0/0/0/4 to 0/0/0/27
- **TenGigE**—0/0/0/4 to 0/0/0/27
- **GigE**—0/0/0/4 to 0/0/0/27
- **GigE**—0/0/0/28 to 0/0/0/31

Port Speed on 100G Ports:

The 100G ports (0-3) support:

- 4x25G QSFP
- 40G QSFP
- 4x10G QSFP

Cisco 8011-32Y8L2H2FH Router

This table shows the interface naming of the Cisco 8011-32Y8L2H2FH router.

Table 3: Port numbers

400G QSFP ports	100G QSFP ports	10/25/50G SFP ports	1/10/25G SFP ports
0, 3	1 to 2	4 to 11	12 to 43

The *interface-path-id* is *rack/slot/module*. The slash between values is required as part of the notation.

- **FourHundredGigE**—0/0/0/0 & 0/0/0/3
- **HundredGigE**—0/0/0/1 to 0/0/0/2
- **FiftyGigE**—0/0/0/4 to 0/0/0/11
- **TwentyFiveGigE**—0/0/0/4 to 0/0/0/43
- **TenGigE**—0/0/0/4 to 0/0/0/43
- **GigE**—0/0/0/12 to 0/0/0/43

This table shows the port speed options for the 400G, 100G, 10/25/50G, and 1/10/25G ports.

Table 4: Port speed

Port speed for 400G ports	Port speed for 100G ports	Port speed for 10/25/50G ports	Port speed for 1/10/25G ports
400G	100G	50G	25G
4x100G	4x25G	25G	10G
8x50G	40G	10G	1G
3x100G	4x10G	—	—
200G	—	—	—
2x100G	—	—	—
100G	—	—	—
4x25G	—	—	—
40G	—	—	—
4x10G	—	—	—



Note Not all 400G, 100G, and 50G ports can be enabled simultaneously at the maximum port speed.

Cisco 8011-12G12X4Y Router

The following table shows the interface naming of the Cisco 8011-12G12X4Y-A and Cisco 8011-12G12X4Y-D routers:

Table 5: Port Numbering

1G CSFP ports	1G SFP ports	1/10G SFP+ ports	1/10/25G SFP28 ports
0/0 to 0/23	0/0, 0/3, 0/4, 0/7, 0/8, 0/11, 0/12, 0/15, 0/16, 0/19, 0/20, and 0/23	0/24 to 0/35	0/36 to 0/39

The *interface-path-id* is *rack/slot/module/port*. The slash between values is required as part of the notation.

- **TwentyFiveGigE**— 0/0/0/36 to 0/0/0/39
- **TenGigE**— 0/0/0/24 to 0/0/0/39
- **GigE**— 0/0/0/0 to 0/0/0/23

Network Timing Interfaces

The router supports these timing interfaces card (TIC):

- Cisco 8010 timing interfaces card (8000-TIC)
- Cisco 8010 timing interfaces card with GNSS (8000-TIC-GNSS)

By default, the router comes with Cisco 8000-TIC. Cisco 8000-TIC-GNSS can also be selected when ordering the router.

Usage guidelines

Some of the guidelines to use network timing interfaces:

- 1PPS input or output and ToD input or output—The interface is used for input or output of Time-of-Day (ToD) and 1PPS pulses. ToD format includes both NTP and IEEE 1588-2008 time formats.
- The same RS422 pins for 1PPS and ToD are shared between the input and output directions. The direction for each are independently configurable through software.
- BITS input or output—The BITS interfaces support clock recovery from either a T1 at 1.544MHz or an E1 at 2.048MHz, configurable by software.

Restrictions

Online insertion and removal (OIR) of a TIC module is not supported when the router is operational.

Cisco 8010 Timing Interfaces Card (8000-TIC)

Cisco 8010 Timing Interfaces Card (8000-TIC) is a pluggable module containing these timing ports:

- 1PPS port
- 10MHz port

8000-TIC is DIN 1.0/2.3 jack connector compliant to DIN 41626 specifications.

Figure 11: Cisco 8011-4G24Y4H-I Router: Ports on Timing Interfaces Card (8000-TIC) Module

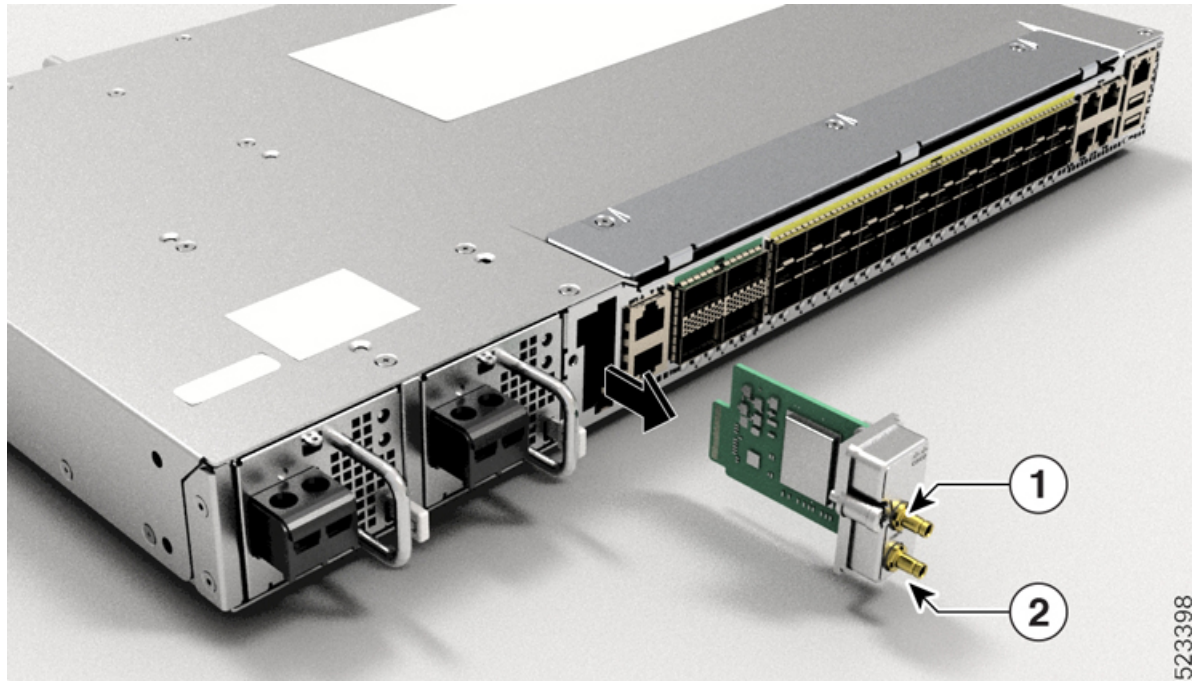


Figure 12: Cisco 8011-32Y8L2H2FH: Ports on Cisco 8010 Timing Interfaces Card (8000-TIC) Module

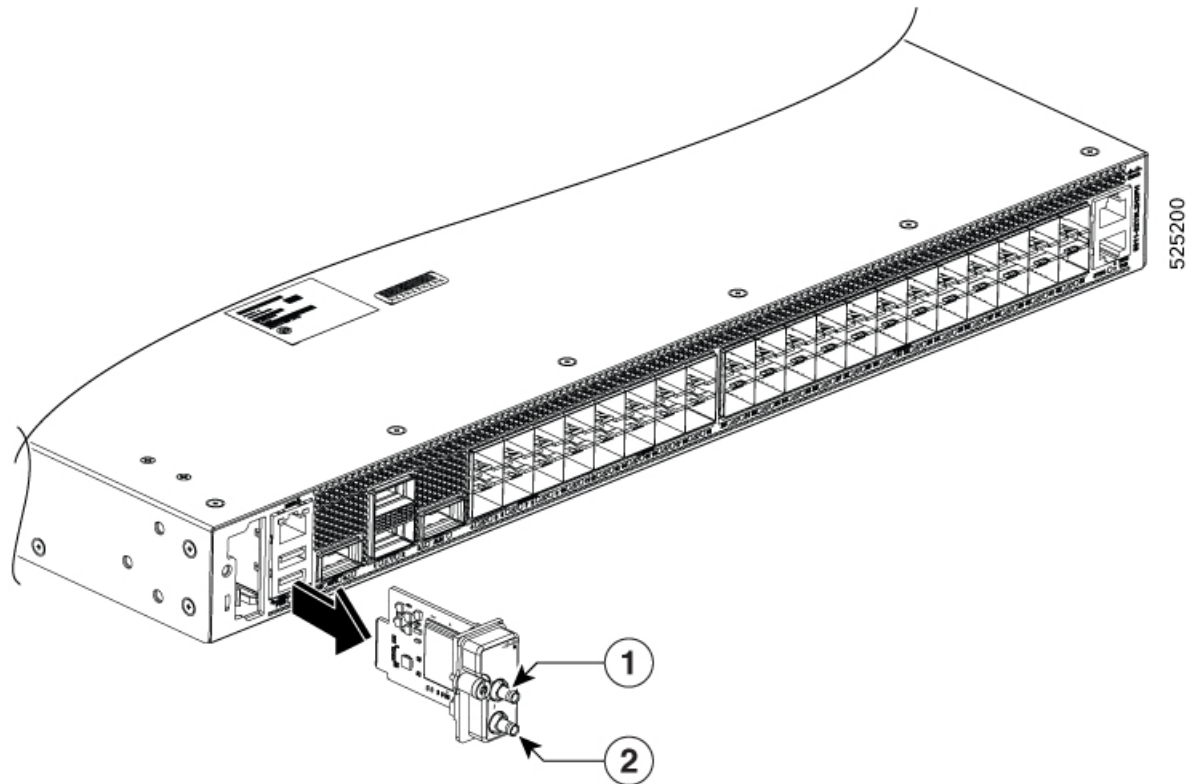
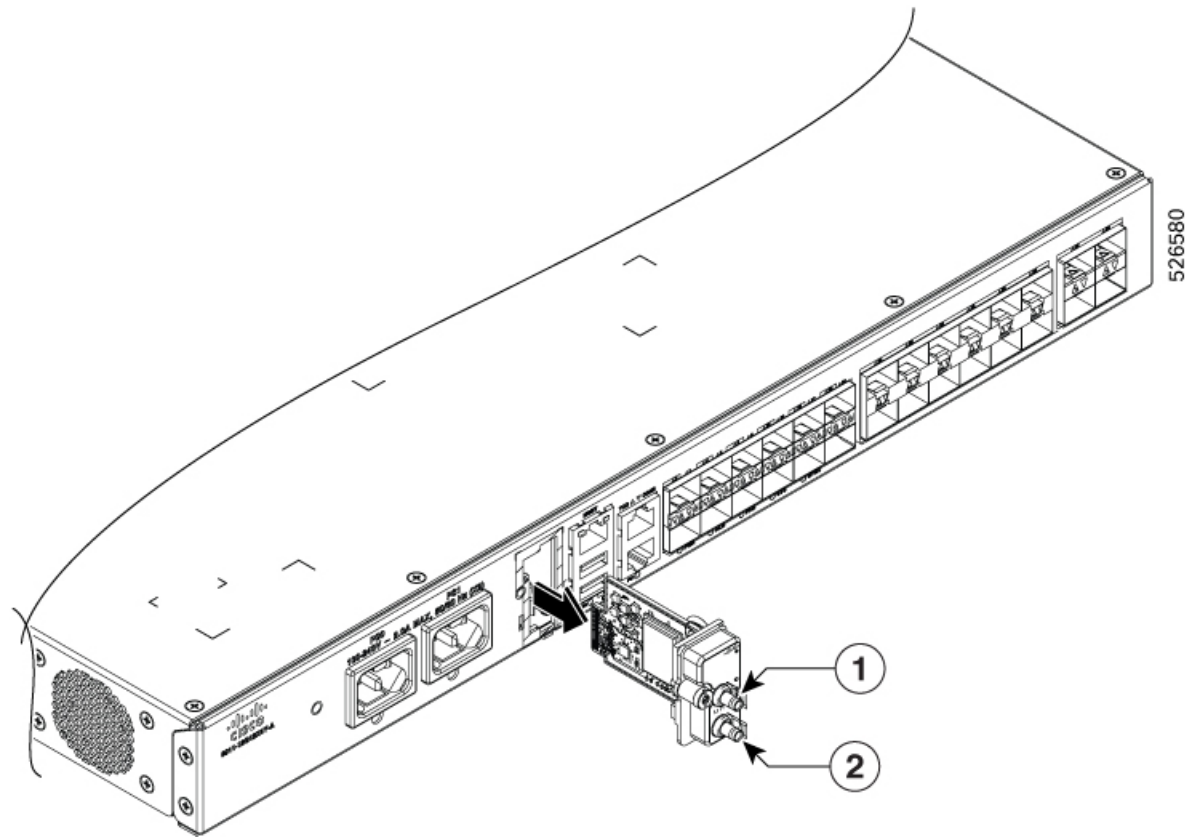


Figure 13: Cisco 8011-12G12X4Y Router: Ports on Timing Interfaces Card (8000-TIC) Module



1	10MHz port
2	1PPS port

Timing Interfaces Card Ports Specification

Table 6: Timing Interfaces Card Ports

Category	10MHz (Input)	1PPS (Input and Output)
Waveform	Input—Sine wave Output—Sine wave	Input—Rectangular pulse Output—Rectangular pulse
Amplitude	Input—Greater than (>) 1.7 volt p-p(+8 to +10 dBm) Output—Greater than (>) 2.4 volts TTL compatible	Input—Greater than (>) 1.2 volts TTL compatible Output—Greater than (>) 2.4 volts TTL compatible
Impedance	50 Ohms	50 Ohms
Pulse Width	50% duty cycle	100 Nanoseconds when configured as Output

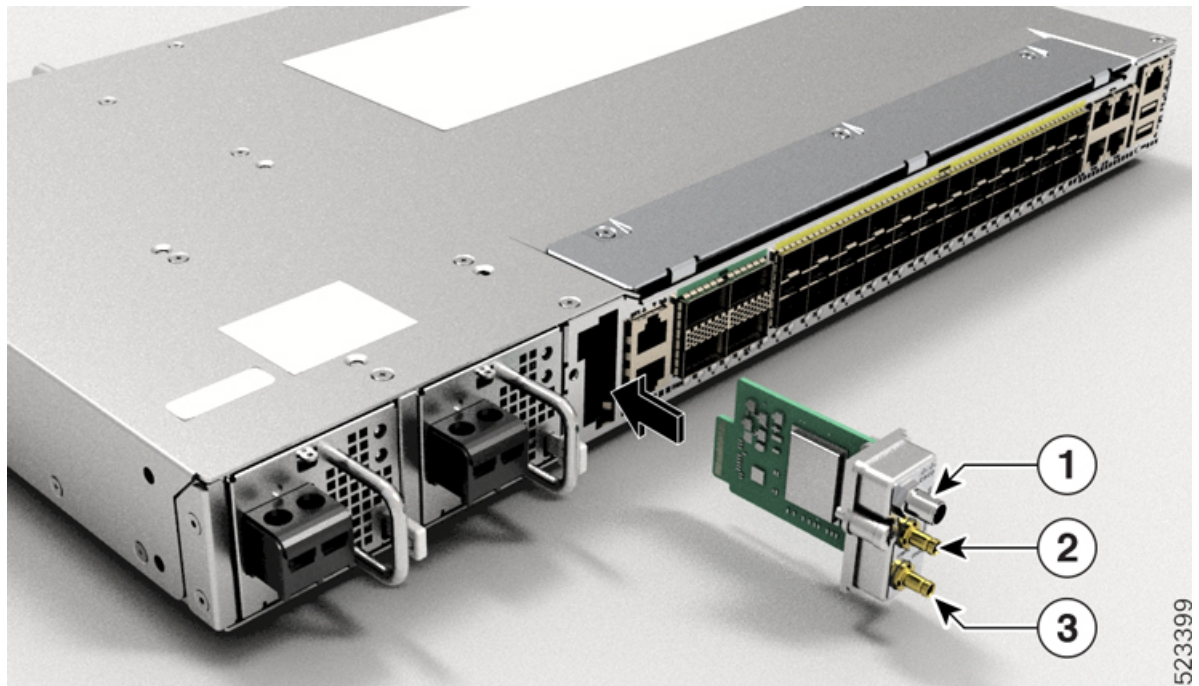
Category	10MHz (Input)	1PPS (Input and Output)
Rise Time	Output—5 Nanoseconds	<5 Nanoseconds

Cisco 8010 Timing Interfaces Card with GNSS (8000-TIC-GNSS)

8000-TIC-GNSS is a Cisco 8010 Timing Interfaces Card with a GNSS module. 8000-TIC-GNSS has:

- an in-built ESD protection on all pins
- these timing ports:
 - GNSS Antenna port
 - 10MHz port
 - 1PPS port

Figure 14: Cisco 8011-4G24Y4H-I Router: Ports on Timing Interfaces Card with GNSS (8000-TIC-GNSS) Module



For the Cisco 8011-4G24Y4H-I Router, an extra surge protection is required if an outdoor antenna is to be connected.

Figure 15: Cisco 8011-32Y8L2H2FH: Ports on Cisco 8010 Timing Interfaces Card with GNSS (8000-TIC-GNSS) Module

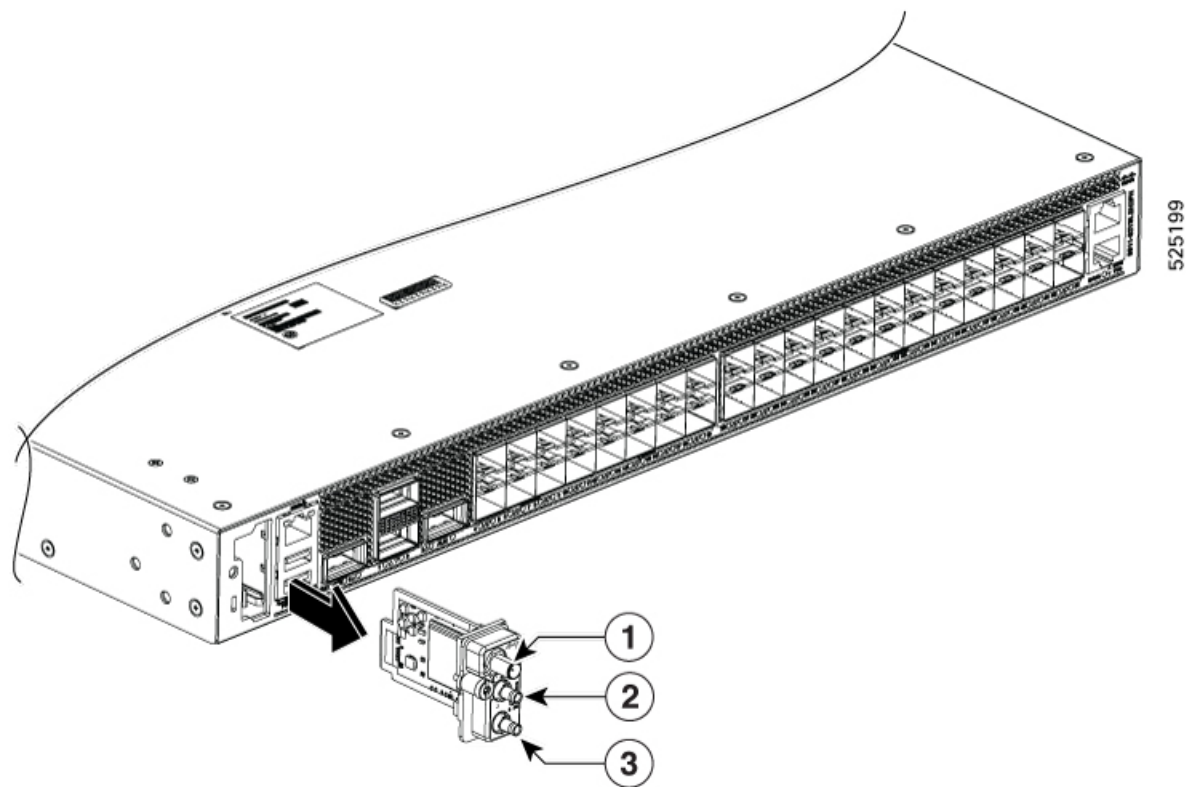
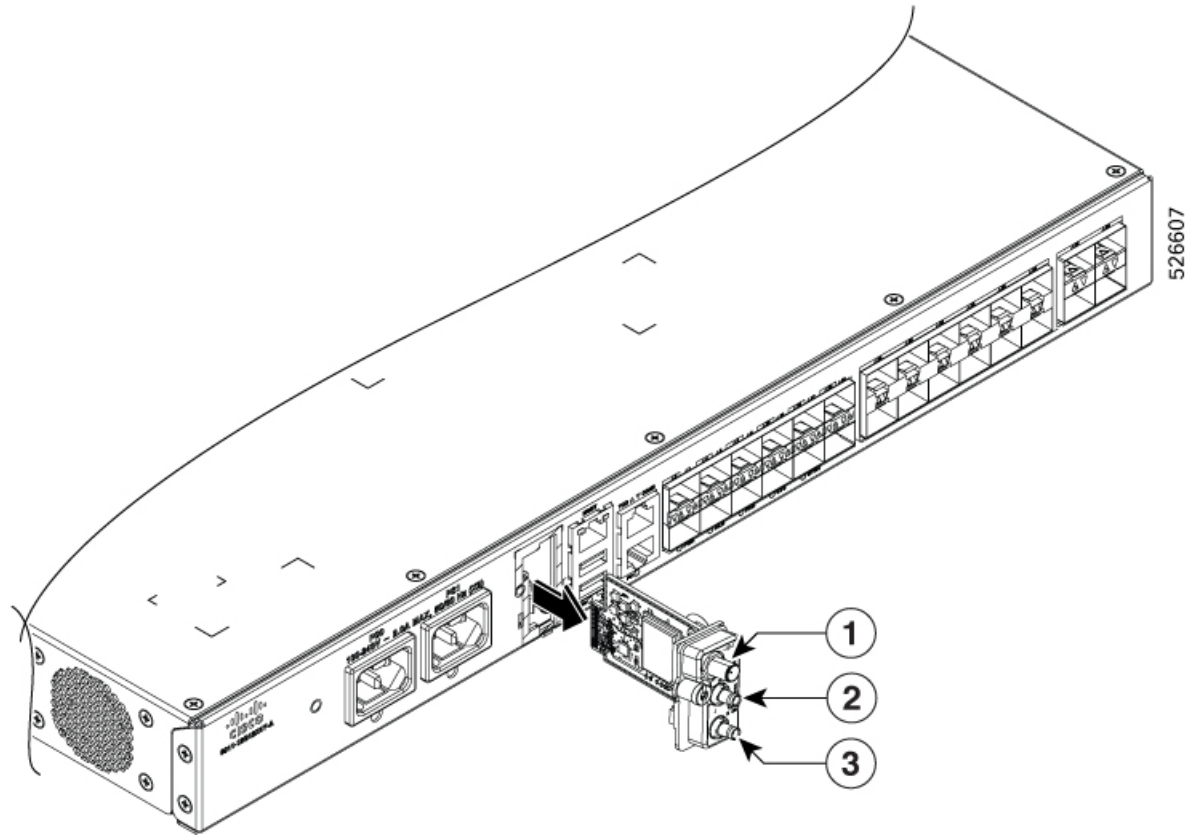


Figure 16: Cisco 8011-12G12X4Y: Ports on Cisco 8010 Timing Interfaces Card with GNSS (8000-TIC-GNSS) Module



For the Cisco 8011-12G12X4Y Router, an extra surge protection is required if an outdoor antenna is to be connected.

1	GNSS Antenna port
2	10MHz port
3	1PPS port

For information on how to insert and secure the Cisco 8000 Timing Interfaces Card with GNSS module, see [Inserting and Securing Timing Interfaces Card with GNSS Module](#).

Timing Interfaces Card with GNSS Module RF Input Specifications

This table provides specifications of the GNSS RF input and connector.

Table 7: Timing Interfaces Card with GNSS Module

Connector Type	SMA Jack
Impedance	50 Ohms

Band	Multi band - L1/L2
Accuracy	PRTC-B

Primary Reference Time Clock

As packet timing requirements emerged, the ITU-T developed the Primary Reference Time Clock (PRTC) standard for time and phase for transport over a packet network and this standard is known as G.8272.



Note The performance specified in the following table also applies to the output of the combined PRTC and T-GM function when integrated into a single piece of equipment. Therefore, for both the PRTC-A and PRTC-B, there's no additional (performance) allowance for the inclusion of the T-GM function.

Table 8: PRTC-A versus PRTC-B

Category	PRTC-A	PRTC-B
Standard	G.8272	G.8272
Time output accuracy (max)	100 ns	40 ns
Phase Error at Output		
Wander in locked mode (MTIE)	100 ns	40 ns
Wander in locked mode (TDEV)	30 ns	5 ns
Holdover	Optional	Optional

There is no standard packet timing requirement and it is based on implementation. Some standalone PRTC devices have very-high quality quartz oscillators or even a Rubidium oscillator as an option to allow an option for extended holdover. Redundancy is provided through network design instead of relying on holdover performance from expensive oscillators.

According to the standards, there are only a few differences between the two classes of clock. Meeting those requirements may result in different implementations, which may influence the cost of the device.

To achieve improved performance, PRTC-B is typically implemented with a two-band GNSS receiver instead of a single-band receiver. The “traditional” band for GPS signals in the L1 band, while the newer devices receive signals in the L2 band.

Table 9: Bands on PRTC-A versus PRTC-B

Category	PRTC-A	PRTC-B
Bands (GPS)	L1 (1575.42 MHz)	L1 + L2 (1227.60 MHz)
Bands (Galileo)	E1 (1575.42 MHz)	E1 (1227.60 MHz)
Bands (BeiDou)	1561.098MHz	1207.140MHz

Category	PRTC-A	PRTC-B
Bands (GLONASS)	1575.42MHz	1207.140MHz
Ionospheric delay model	Basic	Advanced
Antenna	Single band (L1/E1)	Dual band (L1/E1, L2)
Constellation	GPS, Galileo, other	GPS, Galileo, other

Usage Guidelines

These are some guidelines to follow when using the Cisco 8000-TIC-GNSS:

Surge Protection

- The Cisco 8000-TIC-GNSS modules have built-in ESD protections on all pins, including the RF-input pin.
- An extra surge protection may be required if rooftop antennas are to be connected to meet the regulations and standards for lightning protection of countries where the end product is installed.
- The lightning protector must support a low clamping voltage (less than 600V).
- A lightning protection must be mounted at the place where the antenna cable enters the building. The primary lightning protection must be capable of conducting all potentially dangerous electrical energy to protective earth (PE).
- Surge arrestors should support DC-pass and suitable for the Timing Interfaces Card with GNSS frequency range (1227.60MHz and 1.575GHz) with low attenuation.
- Surge arrestors must support DC-pass and suitable for the GNSS frequency range as per the constellation used with low attenuation.
- For optimal performance, the Cisco 8000-TIC-GNSS requires a GNSS antenna with built-in Low-Noise Amplifier (LNA). The antenna LNA amplifies the received satellite signals for the following two purposes:
 - Compensation of losses on the cable.
 - Lifting the signal amplitude to the suitable range for the receiver frontend.

The amplification required is 22dB gain + cable/connector loss + Splitter signal loss.

The recommended range of LNA gain (- cable/connector loss) is 22–30dB with a minimum of 20dB and a maximum of 35dB, at the connector of the receiver module.

Antenna Sky Visibility

- The Cisco 8000-TIC-GNSS provides 5V to the active antenna through the same RF input.
- The Cisco 8000-TIC-GNSS signals can only be received on a direct line of sight between antenna and satellite. The antenna must have a clear view of the sky. For proper timing, minimum of four satellites should be locked.



Note The antenna terminal should be earthed at the building entrance in accordance with the ANSI/NFPA 70, the National Electrical Code (NEC), in particular Section 820.93, Grounding of Outer Conductive Shield of a Coaxial Cable.

- Use a passive splitter if more than one Timing Interfaces Card with GNSS module is fed from a single antenna.



Note The splitter should have at least one RF port capable of DC-pass, and an antenna should be connected to that port, if the antenna needs to feed power from 8000-TIC-GNSS.

External Alarm Inputs

The router supports four dry contact alarm inputs through an RJ45 jack at the back panel.

The alarm condition is normally open, which indicates that no current flows through the alarm circuit, and the alarm is generated when the current is flowing. Each alarm input can be provisioned as being critical, major, or minor.

USB console

USB console is a single USB 2.0 Type-A receptacle on the front panel of the router that:

- provides console access to Uboot flash for Cisco software and diagnostics
- uses the Type-A connector
- operates as a USB peripheral only to connect to an external host computer, and
- uses a Type-A to Type-A connector instead of a standard USB cable.

Restrictions

- The USB console port and RS232 console port are mutually exclusive to each other.

RS232 Console

The RS232 console is a port in RJ45 form factor on the rear panel of the router. This console provides:

- transmission (Tx),
- reception (Rx), and

- ground (Gnd) information.

Online Insertion and Removal

The router supports these online insertion and removal (OIR) operations:

- When an SFP is removed from a port, there is no effect on the traffic flowing on other ports.
- When an SFP is installed on a port:
 - there is no effect on the traffic flowing on other ports.
 - the system initializes that port for operation based on the current configuration.
 - if the inserted SFP is incompatible with the current configuration of that port, the port becomes operational only after the configuration is updated.
- When both the AC and DC power supplies are installed and active:
 - the load may be shared between both the PSUs or a single PSU takes the entire load.
 - when one power supply is not working or the input cable is removed, the other power supply takes the entire load without disruption.
- Both the PSUs are field replaceable units. When one PSU is removed or installed, there is no functional impact to the router.

Supported Transceiver Modules

For more information on the supported transceiver modules, see [Transceiver Module Group \(TMG\) Compatibility Matrix](#). In the **Begin your Search** search box, enter the keyword and click **Enter**.



CHAPTER 3

Prepare for Installation

Before you install the router, you must prepare your site for installation.



Note

The images in this chapter are only for representational purposes, unless specified otherwise. The chassis' actual appearance and size may vary.



Warning

Statement 1071—Warning Definition

IMPORTANT SAFETY INSTRUCTIONS

Before you work on any equipment, be aware of the hazards involved with electrical circuitry and be familiar with standard practices for preventing accidents. Read the installation instructions before using, installing, or connecting the system to the power source. Use the statement number at the beginning of each warning statement to locate its translation in the translated safety warnings for this device.

SAVE THESE INSTRUCTIONS



- [General Precautions, on page 30](#)
- [Compliance and Safety Information, on page 30](#)
- [Laser Safety, on page 31](#)
- [Energy Hazard, on page 32](#)
- [Preventing Electrostatic Discharge Damage, on page 32](#)
- [Cautions and regulatory compliance statements for NEBS, on page 32](#)
- [Power supply considerations, on page 34](#)
- [Site Planning Checklist, on page 35](#)
- [Environmental Requirements, on page 36](#)
- [Temperature, on page 37](#)
- [Dust and Particles, on page 37](#)
- [Air Quality, on page 38](#)
- [Corrosion, on page 38](#)
- [Airflow Guidelines, on page 39](#)

- [Site Power Guidelines, on page 42](#)
- [Site Cabling Guidelines, on page 43](#)
- [Tools and Equipment, on page 44](#)
- [Prepare Your Location , on page 45](#)
- [Prepare Yourself , on page 47](#)
- [Prepare Rack for Router Installation, on page 47](#)
- [Unpack the Cisco 8010 Large and Medium Density Routers, on page 48](#)

General Precautions

Observe the following general precautions when using and working with your router:

- Keep your system components away from radiators and heat sources, and do not block cooling vents.
- Do not spill food or liquids on your system components, and never operate the product in a wet environment.
- Do not push any objects into the openings of your system components. Doing so can cause fire or electric shock by shorting out interior components.
- Position system cables and power supply cable carefully. Route system cables, the power supply cable, and plug so that they are not stepped on or tripped over. Be sure that nothing else rests on your system component cables or power cable.
- Do not modify power cables or plugs. Consult a licensed electrician or your power company for site modifications. Always follow your local and national wiring rules.
- If you turn off your system, wait at least 30 seconds before turning it on again to avoid damage to the system components.

Compliance and Safety Information

The Cisco 8000 Series Routers are designed to meet the regulatory compliance and safety approval requirements. For detailed safety information, see *Regulatory Compliance and Safety Information—Cisco 8010 Series Routers*.



Note **Statement 1089**—Instructed and Skilled Person Definitions

An instructed person is someone who has been instructed and trained by a skilled person and takes the necessary precautions when working with equipment.

A skilled person or qualified personnel is someone who has training or experience in the equipment technology and understands potential hazards when working with equipment.



Warning **Statement 9001**—Product Disposal

Ultimate disposal of this product should be handled according to all national laws and regulations.

**Warning****Statement 1074**—Comply with Local and National Electrical Codes

To reduce risk of electric shock or fire, installation of the equipment must comply with local and national electrical codes.

**Warning****Statement 1090**—Installation by Skilled Person

Only a skilled person should be allowed to install, replace, or service this equipment. See statement 1089 for the definition of a skilled person.

**Warning****Statement 1091**—Installation by an Instructed Person

Only an instructed person or skilled person should be allowed to install, replace, or service this equipment. See statement 1089 for the definition of an instructed or skilled person.

**Warning****Statement 1029**—Blank Faceplates and Cover Panels

Blank faceplates and cover panels serve three important functions: they reduce the risk of electric shock and fire, they contain electromagnetic interference (EMI) that might disrupt other equipment, and they direct the flow of cooling air through the chassis. Do not operate the system unless all cards, faceplates, front covers, and rear covers are in place.

Laser Safety

**Warning****Statement 1051**—Laser Radiation

Invisible laser radiation may be emitted from disconnected fibers or connectors. Do not stare into beams or view directly with optical instruments.

**Warning****Statement 1055**—Class 1/1M Laser

Invisible laser radiation is present. Do not expose to users of telescopic optics. This applies to Class 1/1M laser products.



**Warning****Statement 1255—Laser Compliance Statement**

Pluggable optical modules comply with IEC 60825-1 Ed. 3 and 21 CFR 1040.10 and 1040.11 with or without exception for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.

Energy Hazard

The routers can be configured for a DC power source. Do not touch terminals while they are live. Observe the following warning to prevent injury.

**Warning****Statement 1086—Replace Cover on Power Terminals**

Hazardous voltage or energy may be present on power terminals. To reduce the risk of electric shock, make sure the power terminal cover is in place when the power terminal is not being serviced. Be sure uninsulated conductors are not accessible when the cover is in place.

Preventing Electrostatic Discharge Damage

Many router components can be damaged by static electricity. Not exercising the proper electrostatic discharge (ESD) precautions can result in intermittent or complete component failures. To minimize the potential for ESD damage, always use an ESD-preventive antistatic wrist strap (or ankle strap) and ensure that it makes adequate skin contact.

**Note**

Check the resistance value of the ESD-preventive strap periodically. The measurement should be 1–10 megohms.

Before you perform any of the procedures in this guide, attach an ESD-preventive strap to your wrist and connect the leash to the chassis.

Cautions and regulatory compliance statements for NEBS

The NEBS-GR-1089-CORE regulatory compliance statements and requirements are discussed in this section.

**Note****Statement 7001—ESD Mitigation**

This equipment may be ESD sensitive. Always use an ESD ankle or wrist strap before handling equipment. Connect the equipment end of the ESD strap to an unfinished surface of the equipment chassis or to the ESD jack on the equipment if provided.

**Warning****Statement 7003**—Shielded Cable Requirements for Intrabuilding Lightning Surge

The intrabuilding port(s) of the equipment or subassembly, which is the management Ethernet port, must use shielded intrabuilding cabling/wiring that is grounded at both ends.

**Warning****Statement 7005**—Intrabuilding Lightning Surge and AC Power Fault

The intrabuilding port(s) of the equipment or subassembly, which is the management Ethernet port, must not be metallically connected to interfaces that connect to the OSP or its wiring. These interfaces are designed for use as intrabuilding interfaces only (Type 2 or Type 4 ports as described in GR-1089-CORE) and require isolation from the exposed OSP cabling. The addition of Primary Protectors is not sufficient protection in order to connect these interfaces metallically to OSP wiring.

**Warning****Statement 7011**—Surge Protection Device Requirements for GR-1089 Antenna Ports

Protect equipment antenna ports, that are classified as Type 6 according to GR-1089-CORE, with lightning surge protectors that are rated at a minimum of 600 V peak surge of 1.2/50 uS duration.

Connecting a Cable to the GNSS Antenna Interface

- GNSS modules have built-in ESD protections on all pins, including the RF-input pin. However, additional surge protection is required if an outdoor antenna is being connected. The Lightning Protector must be able to provide a low clamping voltage (less than 600V).
- A lightning protection must be mounted at the place where the antenna cable enters the building. The primary lightning protection must be capable of conducting all potentially dangerous electrical energy to PE (Protective Earth).
- Surge arrestors should support DC-pass and suitable for the GPS frequency range as per the constellation used with low attenuation.

**Warning****Statement 7012**—Equipment Interfacing with AC Power Ports

Connect this equipment to AC mains that are provided with a surge protective device (SPD) at the service equipment that complies with NFPA 70, the National Electrical Code (NEC).

**Note****Statement 7013**—Equipment Grounding Systems—Common Bonding Network (CBN)

This equipment is suitable for installations using the CBN.

**Note****Statement 7016**—Battery Return Conductor

The battery return conductor of this equipment shall be treated as (DC-I).

**Note** **Statement 7018**—System Recover Time

The equipment is designed to boot up in less than 30 minutes provided the neighboring devices are fully operational.

**Note** **Statement 8015**—Installation Location Network Telecommunications Facilities

This equipment is suitable for installation in network telecommunications facilities.

**Note** **Statement 8016**—Installation Location Where the National Electric Code (NEC) Applies

This equipment is suitable for installation in locations where the NEC applies.

Power supply considerations

Check the power at your site to ensure that you are receiving clean power (free of spikes and noise). If necessary, install a power conditioner.

Power Connection Guidelines

This section provides guidelines for connecting the device power supplies to the site power source.

**Warning** **Statement 1024**—Ground Conductor

This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

Guidelines for DC-Powered Systems

Basic guidelines for DC-powered systems include the following:

- Each chassis power supply has its own dedicated input power source. The source must comply with the safety extra-low voltage (SELV) requirements in the UL 60950-1, UL 62368-1, CSA 60950-1, CSA 62368-1, EN 60950-1, EN 62368-1, IEC 60950-1, and IEC 62368-1 standards.
- Protect the circuit by a dedicated two-pole DC circuit breaker. Ensure that the circuit breaker is sized according to the power supply input rating and local or national code requirements.
- The circuit breaker is considered as the disconnect device and is easily accessible.
- The system ground is the power supply and chassis ground.
- Use the grounding lug to attach a wrist strap for ESD protection during servicing.

- Do not connect the DC return wire to the system frame or to the system-grounding equipment.
- Ensure that the DC return is grounded at the source side.
- Ensure that each power feed of the equipment is connected to different power sources.

Guidelines for AC-Powered Systems

Basic guidelines for AC-powered systems include the following:

- Each chassis power supply has its own dedicated branch circuit.
- Ensure that the circuit breaker is sized according to the power supply input rating and local or national code requirements.
- The AC power receptacles that are used to plug in the chassis must be the grounding type. The grounding conductors that connect to the receptacles must connect to protective earth ground at the service equipment.

Prevent Power Loss

Use the following guidelines to prevent power loss to the device:

- To prevent input power loss, ensure that the maximum load on each circuit supplying the power is within the current ratings of the wiring and breakers.
- In some systems, you can use an UPS to protect against power failures at your site. Avoid UPS types that use ferroresonant technology. These UPS types can become unstable with systems such as the device, which can have substantial current-draw fluctuations due to bursty data traffic patterns.

Determining power requirements is useful for planning the power distribution system to support the device.

Site Planning Checklist

Use the following checklist to perform and account for all the site planning tasks described in this chapter:

- The site meets the environmental requirements.
- The site's air conditioning system can compensate for the heat dissipation of the router.
- The floor space that the router occupies can support the weight of the system.
- Electrical service to the site complies with the requirements.
- The electrical circuit servicing the router complies with the requirements.
- Consideration has been given to the console port wiring and limitations of the cabling involved, according to TIA/EIA-232F.
- The router Ethernet cabling distances are within the prescribed limitations.
- The equipment rack in which you plan to install the router complies with prescribed requirements.
- Factors that have been carefully considered when selecting the location of the rack:
 - safety,

- ease of maintenance, and
- proper airflow.

Environmental Requirements

Cisco 8011-32Y8L2H2FH Router

- Temperature must be maintained within -5°C to $+55^{\circ}\text{C}$. The maximum and minimum operating temperatures are supported at 1800m.
- The operating climate, as defined by GR-63-CORE, is between:
 - -5°C to $+55^{\circ}\text{C}$ ($+23^{\circ}\text{F}$ to $+131^{\circ}\text{F}$) at 1800m
 - 5% and 85% RH
 - Airborne particulate concentration level < 20 micro grams per meter cube
 - Altitude: 0 to 4000 meters

For more information on Environmental properties and Regulatory standards, see the [Cisco 8010 Series Large Density Routers Data Sheet](#).

Cisco 8011-12G12X4Y-A and Cisco 8011-12G12X4Y-D Routers

- For outside plant installation (cell site cabinet, hut, and so on), equipment must be protected against airborne contaminants, dust, moisture, insects, pests, corrosive gases, polluted air, or other reactive elements.
- Sealed equipment chamber with air-conditioning or a heat exchanger is recommended for OSP deployments.
- The equipment chamber must comply with the temperature and clearance requirements.

Examples of such cabinets include IP65/IP66 cabinets with heat exchanger complying with Telcordia GR487.

- Temperature must be maintained within:
 - Cisco 8011-12G12X4Y-A: -20°C to $+65^{\circ}\text{C}$ (-4°F to $+149^{\circ}\text{F}$) at MSL
 - Cisco 8011-12G12X4Y-D: -40°C to $+65^{\circ}\text{C}$ (-40°F to $+149^{\circ}\text{F}$) at MSL

The maximum and minimum operating temperatures are supported at Mean Sea Level (MSL).

- Cisco 8011-12G12X4Y-A: The equipment shall be placed inside an enclosure (that is protected from direct outside weather and environmental stresses by the enclosure), and where the operating climate is between:
 - -20°C to $+65^{\circ}\text{C}$ (-4°F to $+149^{\circ}\text{F}$)
 - 5% and 85% RH
 - Airborne particulate concentration level < 20 micro grams per meter cube

- Altitude: 0 to 4000 meters

Cisco 8011-12G12X4Y-A is non-complaint to GR3108 Class 2.

- Cisco 8011-12G12X4Y-D: The equipment shall be placed inside an enclosure (that is protected from direct outside weather and environmental stresses by the enclosure), and where the operating climate, as defined by GR3108 Class 2, is between:
 - -40°C to $+65^{\circ}\text{C}$ (-40°F to $+149^{\circ}\text{F}$)
 - 5% and 85% RH
 - Airborne particulate concentration level < 20 micro grams per meter cube
 - Altitude: 0 to 4000 meters

Temperature

Temperature extremes may cause a system to operate at reduced efficiency and cause various problems, including premature aging and failure of chips, and failure of mechanical devices. Extreme temperature fluctuations may also cause chips to become loose in their sockets.

Observe these guidelines:

- Ensure that the chassis has adequate ventilation.
- Do not place the chassis within a closed-in wall unit or on top of cloth, which can act as thermal insulation.
- Do not place the chassis where it receives direct sunlight, particularly in the afternoon.
- Do not place the chassis next to a heat source of any kind, including heating vents.
- Adequate ventilation is important at high altitudes. Make sure that all the slots and openings on the system remain unobstructed, especially the fan vent on the chassis.
- Clean the installation site at regular intervals to avoid the buildup of dust and debris, which may cause a system to overheat.



Note Failure to observe these guidelines may damage the internal components of the chassis.

Dust and Particles

Fans cool power supplies and system components by drawing in room-temperature air and exhausting heated air out through various openings in the chassis. However, fans also ingest dust and other particles, causing contaminant buildup in the system and increased internal chassis temperature. A clean operating environment can greatly reduce the negative effects of dust and other particles, which act as insulators and interfere with the mechanical components in the system.

PM2.5 and PM10 are some of the commonly available air quality parameters to indicate dust concentration levels. Periodically check dust concentration levels and provide required protection to improve air quality around the equipment.

Air Quality

Dust is everywhere and often invisible to the naked eye. It consists of fine particles in the air that originate from various sources, such as soil dust lifted by weather, from volcanic eruptions, or pollution. Dust at an installation site may contain small amounts of textile, paper fibers, or minerals from outdoor soil. It may also contain natural contaminants, such as chlorine from the marine environment and industrial contaminants such as sulfur. Ionized dust and debris are dangerous and get attracted to electronic equipment.

The accumulation of dust and debris on electronic equipment has the following adverse effects:

- It increases the operating temperature of the equipment. According to the Arrhenius effect, an increase in the operating temperature leads to a decrease in reliability and life of the equipment.
- The moisture and corrosive elements that are present in the dust can corrode the electronic or mechanical components and cause premature board failure.

These adverse effects are further accelerated by the presence of fans in the data networking equipment that ingest dust and other particles into the equipment. Higher the volume of air that is generated by the fans for cooling, the higher the quantity of dust and particulates that get deposited and trapped inside the equipment.

Corrosion

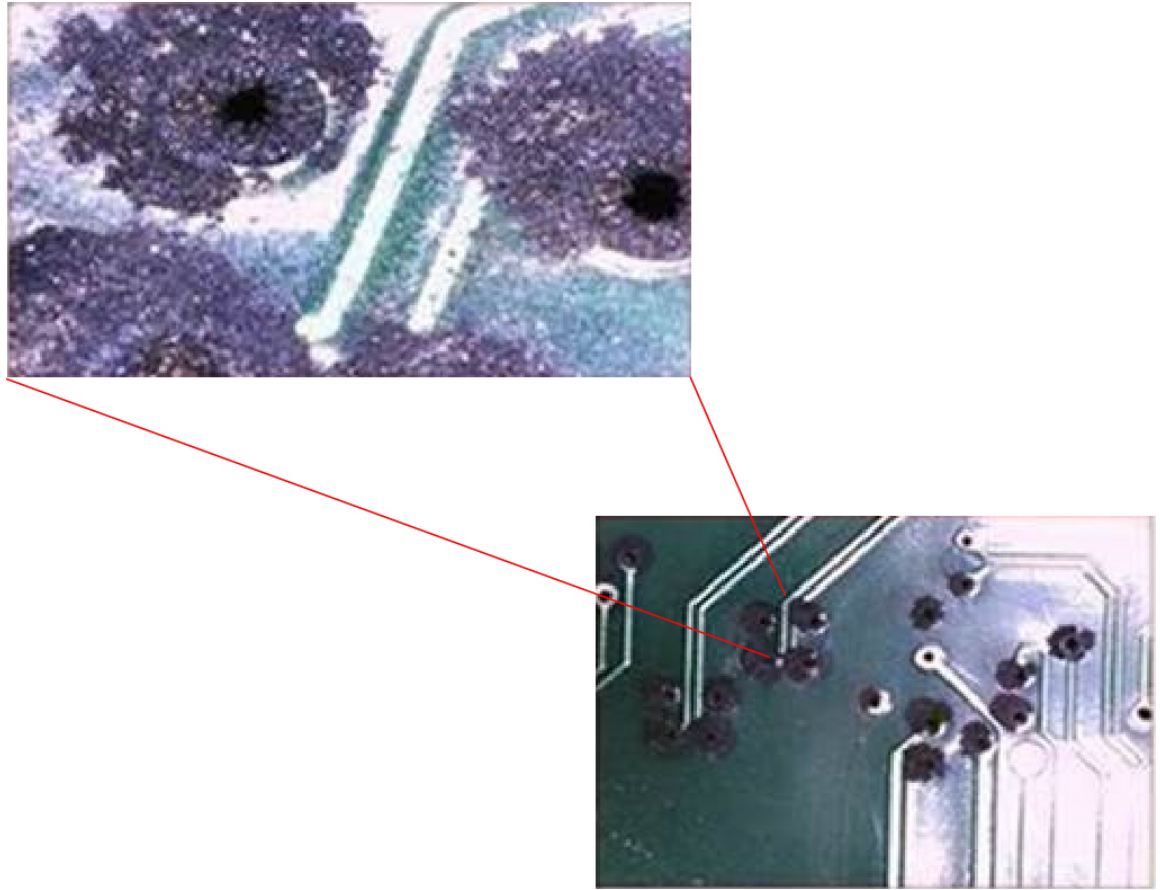
Corrosion is a chemical reaction that occurs between electronic components, gases, and humidity, which results in metal deterioration. Corrosion attacks edge connectors, pin connectors, IC plug-in sockets, wire wraps, and all other metal components. Depending on the type and concentration level of the corrosive gases, performance degradation of the components occurs either rapidly or over a period of time. It also leads to blocked currents, brittle connection points, and overheated electrical systems. Corrosion by-products from insulating layers on circuits and causes electronic failure, short circuits, pitting, and metal loss.

A type of corrosion known as creep corrosion, that primarily affects PCBA (Printed Circuit Board Assembly) occurs when the PCBA is subjected to a harsh, and sulfur-rich (hydrogen sulfide) end-use environment over a prolonged period of time. The corrosion begins on certain exposed metals, such as copper and silver, and then creeps along the remaining metal surface either causing electrical short circuits or creating holes. Creep corrosion also occurs on electronic components such as resistors and PCBs.



Note To prevent corrosion, remove or minimize the presence of dust and particulates at the installation site by following the guidelines mentioned in ANSI 71-04-2013 regulations.

Figure 17: PCB with Corrosion on its Metal Contacts



Airflow Guidelines

Cool air is circulated through the router by fans that are located along the rear side of the router. The internal fans maintain acceptable operating temperatures for the internal components by drawing in cool air through the vents, and circulating the air through the router.

To ensure adequate airflow, we recommend that you always maintain this minimum clearance distance for the router:

- Front clearance:
 - Cisco 8011-4G24Y4H-I—5.5 inches (14 centimeters)
 - Cisco 8011-32Y8L2H2FH—5.5 inches (14 centimeters)
 - Cisco 8011-12G12X4Y—5 inches (12.7 centimeters)
- Rear clearance:
 - Cisco 8011-4G24Y4H-I—2 inches (5.08 centimeters)
 - Cisco 8011-32Y8L2H2FH—9.8 inches (25 centimeters)



Note This clearance is required to replace PSUs.

- Cisco 8011-12G12X4Y—0.5 inches (1.27 centimeters)

Figure 18: Cisco 8011-4G24Y4H-I Router Airflow is from Front to Back

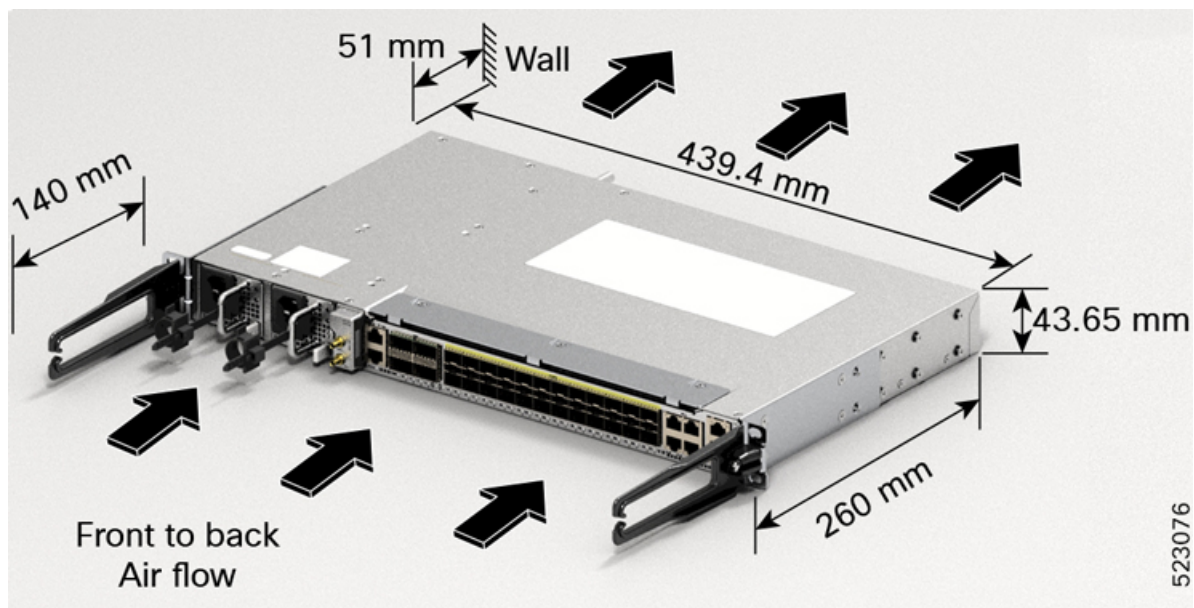


Figure 19: Cisco 8011-32Y8L2H2FH Router Airflow is from Front to Back

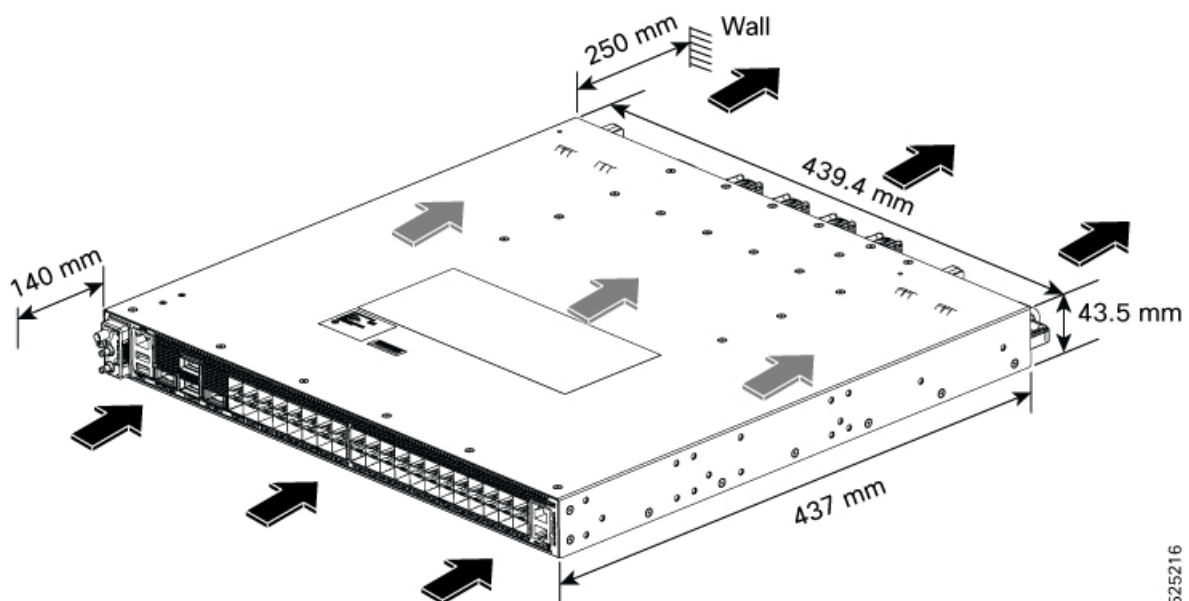
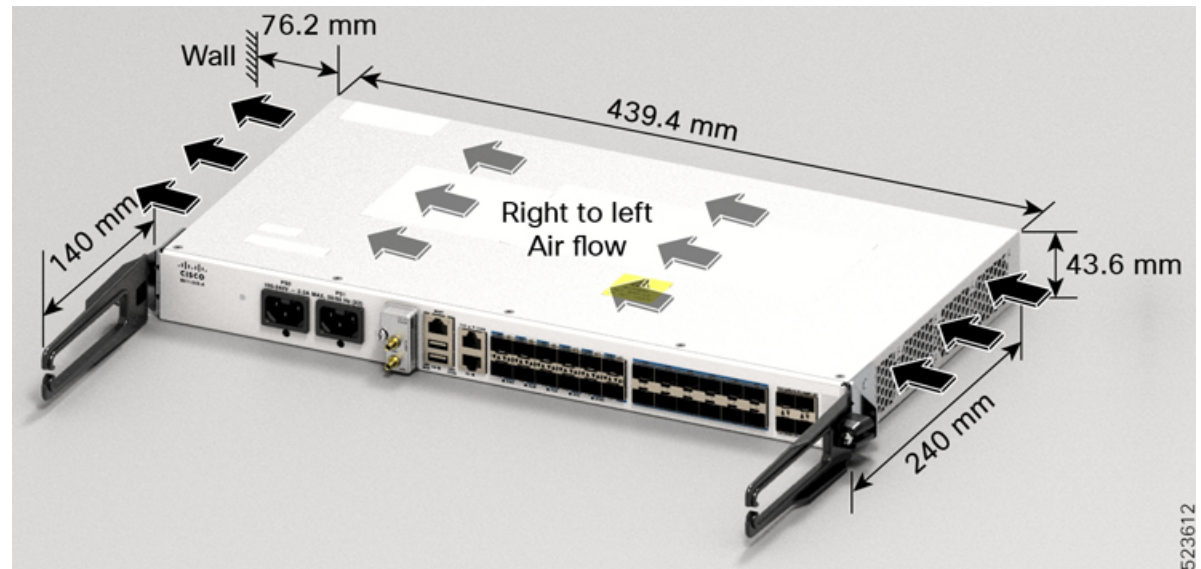


Figure 20: Cisco 8011-12G12X4Y Router Airflow is from Right to Left



Note these points:

- When installing the router in a back-to-back position with another device, ensure that there's a minimum of 3.9 inches (10 centimeters) airflow clearance between the two devices.
- If airflow through the equipment rack and the routers that occupy it's blocked or restricted, or if the ambient air being drawn into the rack is too warm, an overtemperature condition may occur within the rack and the routers that occupy it.
- The site must be as dust-free as possible. Dust tends to clog the router fans, reducing the flow of cooling air through the equipment rack and the routers that occupy it, thereby increasing the risk of an overtemperature condition.
- Enclosed racks must have adequate ventilation. Ensure that the rack isn't congested because each router generates heat. An enclosed rack must have louvered sides and a fan to provide cooling air. The equipment generates heat near the bottom of the rack, which can be drawn upward into the intake ports of the equipment above.
- When mounting a router in an open rack, ensure that the rack frame doesn't block the exhaust fans.
- When rack-installed equipment fails, especially equipment in an enclosed rack, try operating the equipment by itself, if possible. Power off all the other equipment in the rack (and in adjacent racks) to give the router maximum cooling air and clean power.
- Avoid installing the router in a location in which the router air intake vents may draw in the exhaust air from adjacent equipment. Consider how the air flows through the router; the airflow direction is front to back, with ambient air drawn in from the vents located on the front panel of the router.

Site Power Guidelines

The chassis has specific power and electrical wiring requirements. Adhering to these requirements ensures the reliable operation of the system. Follow these precautions and recommendations when planning your site power for the chassis:

- The redundant power option provides a second, identical power supply to ensure uninterrupted power supply.
- Connect each power supply to a separate input power source. Otherwise, it results in a total power failure to the system due to a fault in the external wiring or a tripped circuit breaker.
- To prevent loss of input power, ensure that the maximum load on each circuit is within the current ratings of the wiring and the breakers.
- Circuit breaker specifications—a maximum of 20A (North America) and 16A (Europe) for AC power feed.
- Check the power at your site before installation, and periodically after installation, to ensure that you are receiving clean power. If necessary, install a power conditioner.
- Provide proper grounding to avoid personal injury and damage to the equipment due to power surges or lightning striking power lines. The chassis ground must be attached to a central office or other interior ground system.

**Note**

The chassis installation must comply with all the applicable codes, and is approved for use with only copper conductors. The ground bond-fastening hardware must be compatible and preclude loosening, deterioration, and electrochemical corrosion of hardware and joined material. Attachment of the chassis ground to a central office or other interior ground system must be made with a 6-AWG gauge wire copper ground conductor.

**Caution**

The AC PSU requires external surge protection devices for installations where electrical surges higher than 2KV common mode and 2KV differential mode are expected. Failure to do so can result in permanent damage to the product.

The DC PSU requires external surge protection devices for installations where electrical surges higher than 2KV common mode and 1KV differential mode are expected. Failure to do so can result in permanent damage to the product.

Electrical Circuit Requirements

Each router requires a dedicated electrical circuit. If you equip the router with dual-power feeds, provide a separate circuit for each power supply to avoid compromising the power redundancy feature.

The routers can be powered by a DC source. Ensure that equipment grounding is present and observe the power-strip ratings. Make sure that the total ampere rating of all the products plugged into the power strip does not exceed 80% of the rating.

Site Cabling Guidelines

These are the guidelines for wiring and cabling at your site:

- When preparing your site for network connections to the router, consider:
 - the type of cable required for each component, and the cable limitations.
Possible cable types are fiber, thick or thin coaxial, foil twisted-pair, or unshielded twisted-pair cabling.
 - the distance limitations for signaling, ElectroMagnetic Interference (EMI), and connector compatibility.
 - any additional interface equipment you need, such as transceivers, hubs, switches, modems, Channel Service Units (CSU), or Data Service Units (DSU).
- Before you install the router, keep all the additional external equipments and cables handy. For information about ordering these equipments and cables, contact a Cisco customer service representative.
- The extent of your network and the distances between the network interface connections depend, in part, on the following factors:
 - Signal type
 - Signal speed
 - Transmission medium
- The distance and rate limits referenced in the following sections are the IEEE-recommended maximum speeds and distances for signaling purposes. Use this information as a guideline when planning your network connections *prior to* installing the router.
- If wires exceed the recommended distances, or if wires pass between buildings, give special consideration to the effect of a lightning strike in your vicinity. The electromagnetic pulse caused by lightning or other high-energy phenomena can easily couple enough energy into unshielded conductors to destroy electronic devices. If you have had problems of this sort in the past, you may want to consult experts in electrical surge suppression and shielding.

Asynchronous Terminal Connections

The router provides a console port to connect a terminal or computer for local console access. The router supports RS-232 asynchronous data with distance recommendations specified in the EIA-232 standard.

Interference Considerations

When wires are run for any significant distance, there is a risk that stray signals will be induced on the wires as interference. If interference signals are strong, they may cause data errors or damage to the equipment.

The following sections describe the sources of interference and how to minimize their effects on the router system.

Electromagnetic Interference

All the equipment powered by AC current can propagate electrical energy that can cause EMI and possibly affect the operation of other equipment. The typical sources of EMI are equipment power cords and power service cables from electric utility companies.

Strong EMI can destroy the signal drivers and receivers in the router and even create an electrical hazard by causing power surges through the power lines into installed equipment. These problems are rare, but could be catastrophic.

To resolve these problems, you need specialized knowledge and equipment that could consume substantial time and money. However, you can ensure that you have a properly grounded and shielded electrical environment, paying special attention to the need for electrical surge suppression.

Radio Frequency Interference

When electromagnetic fields act over a long distance, Radio Frequency Interference (RFI) may be propagated. Building wiring can often act as an antenna, receiving the RFI signals and creating more EMI on the wiring.

If you use twisted-pair cable in your plant wiring with a good distribution of grounding conductors, the plant wiring is unlikely to emit radio interference. If you exceed the recommended distances, use a high-quality twisted-pair cable with one ground conductor for each data signal.

Lightning and AC Power Fault Interference

If signal wires exceed the recommended cabling distances, or if signal wires pass between buildings, you should consider the effect that a lightning strike in your vicinity might have on the router.

The Electromagnetic Pulse (EMP) generated by lightning or other high-energy phenomena can couple enough energy into unshielded conductors to damage or destroy electronic equipment. If you have previously experienced such problems, you should consult with RFI and EMI experts to ensure that you have adequate electrical surge suppression and shielding of signal cables in your router operating environment.

Tools and Equipment

You need the following tools and equipment to install and upgrade the router and its components:

- ESD-preventive cord and wrist strap
- Antistatic mat or antistatic foam
- Number 1 and Number 2 Phillips-head screwdrivers
- #12-24 pan-head screws to secure the router to the equipment rack
- Cables for connecting to the network ports (depending on the configuration)
- Ethernet hub, switch, or PC with a network interface card for connecting to the Ethernet ports
- Console terminal (an ASCII terminal or a PC running terminal emulation software) that is configured for 115200 baud, 8 data bits, no parity, no flow control, and 1stop bit
- Console cable for connecting to the console port
- Ratcheting torque screwdriver with a Phillips head that exerts up to 20-pound force per square inch (in-lb) or 0.02-kilograms force per square millimeter (kgf/mm2) of pressure

- Crimping tool as specified by the ground lug manufacturer
- Wire-stripping tools for stripping both 6-AWG and 14-AWG wires
- Tape measure and level

Prepare Your Location

This section illustrates how the building that houses the chassis must be properly grounded to the earth ground.

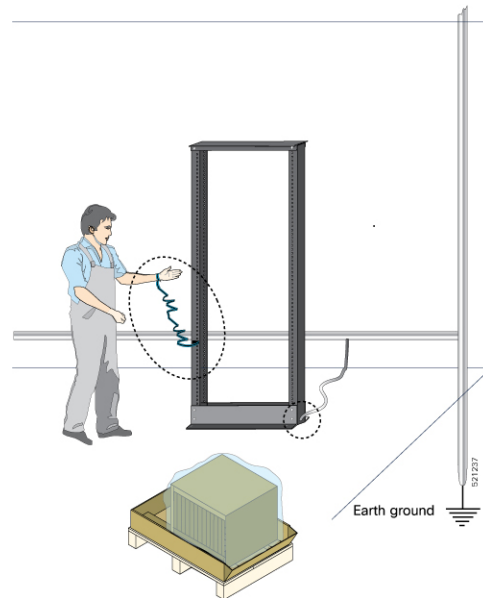
Figure 21: Building with Rack Room Connected to Earth Ground

Prepare Yourself

This section illustrates how to prepare yourself before removing the chassis from the sealed antistatic bag. The figures show how to wear the ESD strap around the wrist and how to connect the other end of the strap to the ground. ESD wrist straps are the primary means of controlling static charge on personnel.

Note: These images are for only representation purposes. The chassis' actual appearance and size would vary.

Figure 22: Wearing the ESD Strap



Prepare Rack for Router Installation

Install the chassis into a two-post standard rack with standard horizontal mounting rails. Before you mount the chassis into the rack, we recommend that you do the following:

Procedure

Step 1 Place the rack where you plan to install the chassis. Ensure that the rack is grounded to earth.

Step 2 Secure the rack to the floor.

To bolt the rack to the floor, a floor bolt kit (also called an anchor embedment kit) is required. For information on bolting the rack to the floor, consult a company that specializes in floor mounting kits (such as Hilti; see [Hilti.com](https://www.hilti.com) for details). Make sure that floor mounting bolts are accessible, especially if an annual retorquing of bolts is required.

Unpack the Cisco 8010 Large and Medium Density Routers

Ensure that there is sufficient room around the chassis pallet for unpacking.

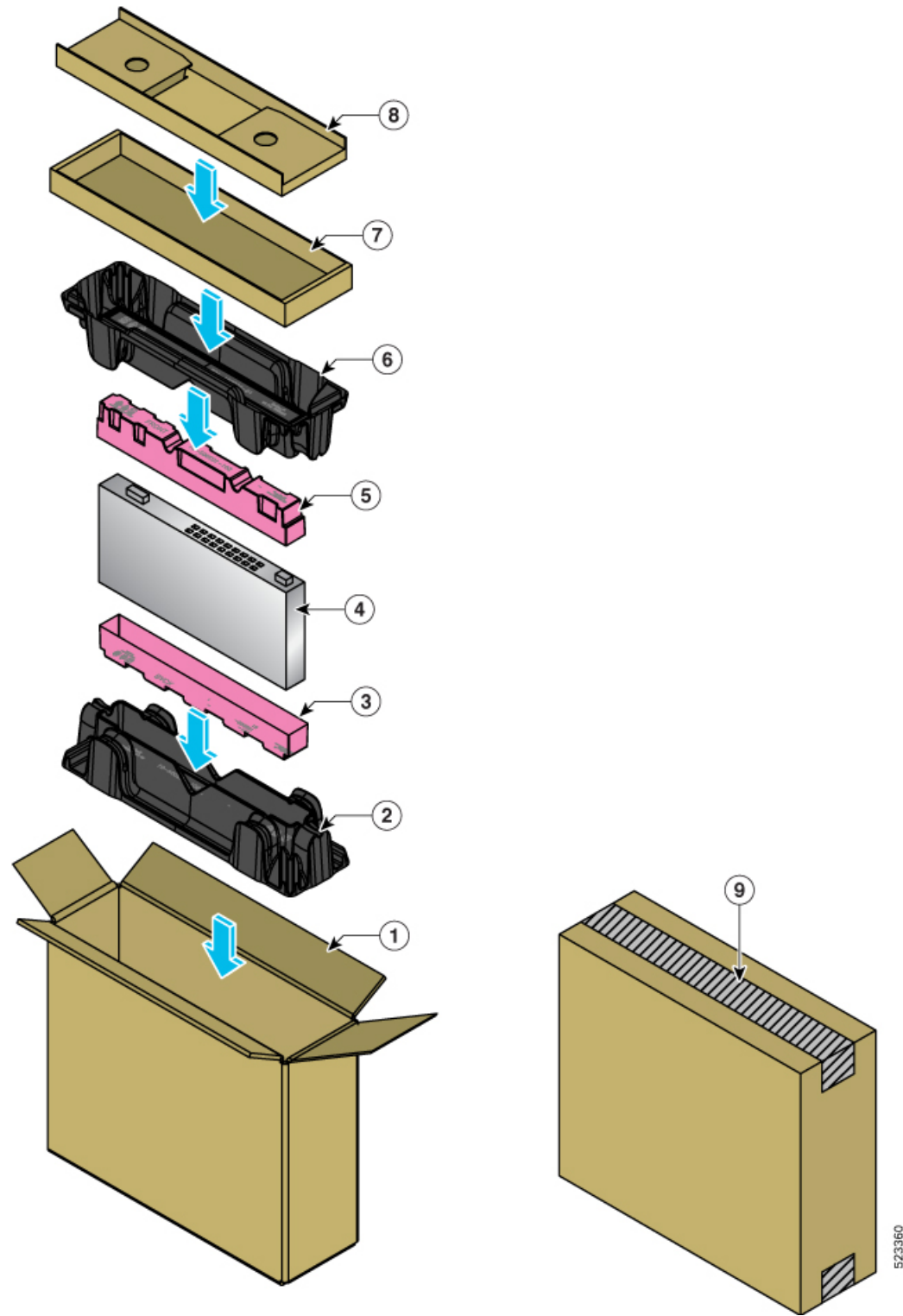
1. Remove the accessory tray and the packing material.
2. Carefully set the packing material aside.



Tip Be sure to save the packaging in case you need to return any of the components products.



Note These images are for only representation purposes. The chassis' actual appearance and size would vary.

Cisco 8011-4G24Y4H-I Router*Figure 23: Unpacking the Cisco 8011-4G24Y4H-I Router*

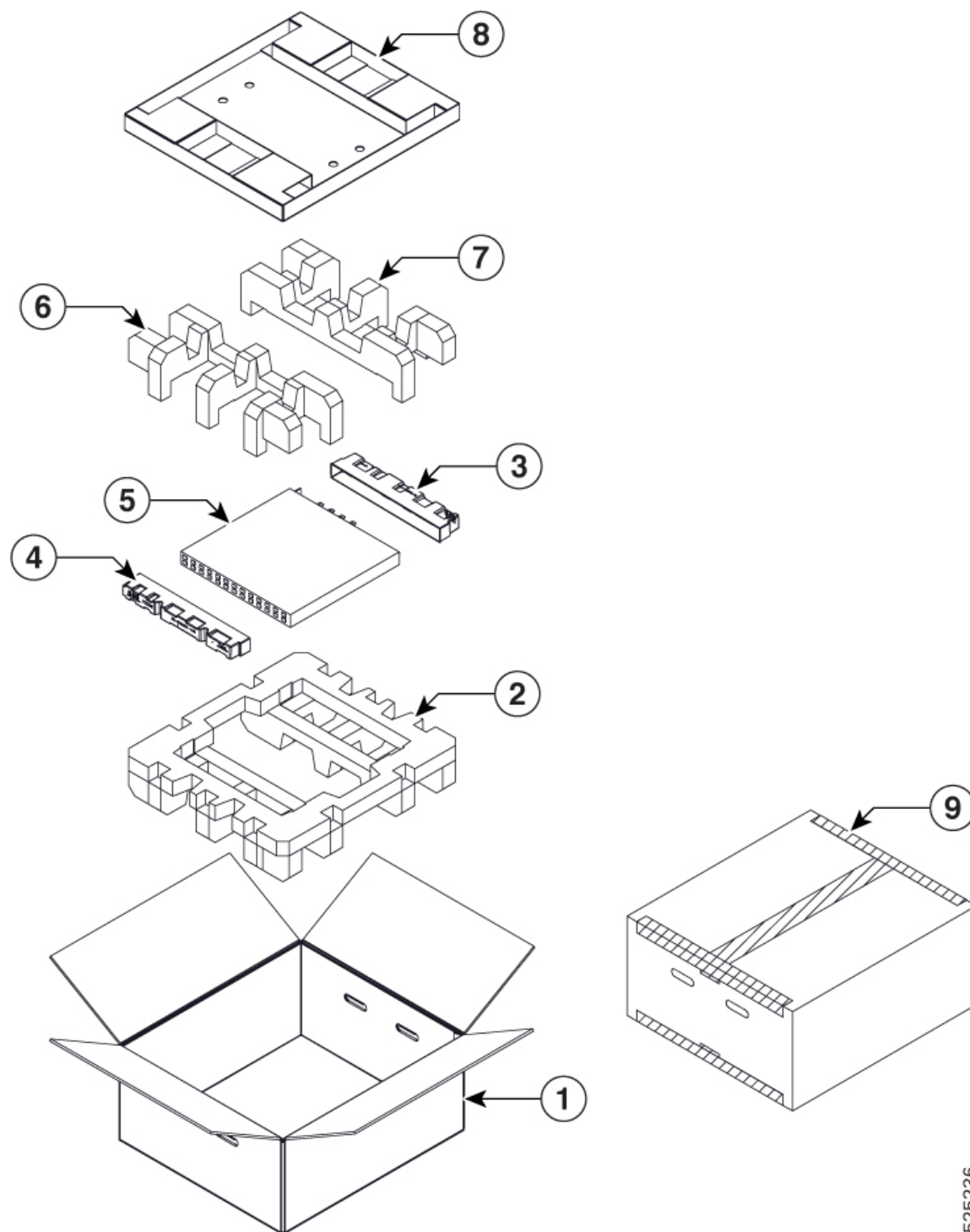
1	Regular Slotted Container (shipping box)	2, 6	Foam end caps
3, 5	Corrugated caps	4	Front end of the product
7	Accessory tray	8	Corrugated insert
9	Carton sealing tape		

Table 10: Accessories Kit for Cisco 8011-4G24Y4H-I Router

19- inch Rack Mount Brackets	23- inch Rack Mount Brackets	ETSI Rack Mount Brackets	Wall Mount Brackets	Desktop Brackets	Cable Management Brackets
RCKMT-19-V1	RCKMT-23-V1	RCKMT-ETSI-V1	53-101650-01	NA	CBL-BRKT-V1

Cisco 8011-32Y8L2H2FH Router

Figure 24: Unpacking the Cisco 8011-32Y8L2H2FH Router



1	Regular slotted container (shipping box)	2	Bottom foam cap
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3	Rear thermofoam-end cap	4	Front thermofoam-end cap
5	Top view of the product	6, 7	Top foam caps
8	Accessory tray	9	Carton sealing tape

Table 11: Accessories Kit for Cisco 8011-32Y8L2H2FH Router

19- inch Rack Mount Brackets	23- inch Rack Mount Brackets	ETSI Rack Mount Brackets	Desktop Brackets	Cable Management Brackets
RCKMT-19-KIT1 (2-post variant)	RCKMT-23-KIT1 (2-post variant)	RCKMT-ETSI-KIT1 (2-post variant)	NA	CBL-BRKT-V2
RCKMT-19-KIT2 (4-post variant)	RCKMT-23-KIT2 (4-post variant)	RCKMT-ETSI-KIT2 (4-post variant)	NA	NA

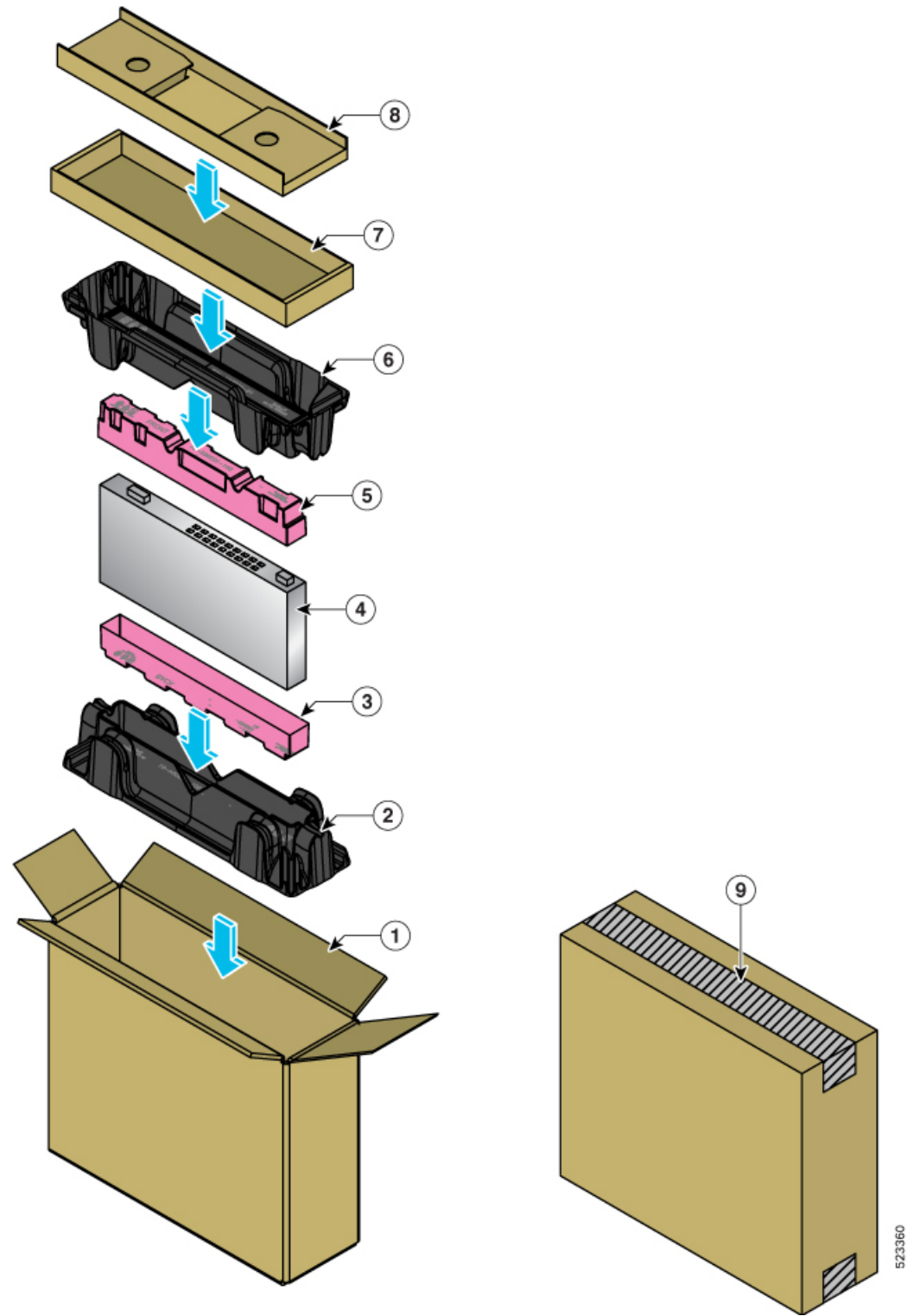


Caution

If the product is not in use, store the router in the initial packaged condition or in an ESD PE sealed bag with silica gel.

Cisco 8011-12G12X4Y Router

Figure 25: Unpacking the Cisco 8011-12G12X4Y Router



1	Regular Slotted Container (shipping box)	2, 6	Foam end caps
3, 5	Corrugated caps	4	Front end of the product
7	Accessory tray	8	Corrugated insert
9	Carton sealing tape		

Table 12: Accessories Kit for 8010 Router

19- inch Rack Mount Brackets	23- inch Rack Mount Brackets	ETSI Rack Mount Brackets	Wall Mount Brackets	Desktop Brackets	Cable Management Brackets
RCKMT-19-A RCKMT-19-D	RCKMT-23-A RCKMT-23-D	RCKMT-ETSI-A RCKMT-ETSI-D	NA	NA	CBL-BRKT-V1



CHAPTER 4

Install the Router

Before you begin this task, ensure that you have read and understood the safety warnings.



Note The installation instructions of the Cisco 8010 Large and Medium Density Routers are similar, and any differences between the router variants are specifically called out.

The illustrations are for reference purpose only, the actual router may vary based on your variant of Cisco 8010 Large and Medium Density Routers.

Installing the the Cisco router involves the following tasks:

- [Rack Mounting the Router, on page 55](#)
- [Ground the Router, on page 84](#)
- [Connect to an AC Power Source, on page 88](#)
- [Connect to a DC Power Source, on page 92](#)
- [Port Connection Guidelines, on page 98](#)
- [Console Port Connection, on page 98](#)
- [Management Ethernet Port Connection, on page 101](#)
- [Connecting Cables to Timing Interfaces Card, on page 102](#)
- [Inserting and Securing Timing Interfaces Card with GNSS Module, on page 103](#)
- [Install and Remove Transceiver Module, on page 104](#)
- [Connect Interface Ports, on page 110](#)
- [Maintain Transceivers and Optical Cables, on page 110](#)

Rack Mounting the Router

You can choose to either set up the router on a rack or wall mount it.

We recommend that you use the following racks while mounting the router.

Figure 26: Rack specification EIA (19 inches and 23 inches)

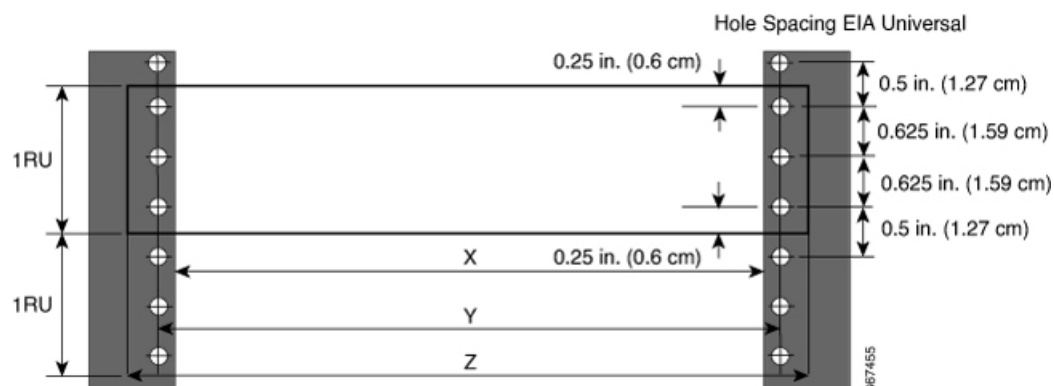
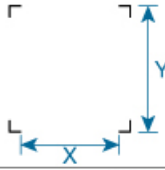
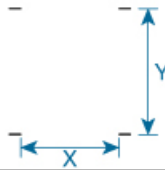
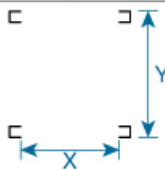


Table 13: Rack specification EIA (19 inches and 23 inches)

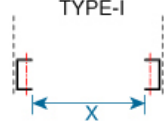
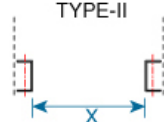
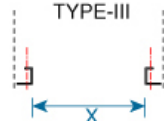
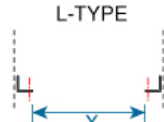
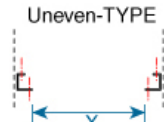
Post Type	Rack Type	Rack Front Opening (X)	Rack Mounting Hole Centre-Centre (Y)	Mounting Flange Dimension (Z)
4 Post	19 inches (48.3 centimeters)	17.75 inches (45 centimeters)	18.31 inches (46.5 centimeters)	19 inches (48.2 centimeters)
2 Post				
4 Post	23 inches (58.4 centimeters)	21.75 inches (55.24 centimeters)	22.31 inches (56.6 centimeters)	23 inches (58.4 centimeters)
2 Post				

Figure 27: Four Post Rack Type

4 – Post Type (Hole EIA Universal)		Width Available (X)	Compatibility
All 23" Type rack		552.45mm (21.75")	Yes
All ETSI rack (21" rack)		500.0mm (19.68")	Yes
19" Type rack L-Type Post		17.75" (450.8 mm)	Yes
		17.50" (444.5 mm)	No
19" Type Racks Flat-Post		17.75" (450.8 mm)	Yes
		17.50" (444.5 mm)	No
19" Type racks C- Type Post		17.75" (450.8 mm)	Yes
		17.50" (444.5 mm)	No

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Figure 28: Two Post Rack Type

2 – Post Type (Hole EIA Universal)	X – 19" Rack	Compatibility	X-23" Rack	Compatibility
TYPE-I 	17.75" (450.8 mm)	Yes	21.75" (552.45mm)	Yes
	17.50" (444.5 mm)	No	21.75" (552.45mm)	Yes
TYPE-II 	17.75" (450.8 mm)	Yes	21.75" (552.45mm)	Yes
	17.50" (444.5 mm)	No	21.75" (552.45mm)	Yes
TYPE-III 	17.75" (450.8 mm)	Yes	21.75" (552.45mm)	Yes
	17.50" (444.5 mm)	No	21.75" (552.45mm)	Yes
L-TYPE 	17.75" (450.8 mm)	Yes	21.75" (552.45mm)	Yes
	17.50" (444.5 mm)	No	21.75" (552.45mm)	Yes
Uneven-TYPE 	17.75" (450.8 mm)	Yes	21.75" (552.45mm)	Yes
	17.50" (444.5 mm)	No	21.75" (552.45mm)	Yes

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Rack Mounting Brackets

The router is shipped with rack mounting brackets that are to be secured on the sides of the router.



Caution If the rack is on wheels, ensure that the brakes are engaged or the rack is otherwise stabilized.

Cisco 8011-4G24Y4H-I Router

Table 14: Rack-Mount Kit Product Identification Number

Product Identification Number	Description
RCKMT-19-V1	19 inch rack mounting kit
RCKMT-23-V1	23 inch rack mounting kit
RCKMT-ETSI-V1	ETSI rack mounting kit
53-101699-01	Grounding lug kit
CBL-BRKT-V1	Cable management

Product Identification Number	Description
53-101650-01	Wall mount bracket

Cisco 8011-32Y8L2H2FH Router

Table 15: Rack-mount kit product identification number

Product identification number	Description
RCKMT-19-KIT1	19-inch 2-post rack mounting kit
RCKMT-19-KIT2	19-inch 4-post rack mounting kit
RCKMT-23-KIT1	23-inch 2-post rack mounting kit
RCKMT-23-KIT2	23-inch 4-post rack mounting kit
RCKMT-ETSI-KIT1	ETSI 2-post rack mounting kit
RCKMT-ETSI-KIT2	ETSI 4-post rack mounting kit
CBL-BRKT-V2	Cable management

Cisco 8011-12G12X4Y Router

Table 16: Rack-Mount Kit Product Identification Number

Product Identification Number	Description
RCKMT-19-A RCKMT-19-D	19 inch rack mounting kit
RCKMT-23-A RCKMT-23-D	23 inch rack mounting kit
RCKMT-ETSI-A RCKMT-ETSI-D	ETSI rack mounting kit
CBL-BRKT-V1	Cable management

Table 17: Router Rack-Mount Kit

Quantity	Part Description
2	Rack-mount brackets
8 (48-101850-01)	M3 x 0.5 x 6-mm Phillips flat-head screws
2 (48-101620-01)	SCR, M, PAN, PH, SEX, 10-32 x 0.365"L, CSwZN, nickel alloy

Quantity	Part Description
1 (32-0619-01)	LUG, FAST, UNIN, #6AWG, #10, 2 HOLES
4 (48-101690-01)	SCR, M, PAN, PH, 12-24 x 0.49"L, CSwZN, nickel alloy

**Caution**

- Ensure that you secure the lugs only with the Cisco-provided screw or a Phillips head screw with an integrated washer of 10-32 x 0.3125 inch. Secure the screws only while assembling the lugs.
- Using excessive length screw for mounting the grounding lug may cause short-circuit as it may come in contact with parts inside the router.

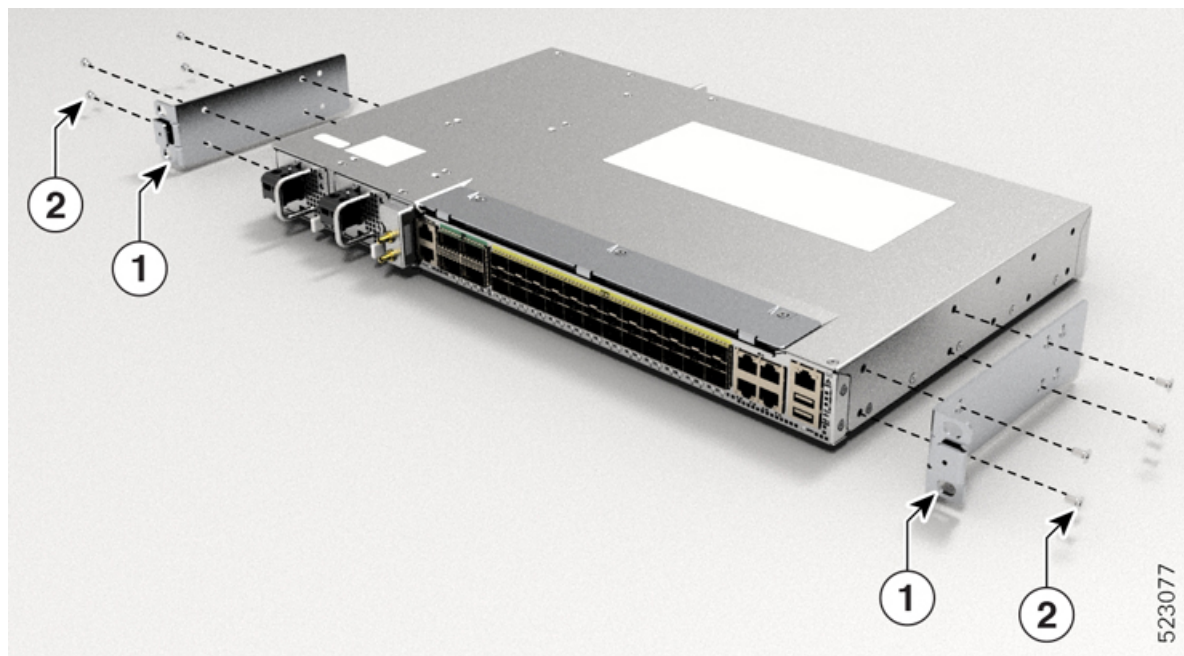
Mounting the Router on the Rack

Cisco 8011-4G24Y4H-I Router

To mount the router on the rack:

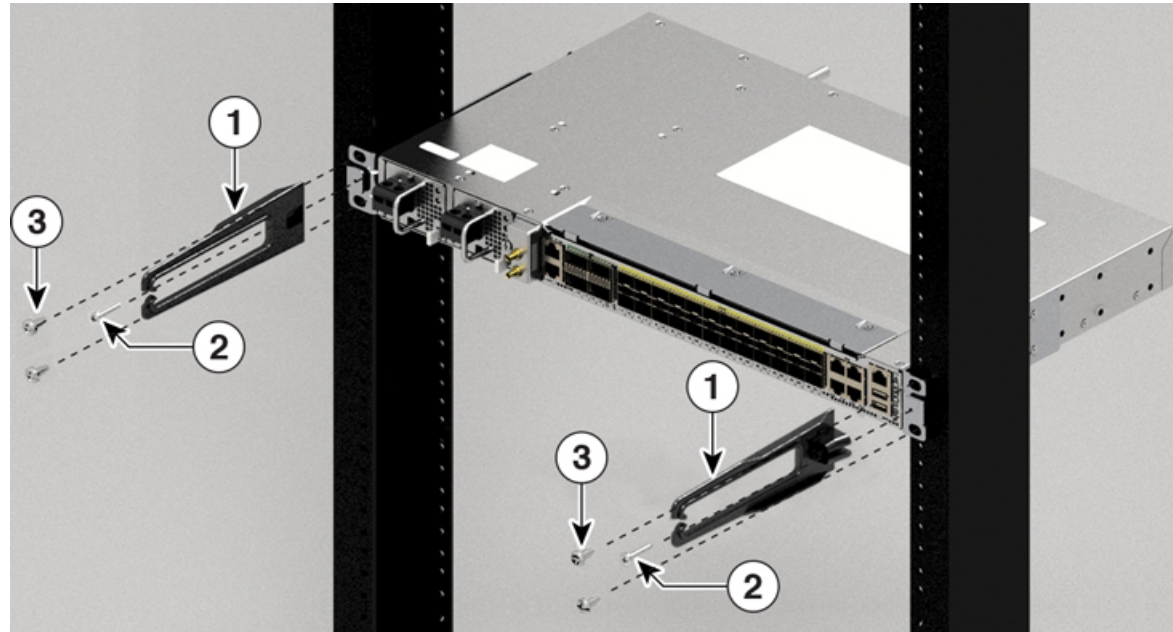
1. Attach the rack-mount brackets and the cable guides to the router as follows:
 - a. The router has port-side intake modules, position the router so that its ports are facing the cold aisle.
 - b. Position the bracket ears facing front or middle rack-mount, on the side of the chassis so that the holes are aligned.
 - c. Use four M4 screws to attach the brackets to the chassis. Tighten the M4 screws to 13.3 inch-pounds (1.5 Nm).

Figure 29: Installing 19 inch Rack-Mount Brackets in the Front



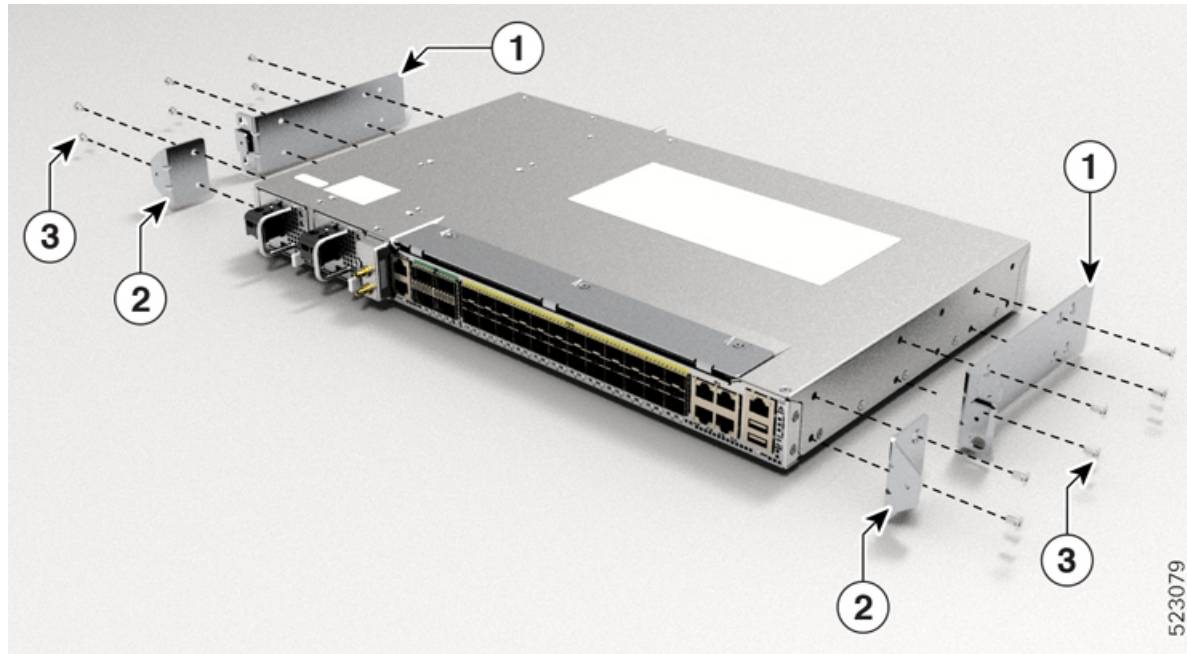
1	Mount Bracket
2	Screw

Figure 30: Installing Cable Management and 19 inch Rack-Mount Brackets in the Front



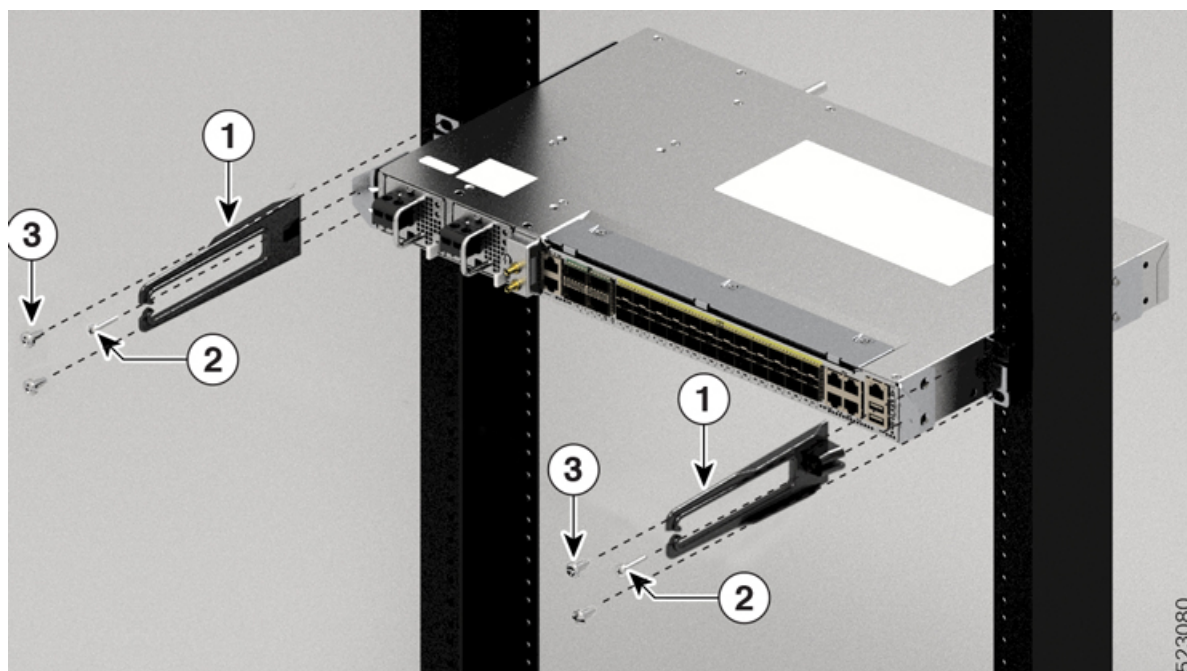
1	Cable Management Bracket
2	Cable Management Screw
3	Screw

Figure 31: Installing 19 inch Rack-Mount Brackets in the Middle



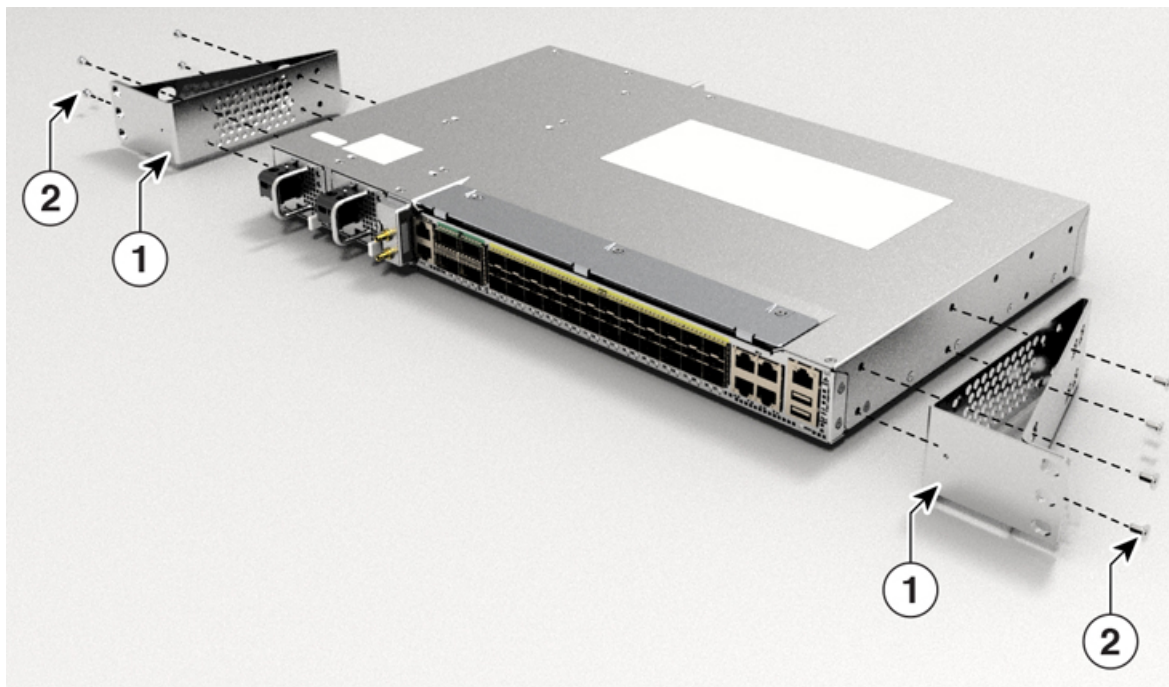
1	Mount Bracket
2	Cable Management Bracket
3	Screw

Figure 32: Installing Cable Management and 19 inch Rack-Mount Brackets in the Middle



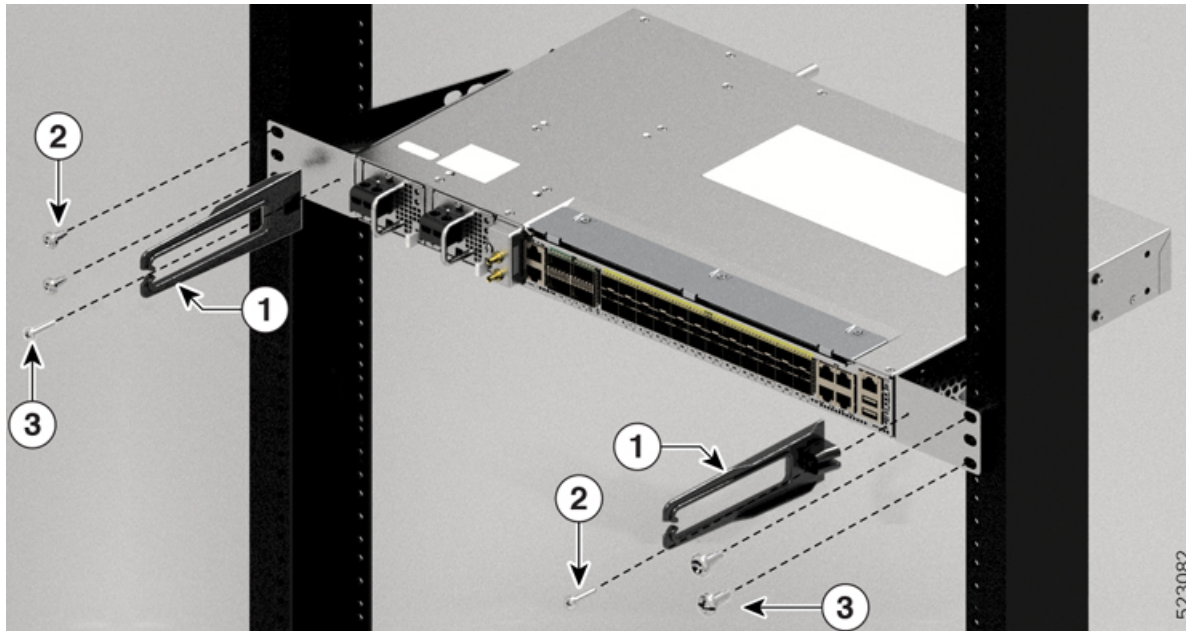
1	Cable Management Bracket
2	Cable Management Screw
3	Screw

Figure 33: Installing 23 inch Rack-Mount Brackets in the Front



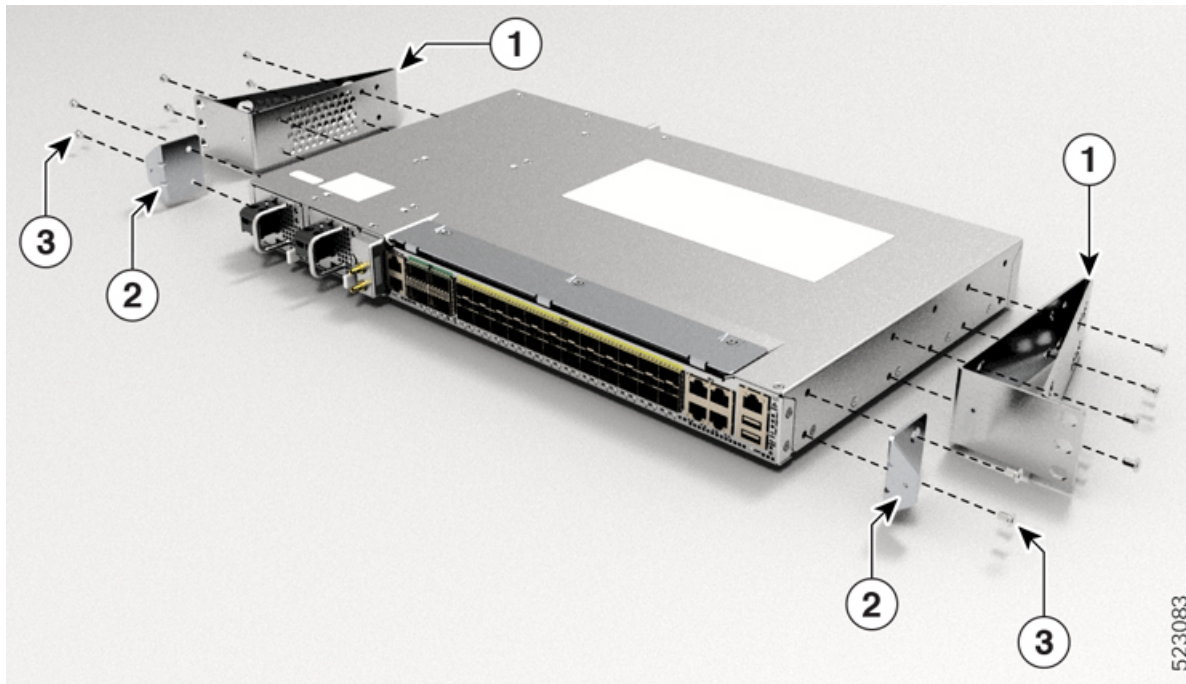
1	Mount Bracket
2	Screw

Figure 34: Installing Cable Management and 23 inch Rack-Mount Brackets in the Front



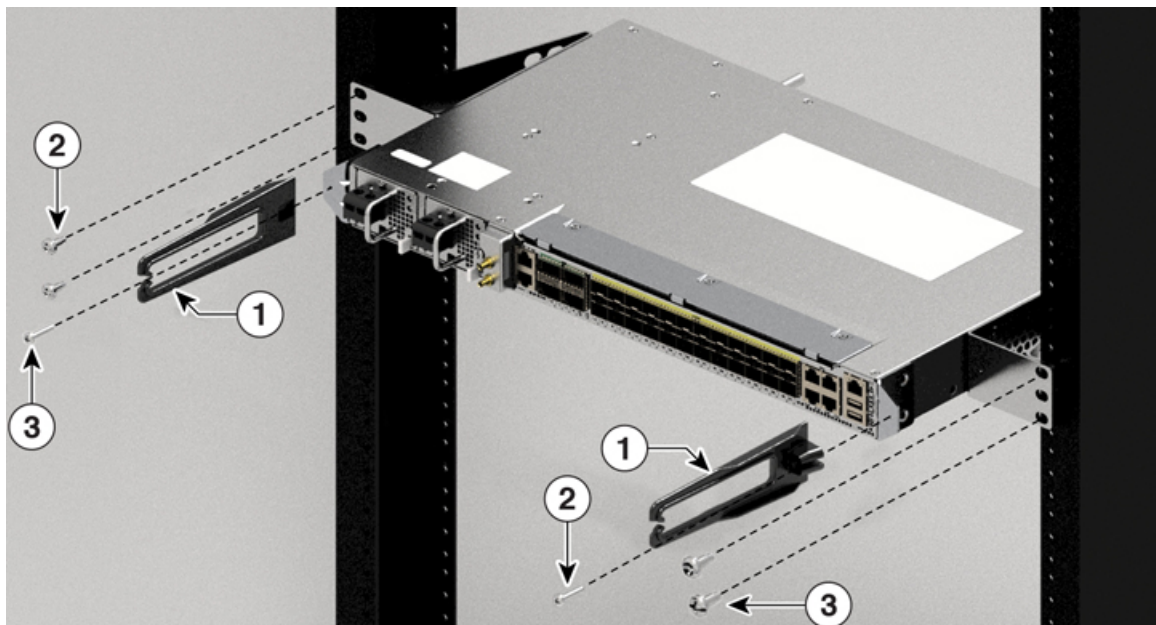
1	Cable Management Bracket
2	Cable Management Screw
3	Screw

Figure 35: Installing 23 inch Rack-Mount Brackets in the Middle



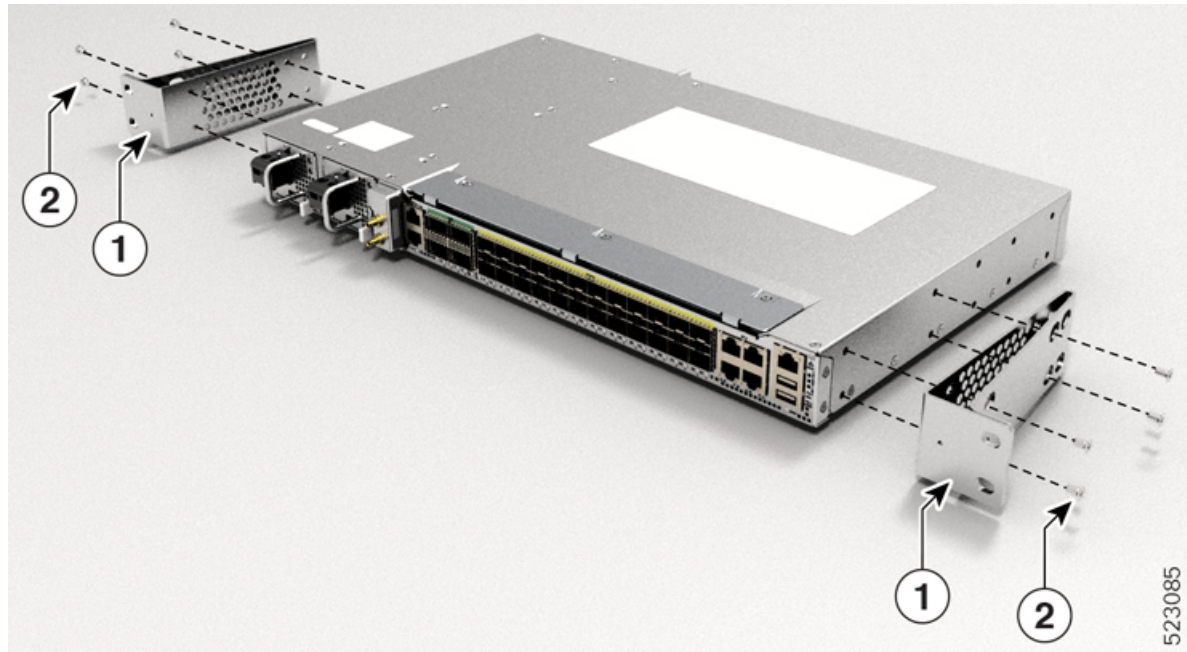
1, 2	Mount Bracket
3	Screw

Figure 36: Installing Cable Management and 23 inch Rack-Mount Brackets in the Middle



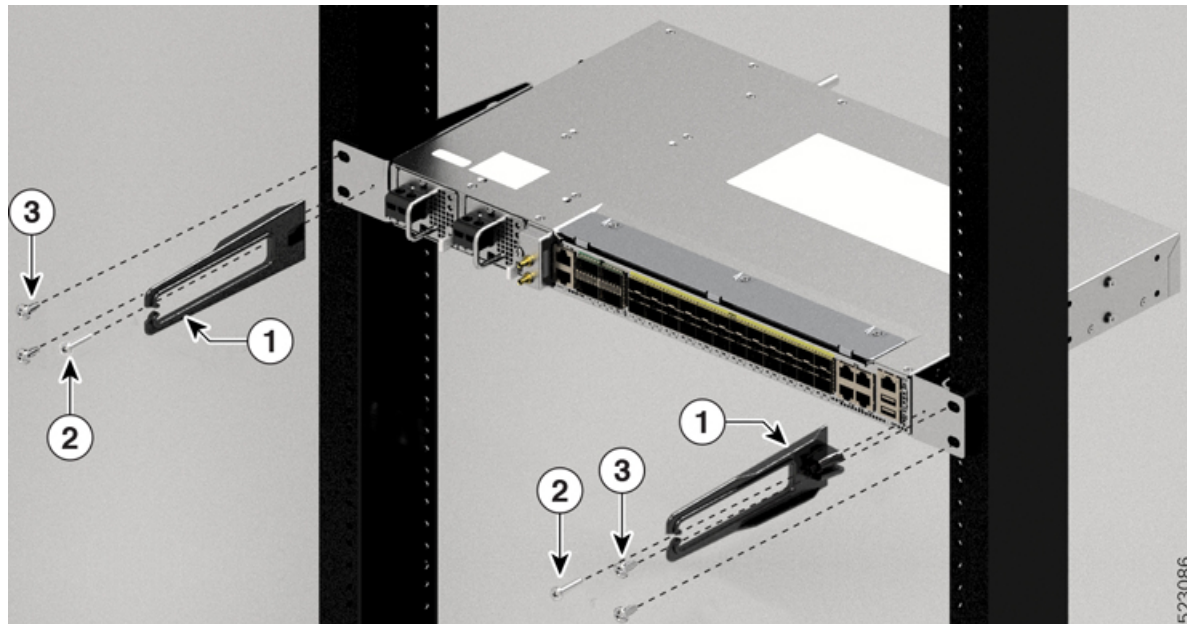
1	Cable Management Bracket
2	Cable Management Screw
3	Screw

Figure 37: Installing ETSI Rack-Mount Brackets in the Front



1	Mount Bracket
2	Screw

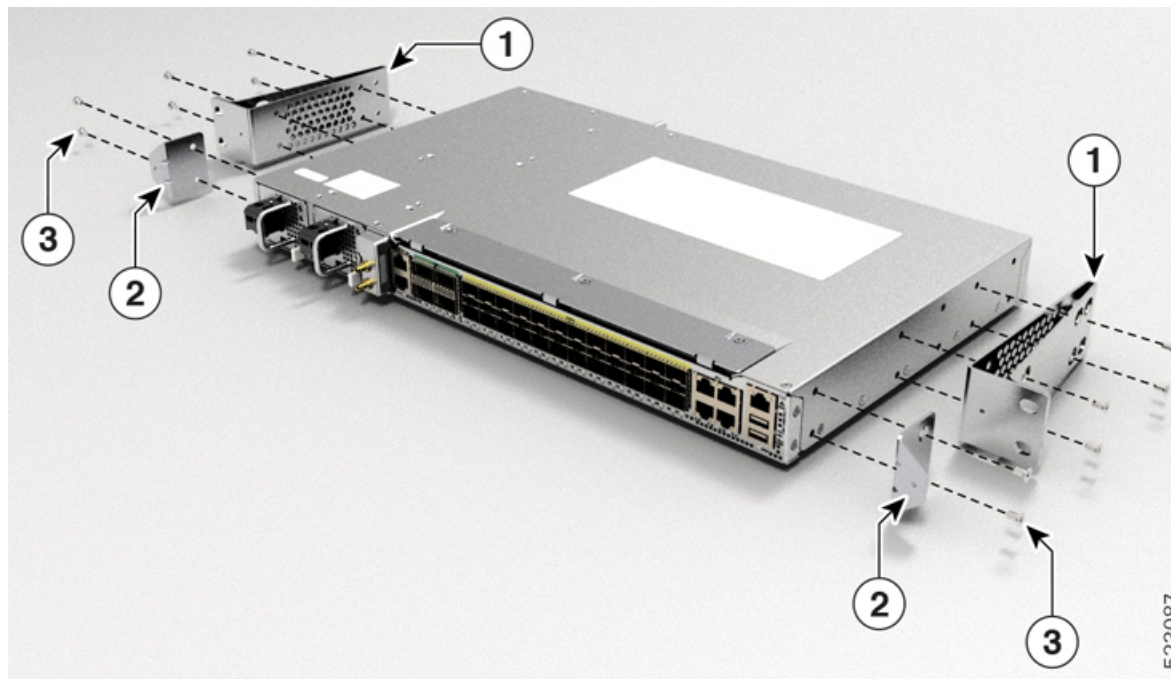
Figure 38: Installing Cable Management and ETSI Rack-Mount Brackets in the Front



1	Cable Management Bracket
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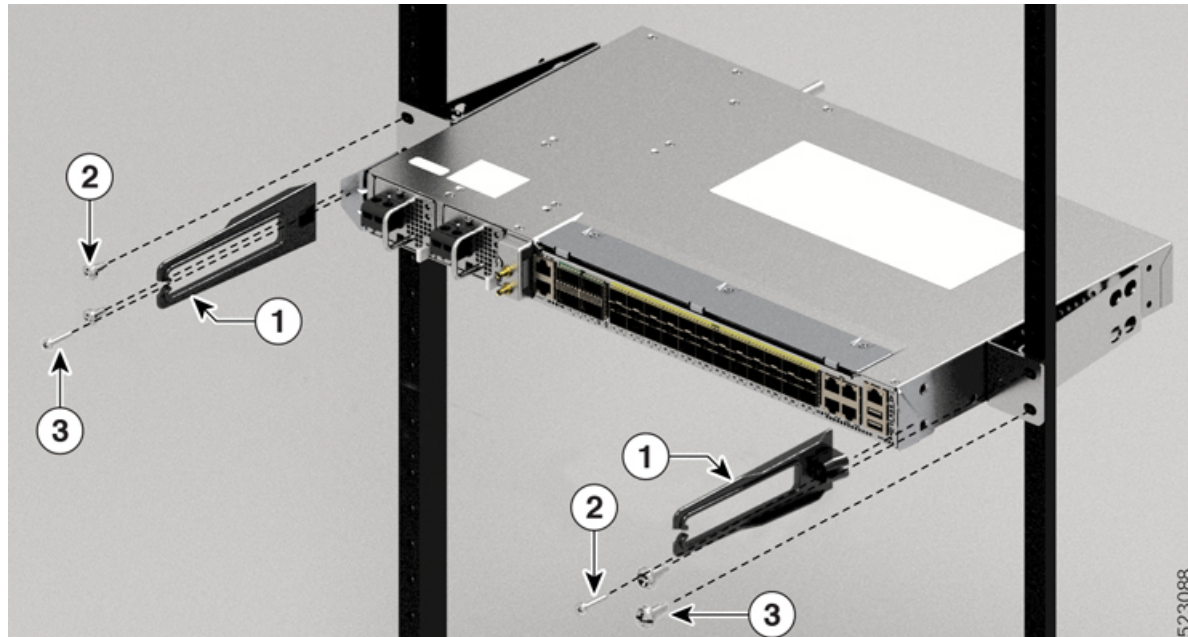
2	Cable Management Screw
3	Screw

Figure 39: Installing ETSI Rack-Mount Brackets in the Middle



1,2	Mount Bracket
3	Screw

Figure 40: Installing Cable Management and ETSI Rack-Mount Brackets in the Middle



1	Cable Management Bracket
2	Cable Management Screw
3	Screw

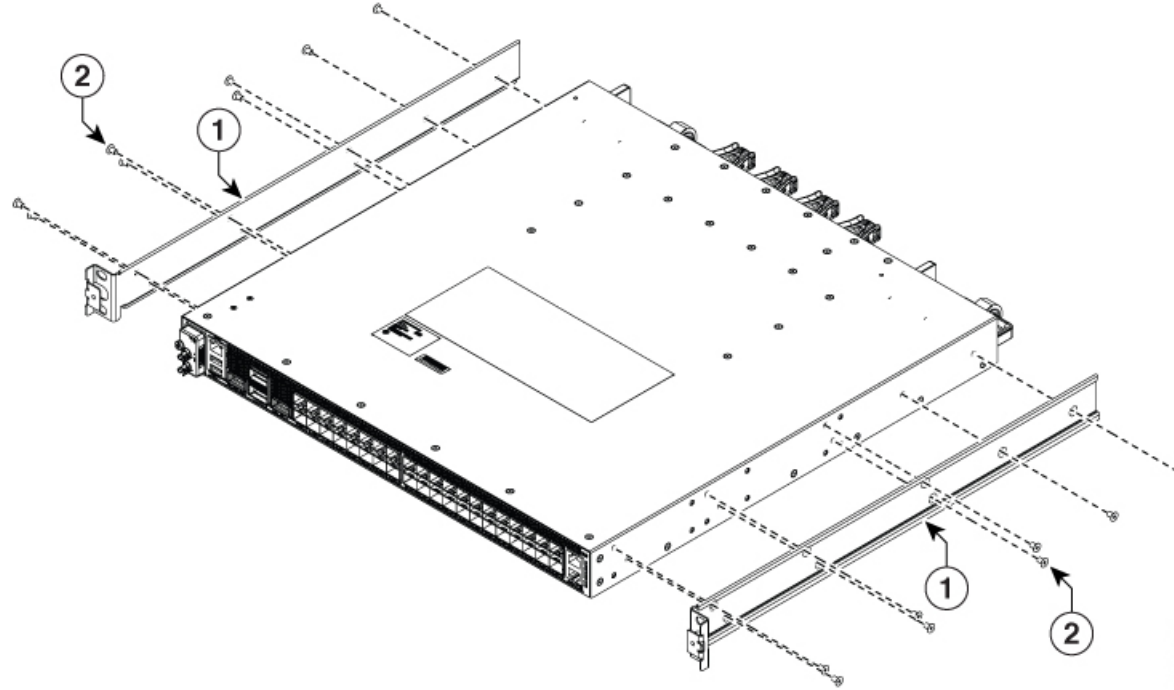
- d. Repeat Steps 1b and 1c with the other rack-mount bracket on the other side of the router.
 - e. Use four 12-24 screws and mount the router to the rack.
2. Install the router onto the 2-post rack as follows:
 - a. Lift and position the router into position between the two rack posts.
 - b. Move the router until the rack-mount brackets come in contact with the two rack posts.
 - c. Hold the chassis at level and have another while the second person inserts two screws 12-24 in each of the two rack-mount brackets (using a total of four screws) and into the cage nuts or threaded holes in the vertical rack-mounting rails.
 - d. Tighten the 12-24 screws to 30 in-lb (3.39 N.m).

Cisco 8011-32Y8L2H2FH Router

1. Attach the rack-mount brackets and the cable guides to the router as follows:
 - a. The router has port-side intake modules, position the router so that its ports are facing the cold aisle.
 - b. Position the bracket ears facing front or middle rack-mount, on the side of the chassis so that the holes are aligned.

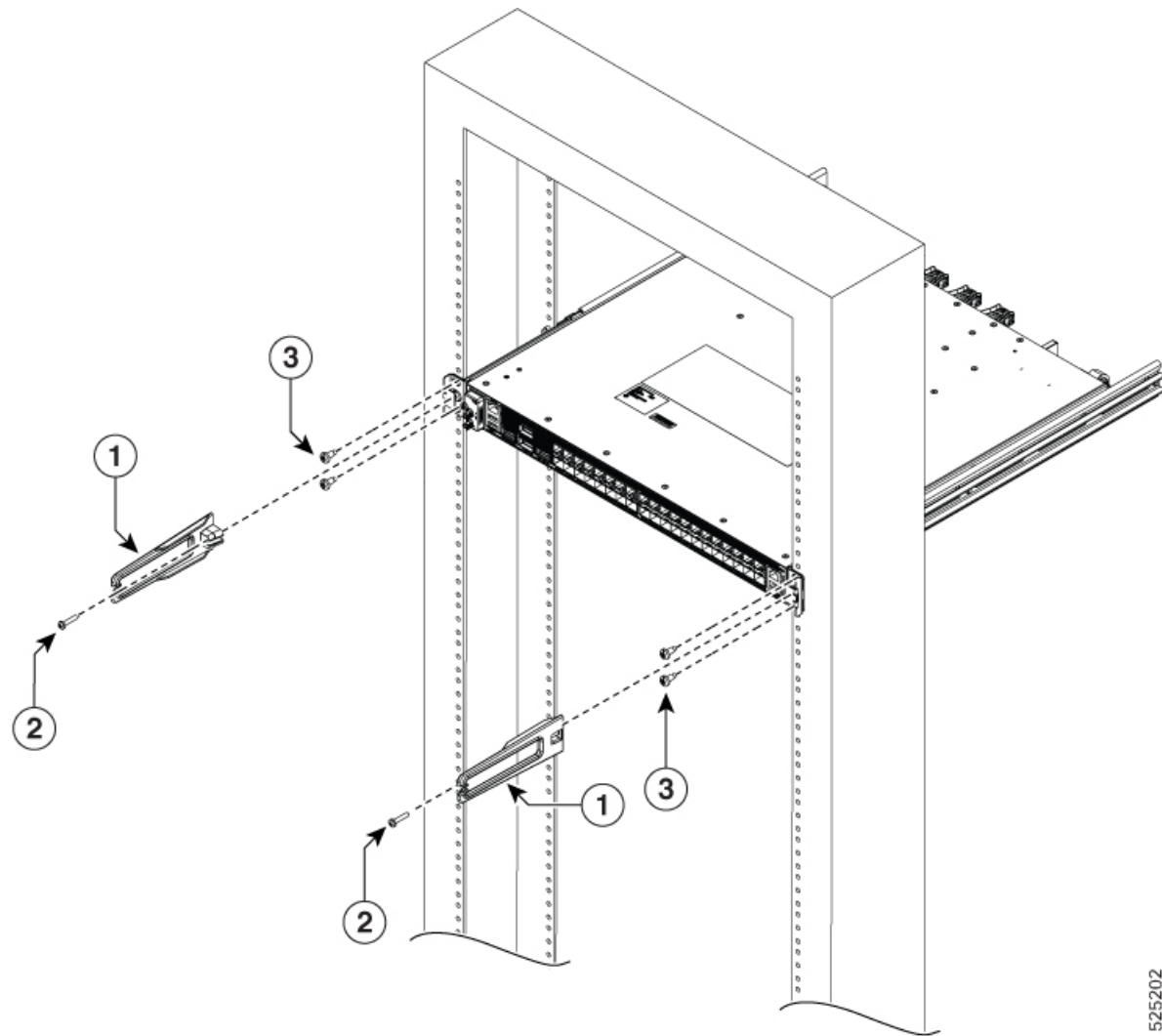
- c. Use sixteen M4 screws to attach the brackets to the chassis. Tighten the M4 screws to 13.3 inch-pounds (1.5 Nm).

Figure 41: Installing rack-mount brackets in the front



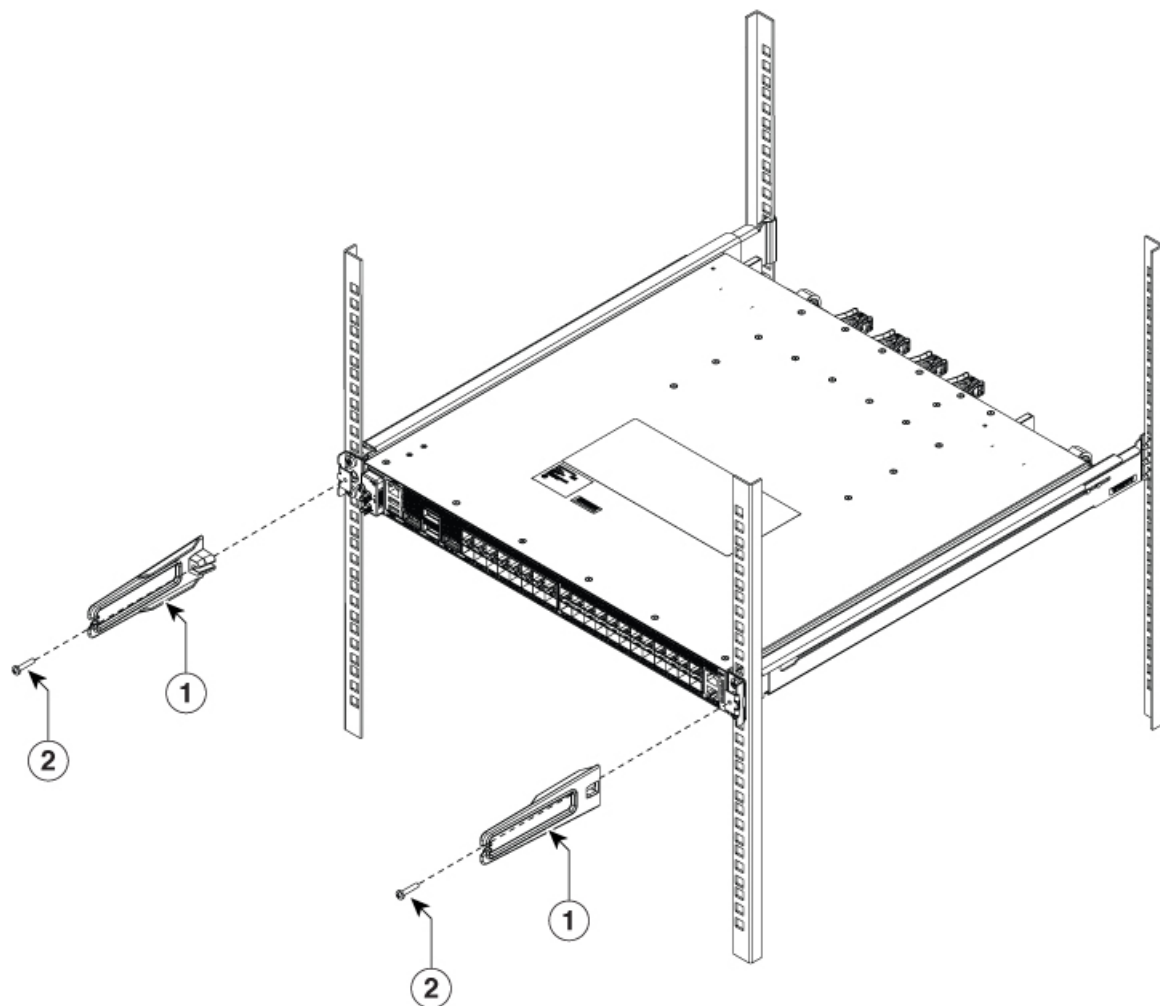
1	Bracket
2	Screw

Figure 42: Installing cable management with 19-inch 2-post rack-mount brackets in the front

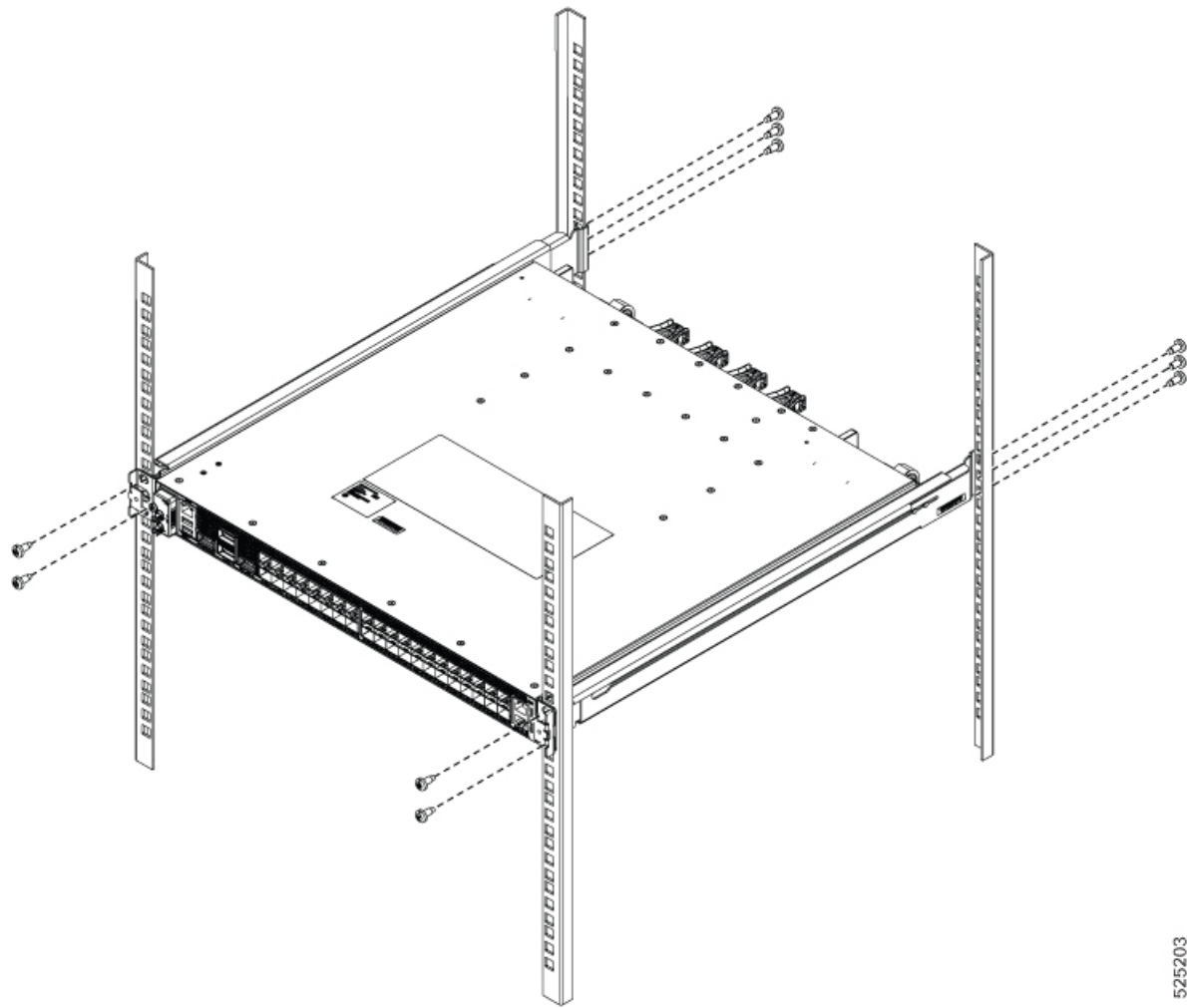


1	Cable management bracket
2	Cable management screw
3	Screw

Figure 43: Installing cable management with 19-inch 4-post rack-mount brackets in the front

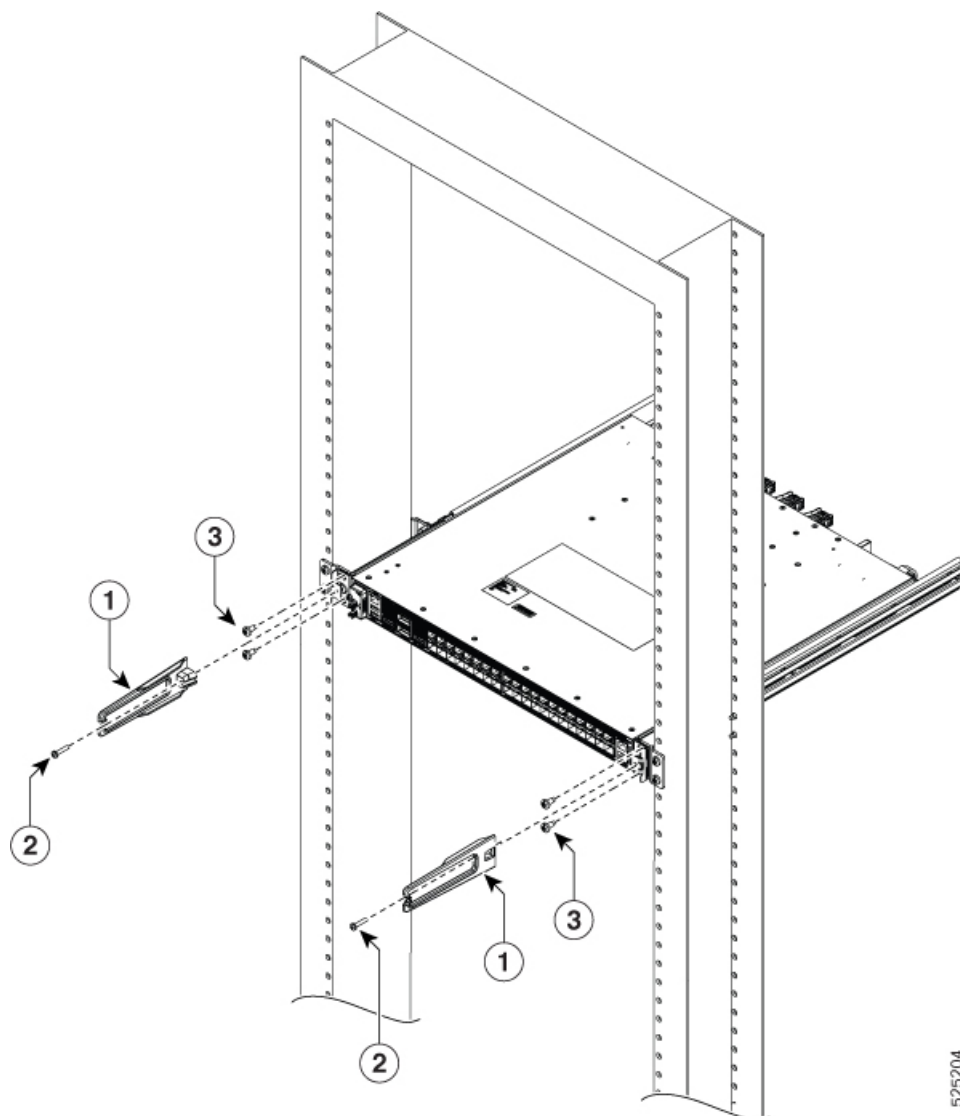


1	Cable management bracket
2	Cable management screw



525203

Figure 44: Installing cable management with 21-inch 2-post rack-mount brackets in the front



525204

1	Cable management bracket
2	Cable management screw
3	Screw

Figure 45: Installing cable management with 21-inch 4-post rack-mount brackets in the front

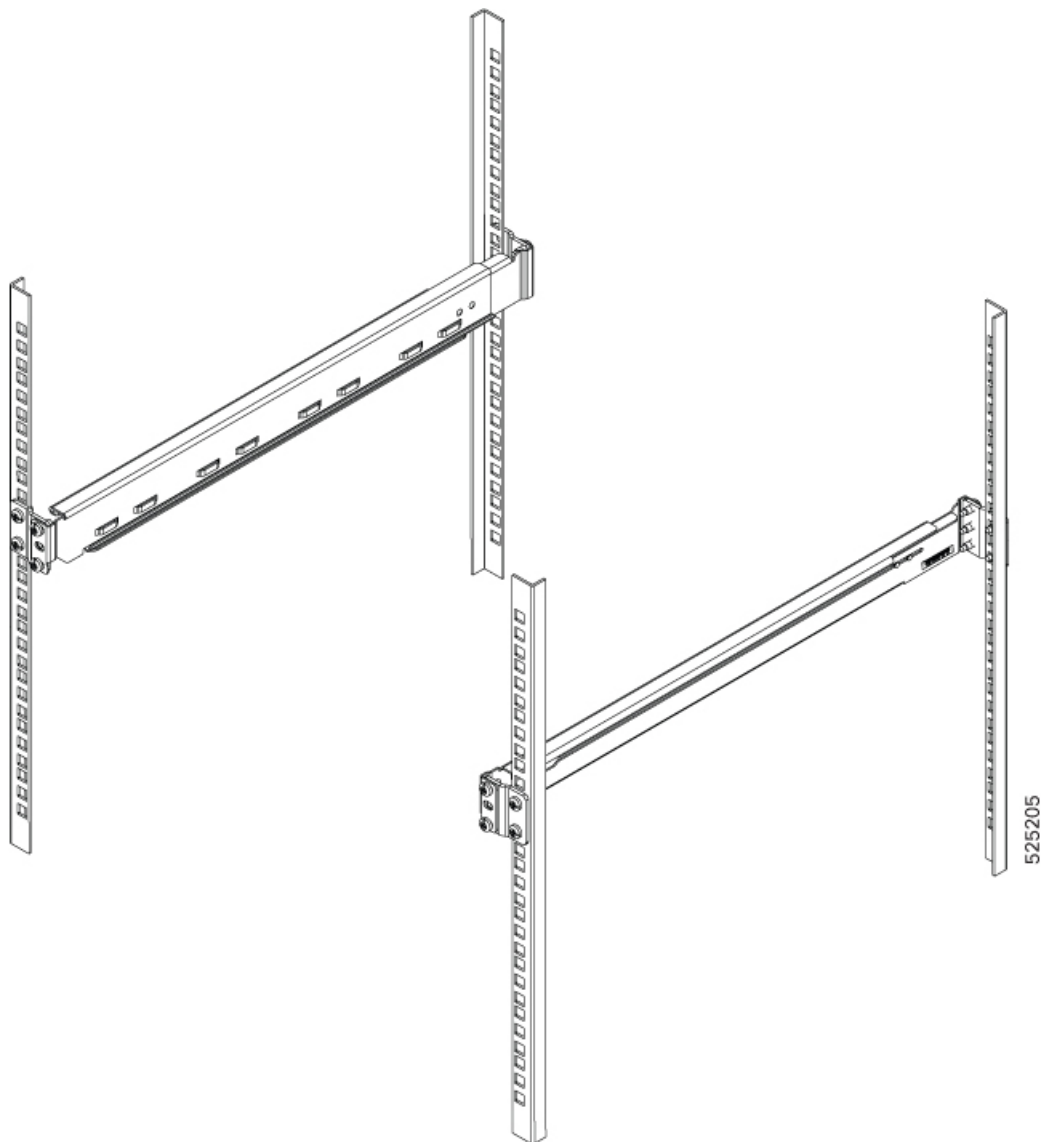
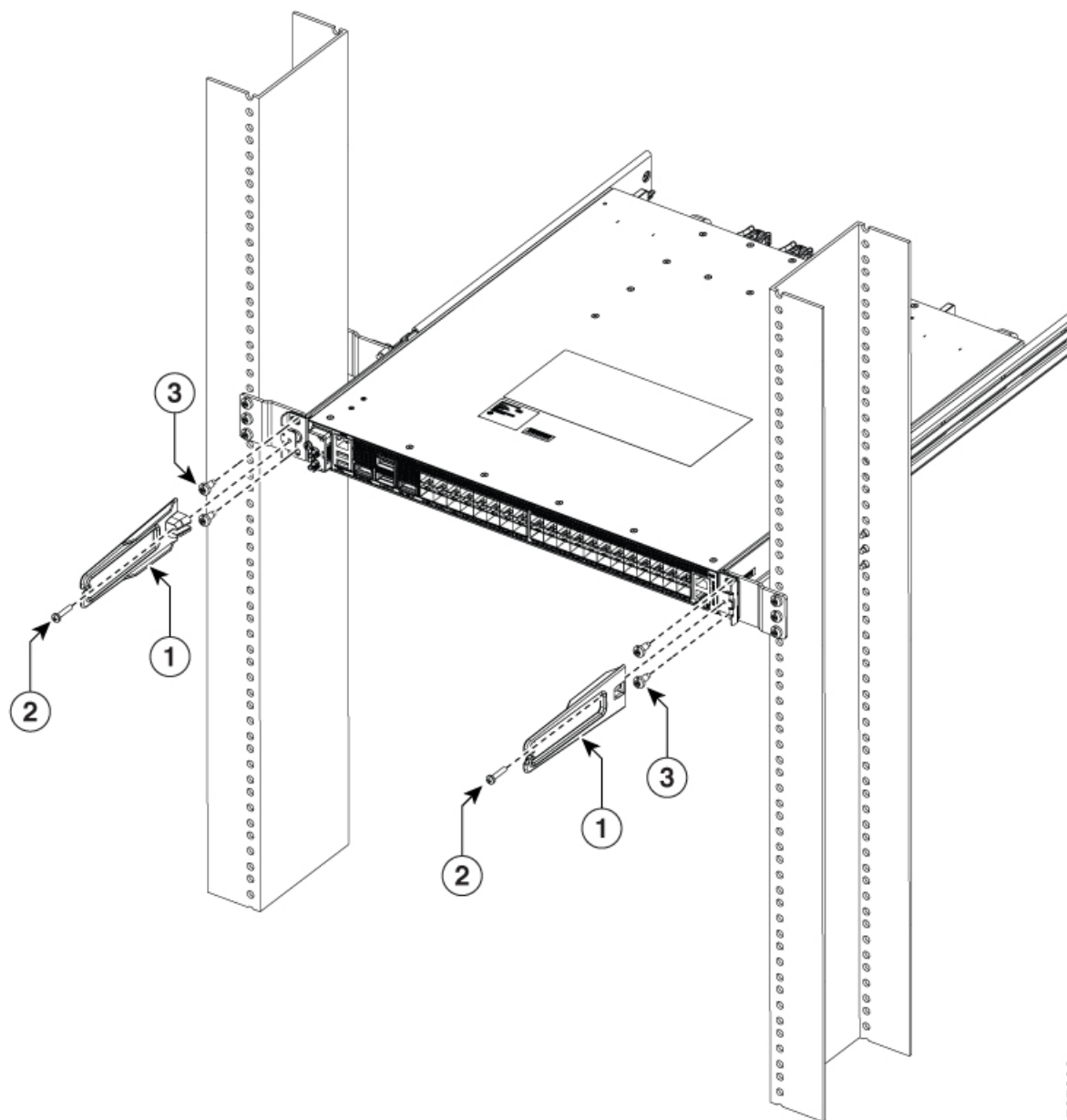
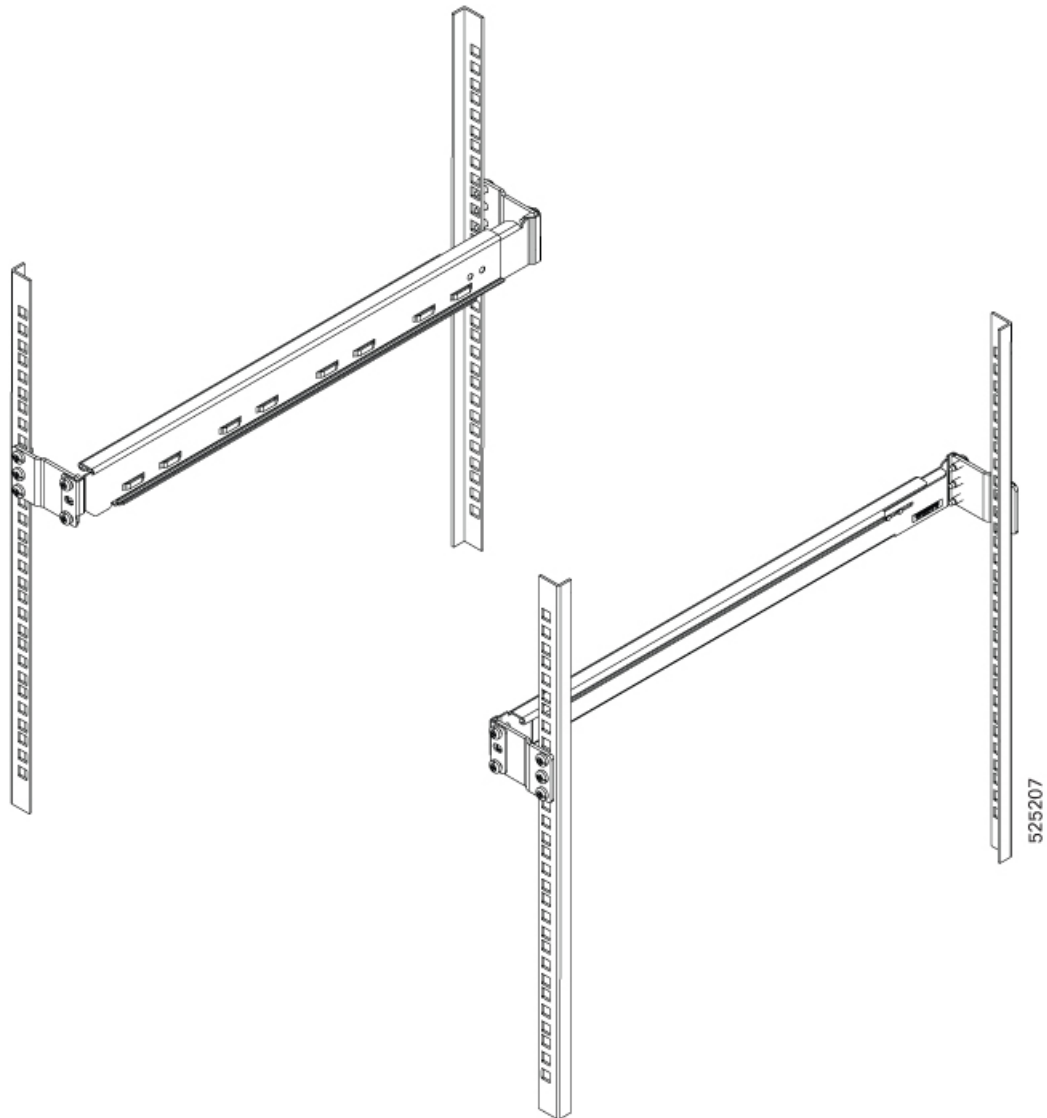


Figure 46: Installing cable management with 23-inch 2-post rack-mount brackets in the front



1	Cable Management Bracket
2	Cable Management Screw
3	Screw

Figure 47: Installing cable management with 23-inch 4-post rack-mount brackets in the front



- d. Repeat Steps 1b and 1c with the other rack-mount bracket on the other side of the router.
 - e. Use four 12-24 screws and mount the router to the rack.
2. Install the router onto the 2-post rack as follows:
 - a. Lift and position the router into position between the two rack posts.
 - b. Move the router until the rack-mount brackets come in contact with the two rack posts.
 - c. Hold the chassis at level and have another while the second person inserts two screws 12-24 in each of the two rack-mount brackets (using a total of four screws) and into the cage nuts or threaded holes in the vertical rack-mounting rails.
 - d. Tighten the 12-24 screws to 30 in-lb (3.39 N.m).

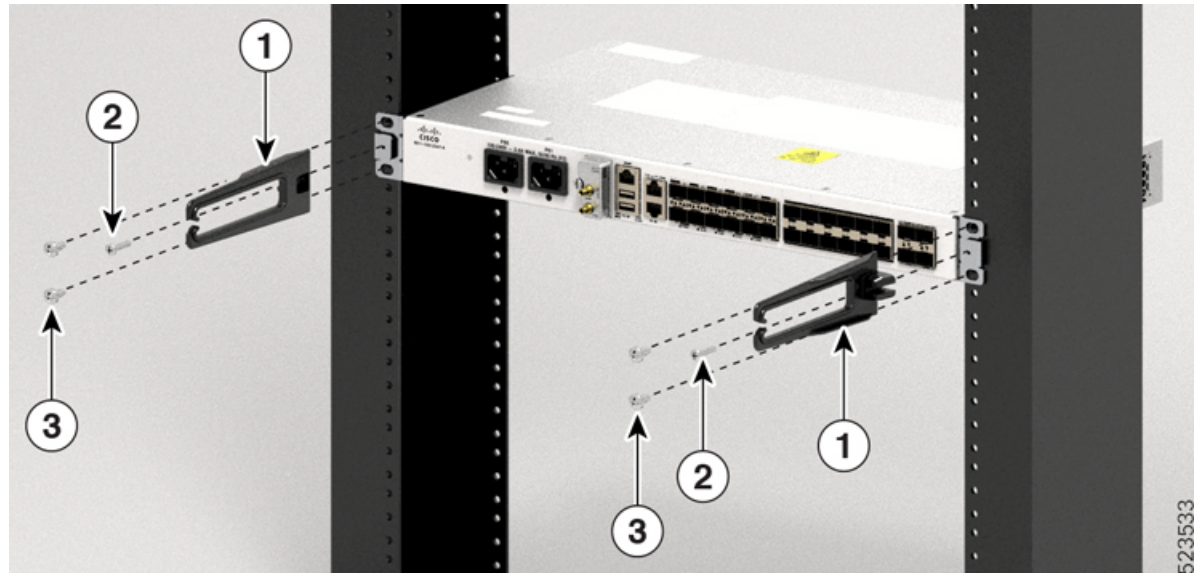
Cisco 8011-12G12X4Y Router

1. Attach the rack-mount brackets and the cable guides to the router as follows:
 - a. The router has port-side intake modules, position the router so that its ports are facing the cold aisle.
 - b. Position the bracket ears facing front or middle rack-mount, on the side of the chassis so that the holes are aligned.
 - c. Use four M3 screws to attach the brackets to the chassis.
 - d. Repeat Steps 1b and 1c with the other rack-mount bracket on the other side of the router.
 - e. Use four No. 12-24 screws and mount the router to the rack.

Figure 48: Installing 19 inch Rack-Mount Brackets in the Front

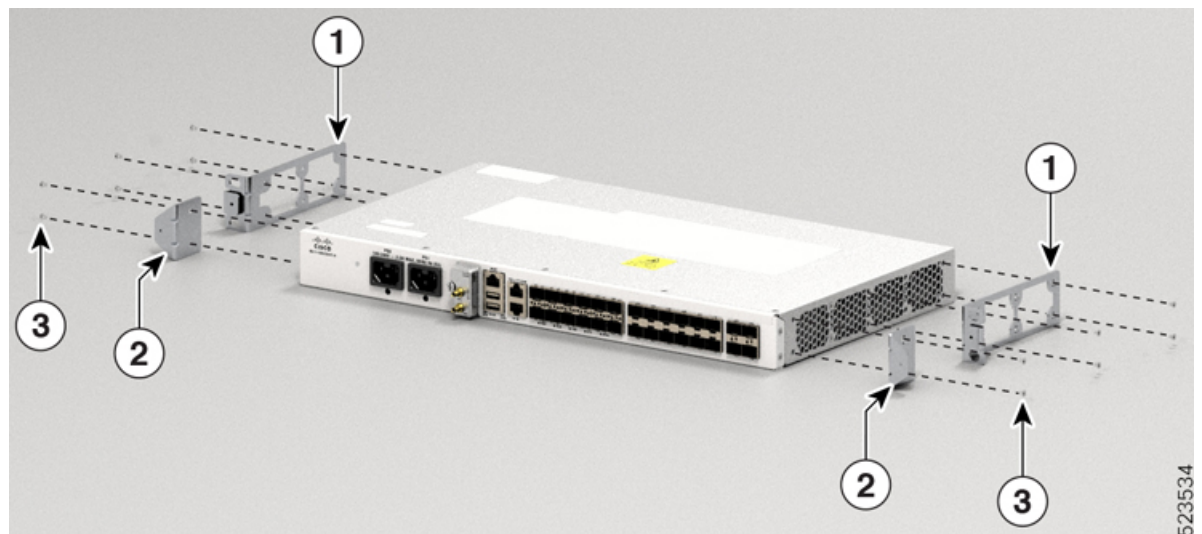
1	Mount Bracket
2	Screw

Figure 49: Installing Cable Management and 19 inch Rack-Mount Brackets in the Front



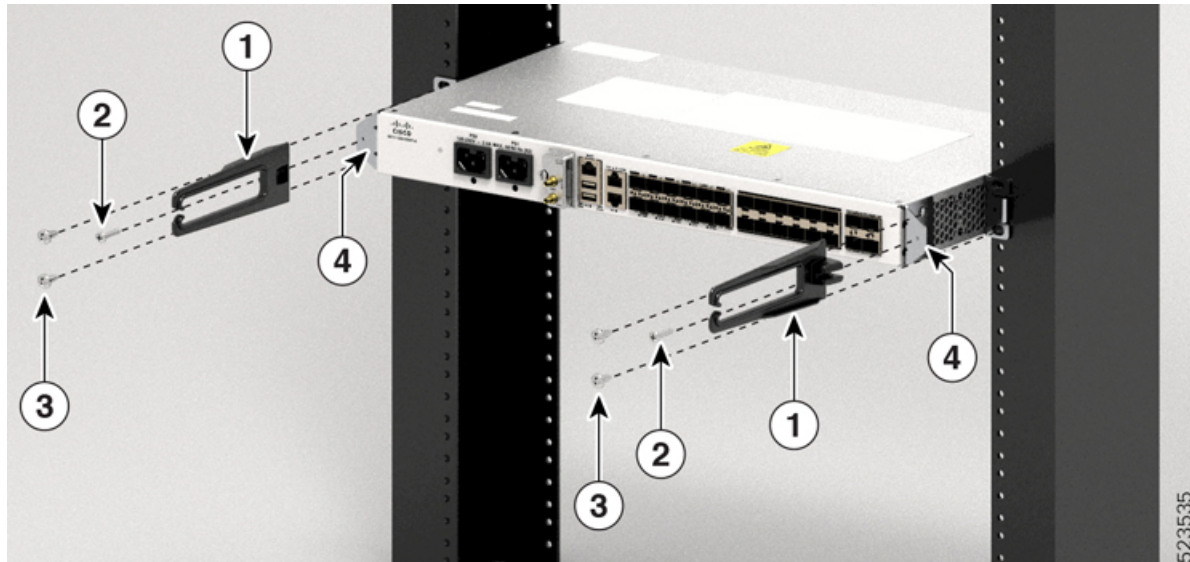
1	Cable Management Bracket
2	Cable Management Screw
3	Screw

Figure 50: Installing 19 inch Rack-Mount Brackets in the Middle



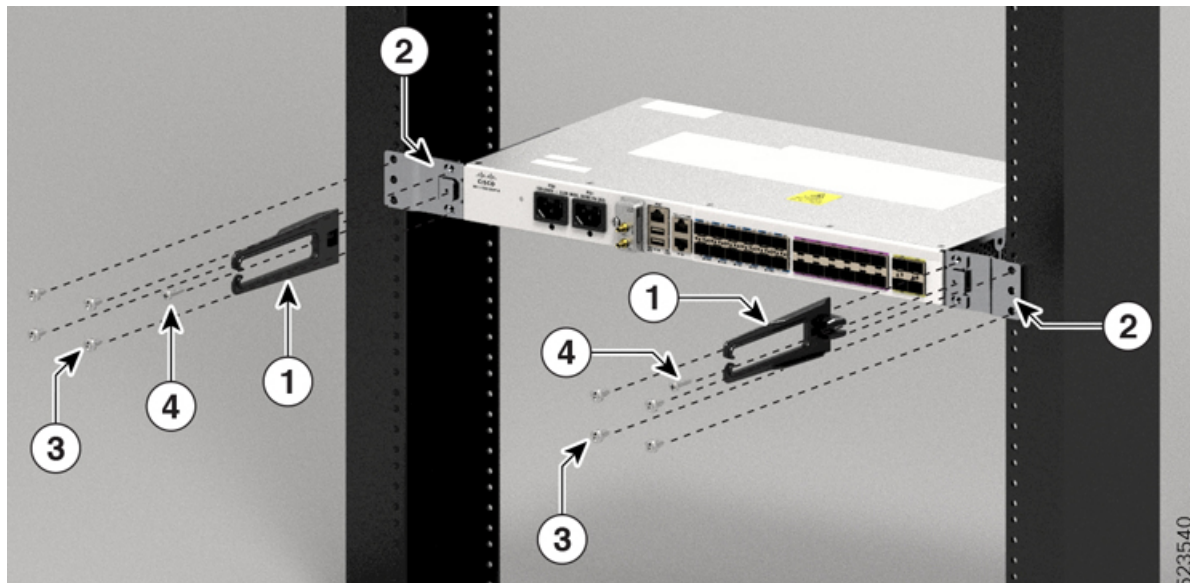
1	Mount Bracket
2	Cable Management Bracket
3	Screw

Figure 51: Installing Cable Management and 19 inch Rack-Mount Brackets in the Middle



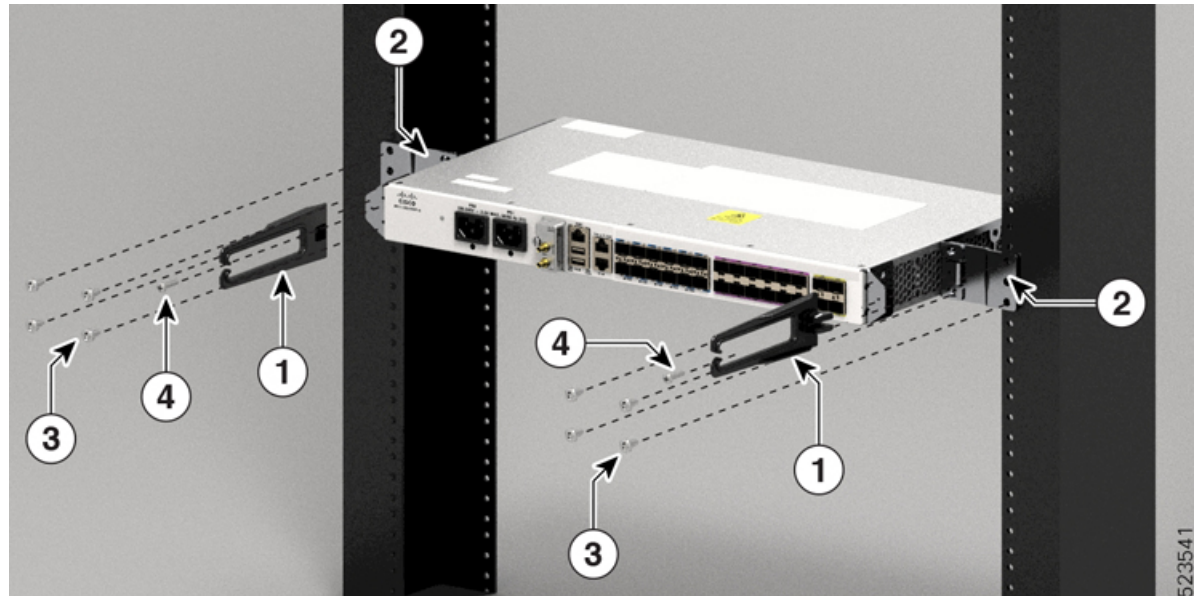
1	Cable Management Bracket
2	Cable Management Screw
3	Screw
4	Bracket

Figure 52: Installing 23 inch Rack-Mount Brackets in the Front



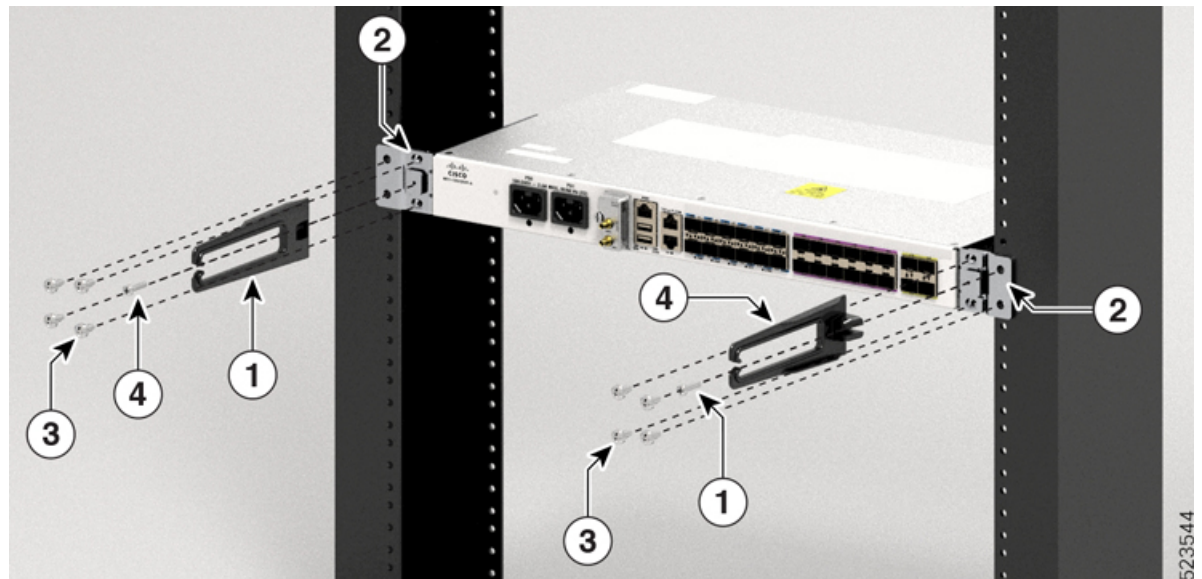
1	Mount Bracket
2	Screw

Figure 53: Installing 23 inch Rack-Mount Brackets in the Middle



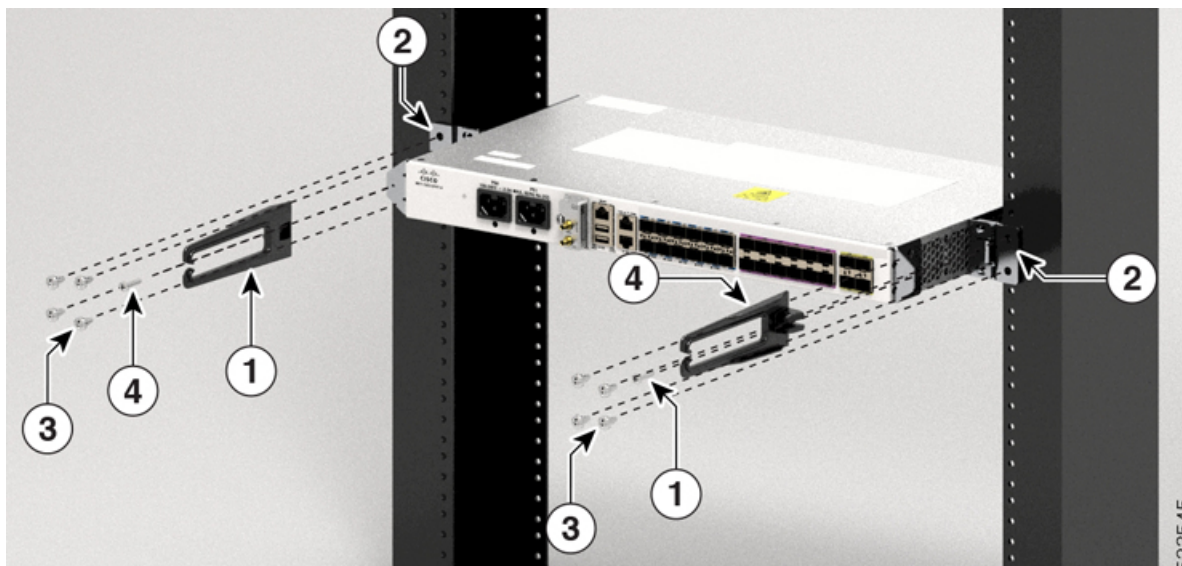
1, 2	Mount Bracket
3	Screw

Figure 54: Installing Cable Management and ETSI Rack-Mount Brackets in the Front



1	Cable Management Bracket
2	Cable Management Screw
3	Screw

Figure 55: Installing Cable Management and ETSI Rack-Mount Brackets in the Middle



1	Cable Management Bracket
2	Cable Management Screw
3	Screw

2. Install the router onto the 2-post rack as follows:
 - a. Lift and position the router into position between the two rack posts.
 - b. Move the router until the rack-mount brackets come in contact with the two rack posts.
 - c. Hold the chassis at level and have another while the second person inserts two screws 12-24 in each of the two rack-mount brackets (using a total of four screws) and into the cage nuts or threaded holes in the vertical rack-mounting rails.
 - d. Tighten the 12-24 screws to 30 in-lb (3.39 N.m).

Wall Mount



Note This feature is supported only for the Cisco 8011-4G24Y4H-I Router.

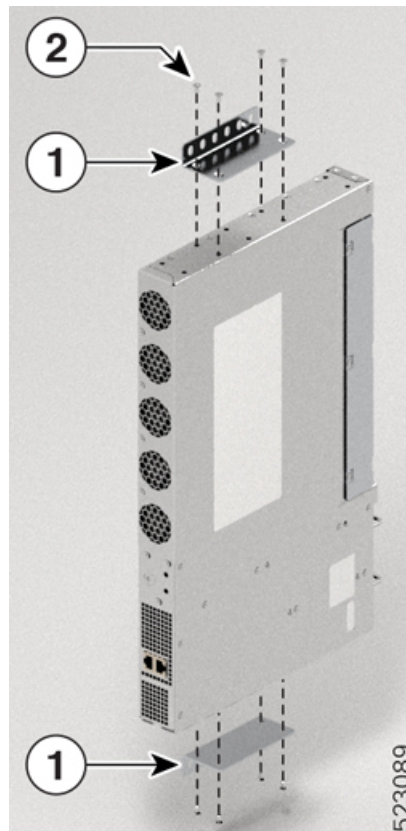
The router is shipped with wall mounting brackets that are to be secured on the sides of the router.

Install the wall mounting brackets and cable guides on to the chassis before you mount the chassis on the wall.

Install Wall Brackets

1. Remove the wall mounting brackets from the accessory kit and position them beside the router. You can install the brackets as shown in the figure.

Figure 56: Install Wall Mount Brackets



1	Wall Mount Bracket
2	Screw

- Secure the bracket to the router with the recommended maximum torque of 13.3 inch-pounds (1.5 newton meters) using M4 flat head screws.

Wall Mount the Router



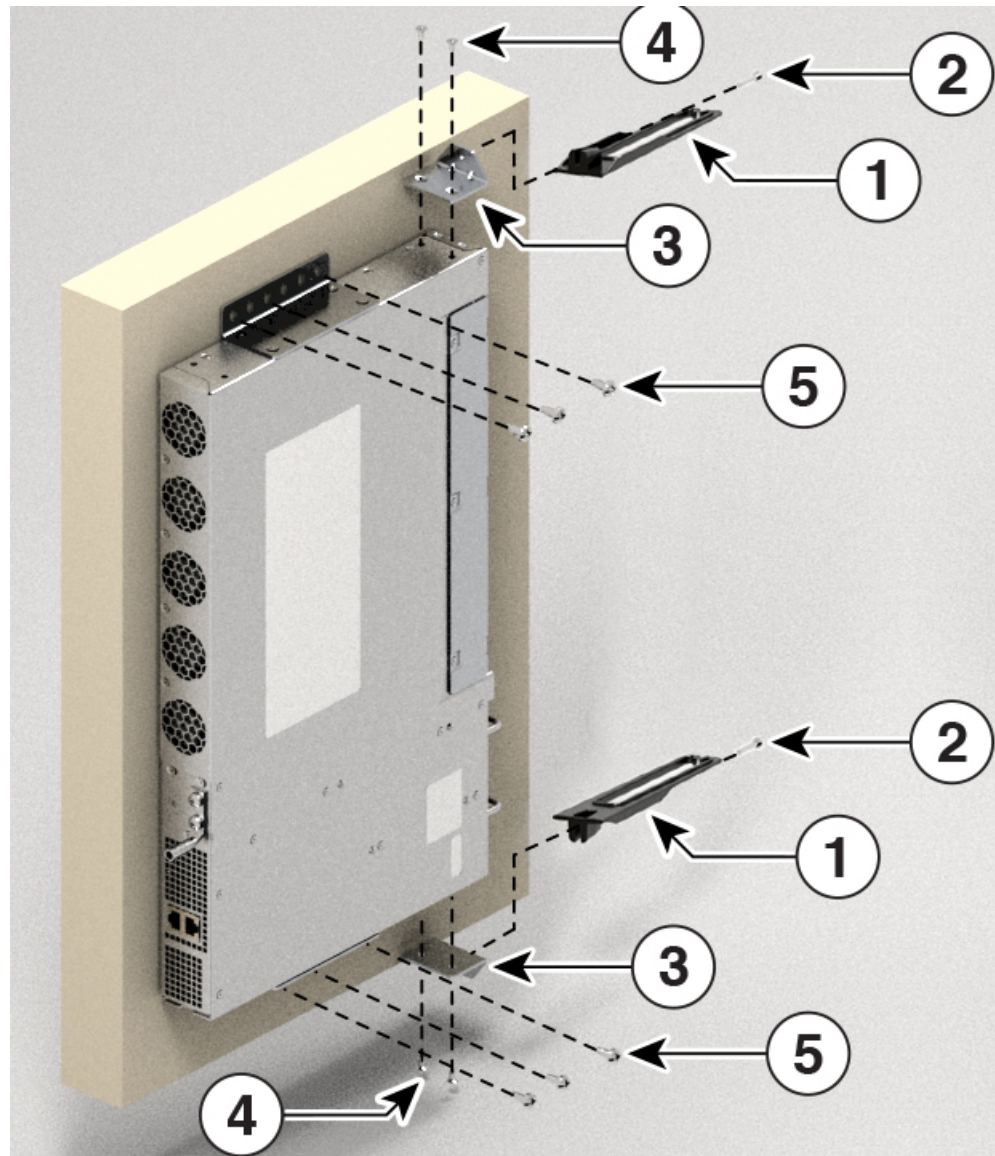
Caution Before mounting the router, ensure that all unused holes at the sides of the router are always protected by screws.



Note While you mount the router, always ensure that the power supplies are at the bottom position.

For the support of the router and cables, ensure that the router is attached securely to wall studs or to a firmly attached plywood mounting backboard.

Figure 57: Install Brackets to Wall



1	Cable Management
2	Cable Management Screw
3	Cable Management Bracket
4	Cable Management Bracket Screw
5	Wall Mount Bracket Screw

Ground the Router

Before you begin this task, ensure that you have read and understood the safety warnings in the Preventing ESD Damage section of the *Safety Warnings* handout.

Before you connect the power or turn on the power to the router, you must provide an adequate router ground (earth) connection to your router.

This section describes how to ground the router. The grounding lug location is on the back panel of the router.



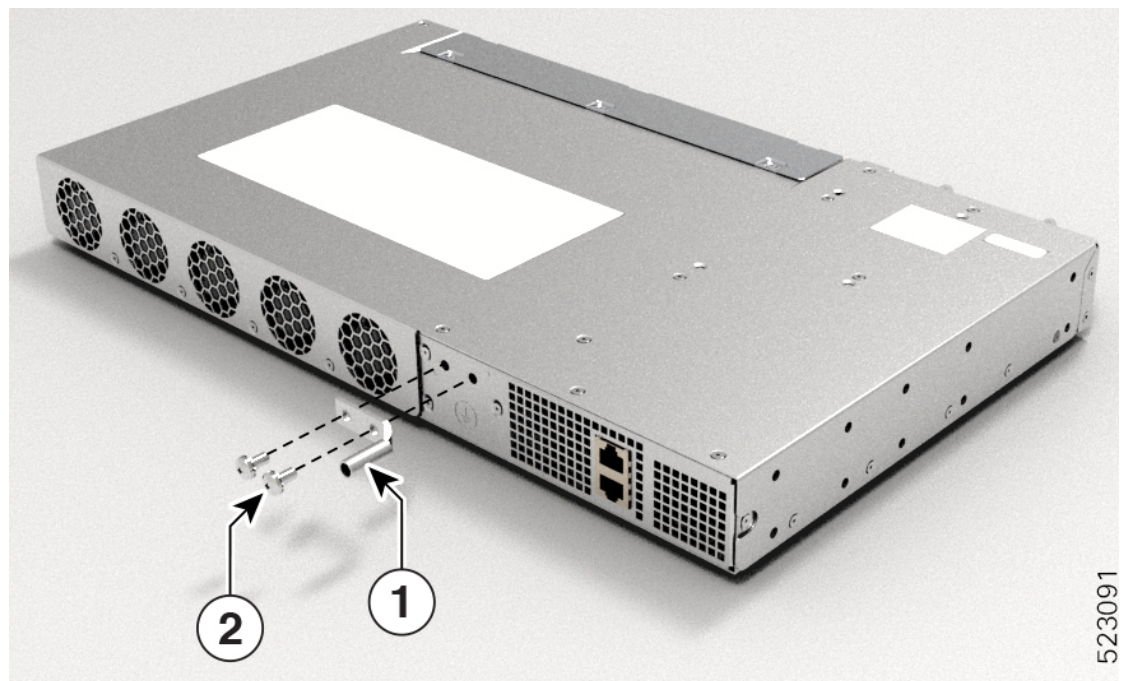
Note A minimum of 5 inches (127 mm) space must be provided at the rear to accommodate 90 degree grounding lug with a minimum bend radius for 6 AWG ground cable.

Cisco 8011-4G24Y4H-I Router

To ground the router:

1. Verify that the ground cable is connected to the top of the rack and according to local site practice.

Figure 58: Ground Lug



1	Lug (Part number is 32-0608-01)
2	Screw (Part number is 48-101620-01)

**Caution**

Use only Cisco supplied screws for mounting the grounding lugs. If procured from local market or from outside other source, ensure that the grounding lug screw length should not exceed 0.365 inches (9.27 mm). Using excessive length screw for mounting the grounding lug may cause short-circuit as it may come in contact with parts inside the router.

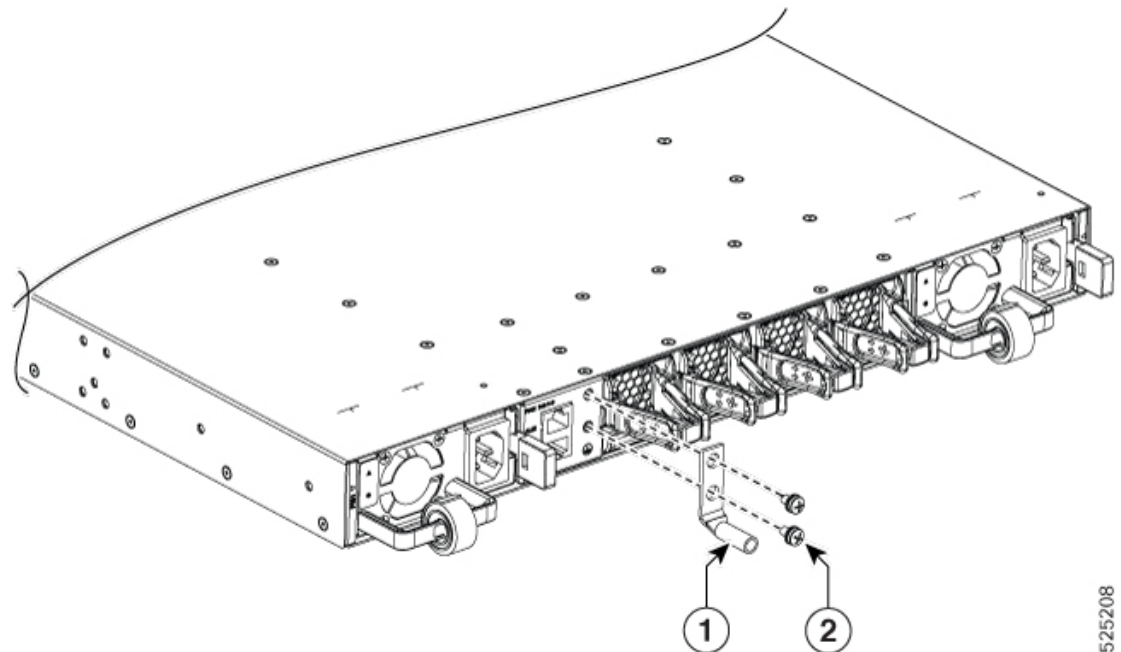
2. Attach one end of the shelf ground cable (No. 6 AWG cable) to the ground point on the rear of the router using the specified dual-hole lug connector.
 - a. Use a wire-stripping tool to remove approximately 0.75 inches (19 mm) of the covering from the end of the grounding cable.
 - b. Insert the stripped end of the grounding cable into the open end of the grounding lug.
 - c. Use the crimping tool to secure the grounding cable in the grounding lug.
 - d. Remove the adhesive label from the grounding pad on the chassis.
 - e. Place the grounding lug against the grounding pad so that there is solid metal-to-metal contact, and insert the two 10-32 screws with washer through the holes in the grounding lug and into the grounding pad. Tighten the 10-32# screws to 27.5 inch-pounds (3.1 Nm).
 - f. Ensure that the lug and cable do not interfere with other equipment.
 - g. Prepare the other end of the grounding cable and connect it to an appropriate grounding point in your site to ensure adequate earth ground.

Cisco 8011-32Y8L2H2FH Router

To ground the router:

1. Verify that the ground cable is connected to the top of the rack and according to local site practice.

Figure 59: Ground Lug



1	Lug (Part number is 32-0670-01)
2	Screw (Part number is 48-0418-01)

**Caution**

Use only Cisco supplied screws for mounting the grounding lugs. If procured from local market or from outside other source, ensure that the grounding lug screw length should not exceed 0.315 inches (8 mm). Using excessive length screw for mounting the grounding lug may cause short-circuit as it may come in contact with parts inside the router.

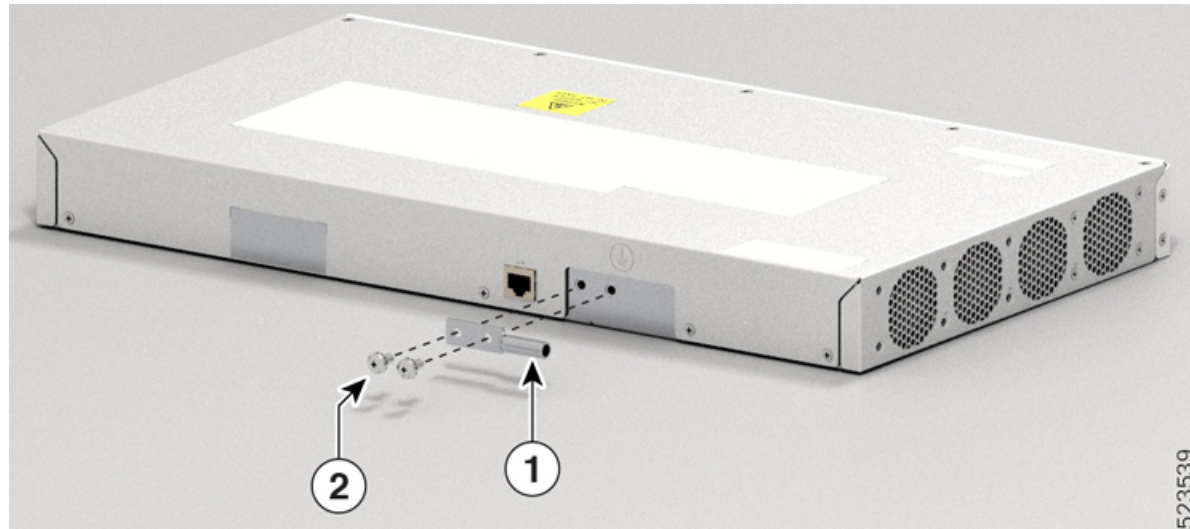
2. Attach one end of the shelf ground cable (No. 6 AWG cable) to the ground point on the rear of the router using the specified dual-hole lug connector.
 - a. Use a wire-stripping tool to remove approximately 0.75 inches (19 mm) of the covering from the end of the grounding cable.
 - b. Insert the stripped end of the grounding cable into the open end of the grounding lug.
 - c. Use the crimping tool to secure the grounding cable in the grounding lug.
 - d. Remove the adhesive label from the grounding pad on the chassis.
 - e. Place the grounding lug against the grounding pad so that there is solid metal-to-metal contact, and insert the two M4 screws with washer through the holes in the grounding lug and into the grounding pad. Tighten the M4 screws to 13.5 inch-pounds (1.5 Nm).
 - f. Ensure that the lug and cable do not interfere with other equipment.
 - g. Prepare the other end of the grounding cable and connect it to an appropriate grounding point in your site to ensure adequate earth ground.

Cisco 8011-12G12X4Y Router

To ground the device:

1. Verify that the ground cable is connected to the top of the rack and according to local site practice.

Figure 60: Ground Lug



1	Lug (32-0619-01)
2	Screw (48-101620-01) (SCR,M,PAN,PH,SEX,10-32x0.365"L,CSwZN,NICKEL ALLOY)

**Warning**

Use only Cisco supplied screws for mounting the grounding lugs. If procured from local market or from outside other source, ensure that the grounding lug screw length should not exceed 0.365 inches (9.27 mm). Using excessive length screw for mounting the grounding lug may cause short-circuit as it may come in contact with parts inside the router.

2. Attach one end of the shelf ground cable (No. 6 AWG cable) to the ground point on the rear of the router using the specified dual-hole lug connector.
 - a. Use a wire-stripping tool to remove approximately 0.75 inches (19 mm) of the covering from the end of the grounding cable.
 - b. Insert the stripped end of the grounding cable into the open end of the grounding lug.
 - c. Use the crimping tool to secure the grounding cable in the grounding lug.
 - d. Remove the adhesive label from the grounding pad on the chassis.
 - e. Place the grounding lug against the grounding pad so that there is solid metal-to-metal contact, and insert the two M4 screws with washers through the holes in the grounding lug and into the grounding pad.
 - f. Ensure that the lug and cable do not interfere with other equipment.

- g. Prepare the other end of the grounding cable and connect it to an appropriate grounding point in your site to ensure adequate earth ground.

Connect to an AC Power Source



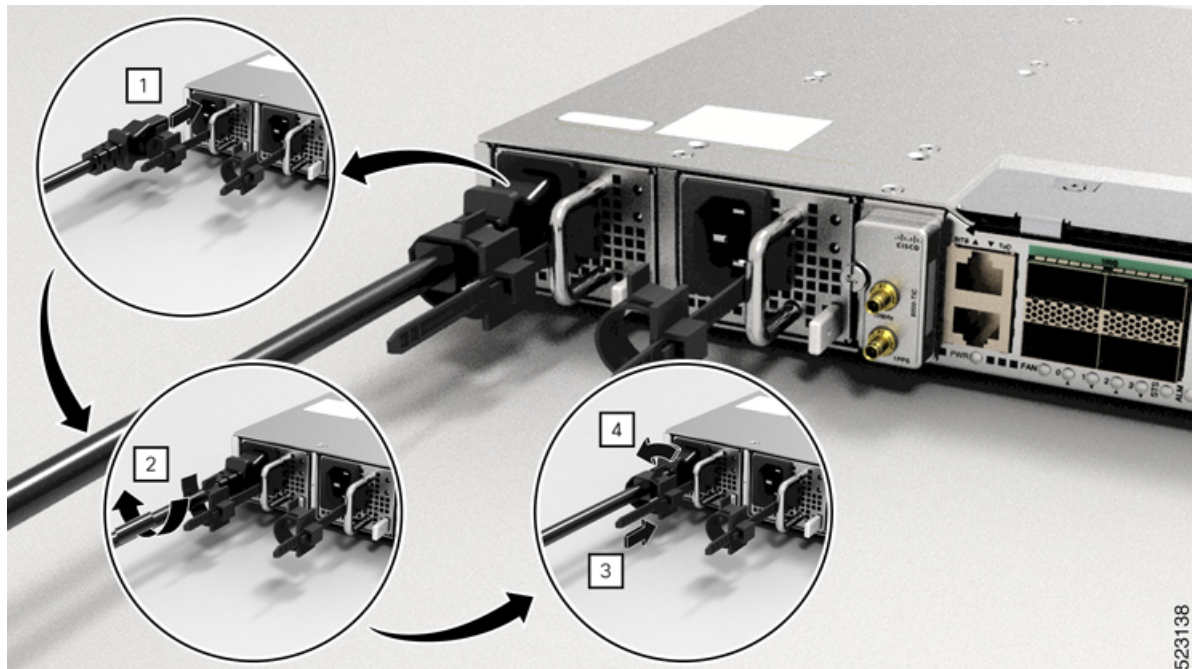
Note A dual pole breaker is needed for the installation. The rating of the dual pole breaker for 110V is 20A and for 220V is 16A. The minimum cable size is 14AWG for 110V and 16AWG for 220V.

Cisco 8011-4G24Y4H-I Router

To connect to a power source:

1. Plug the power supply cord in to the power supply module.
2. Wrap the tie around the power supply cord.
3. Ensure that the power supply cord is secured to the power supply module.
4. Tighten the tie around the power supply cord as shown.
5. Ensure that the power cord is secured always to a cable support to ensure the cable load doesn't act on the PSU.

Figure 61: Attach the AC Power Cables





Note These images are for representation purpose only. Certain Cisco 8010 Series Routers may not include a tie for the power supply cord.

1	Insert power cord
2	Wrap power cord with tie
3	Attach the power cord
4	Secure power cord with tie

Cisco 8011-32Y8L2H2FH Router



Note Prior to connecting the power supply to a power source, ensure that the chassis is properly grounded.

To connect to a power source, follow these steps:

1. Plug the power cable into the power supply.
2. Plug the other end of the power cable into a power source supplied by the data center.

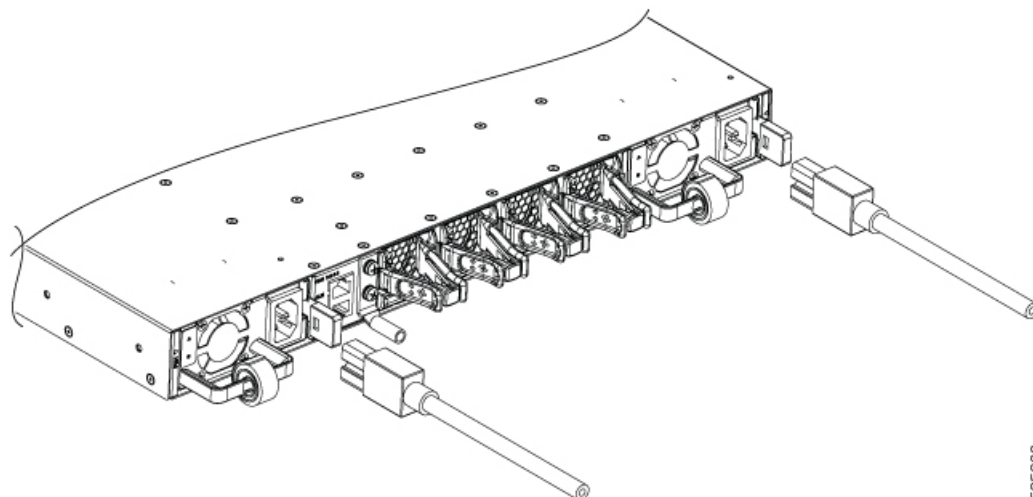


Note When using redundant mode, connect each power supply to a separate power source.

3. Verify that the power supply is receiving power by checking that the LED is on and is amber or red. For more information about the power supply LEDs and the conditions that they indicate, see Power status LEDs.

When you first activate the power supply, you can verify the functionality of the LED by checking that LED turns on for a couple of seconds. If the LED is flashing amber or red, check the power connections on the power supply and the power source.

Figure 62: Attach the AC Power Cables



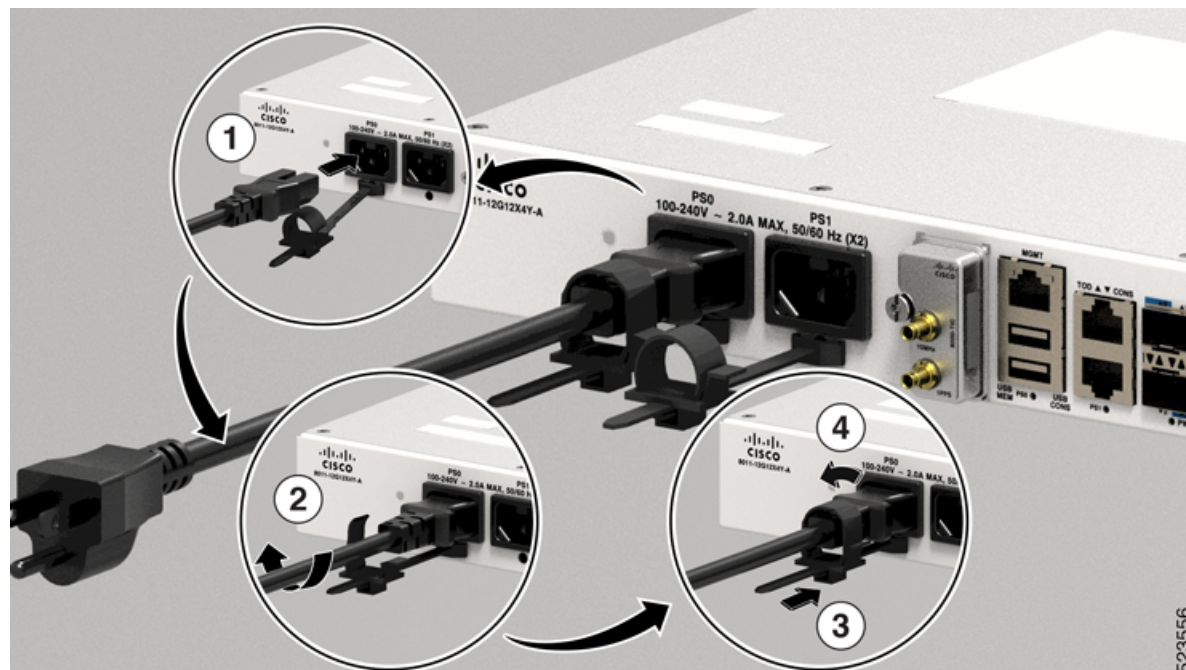
Note These images are for representation purpose only. Certain Cisco 8010 Series Routers may not include a tie for the power supply cord.

Cisco 8011-12G12X4Y Router

To connect to a power source:

1. Plug the power supply cord in to the power supply module.
2. Wrap the tie around the power supply cord.
3. Ensure that the power supply cord is secured to the power supply module.
4. Tighten the tie around the power supply cord as shown.
5. Ensure that the power cord is secured always to a cable support to ensure the cable load doesn't act on the PSU.

Figure 63: Attach the AC Power Cables



Note These images are for representation purpose only. Certain Cisco 8010 Series Routers may not include a tie for the power supply cord.

1	Insert power cord
2	Wrap power cord with tie
3	Attach the power cord
4	Secure power cord with tie

Turn On an AC Power Supply Module

Perform the following procedure to activate an AC power supply:

1. Plug the power cord into the power supply.
2. Connect the other end of the power cord to an AC-input power source.
3. Verify power supply operation by checking if the respective power supply front panel LED (PM0 or PM1) is green.
4. If the LEDs indicate a power problem, see *Appendix* for troubleshooting information.
5. If you're also connecting a redundant AC power supply, repeat these steps for the second power source.



Note If you're connecting a redundant AC power supply, ensure that each power supply is connected to a separate power source in order to prevent power loss in the event of a power failure.

Operating Voltage Range

- **Cisco 8011-4G24Y4H-I:** 100V-240VAC, 50/60Hz, 5-2.2A maximum.
- **Cisco 8011-32Y8L2H2FH:** 100V-240VAC, 50/60Hz, 7.6-3.65A maximum.
- **Cisco 8011-12G12X4Y:** 100V-240VAC, 50/60Hz, 2A maximum.

Connect to a DC Power Source

Cisco 8011-4G24Y4H-I Router

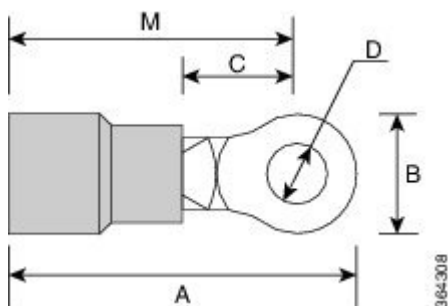


Note When installing DC power supply, use 14AWG for longer cables and 14-16AWG for shorter cables, 90°C temperature rated cable. The recommended cable length is three meters maximum from source.



- Note**
- Always ensure that the building's installation for short-circuit (overcurrent) protection does not exceed 15A.
 - We recommend you to use a circuit breaker or a fast acting fuse with a maximum DC rating, based on the router variants for over current protection.

Figure 64: DC Lug Dimensions

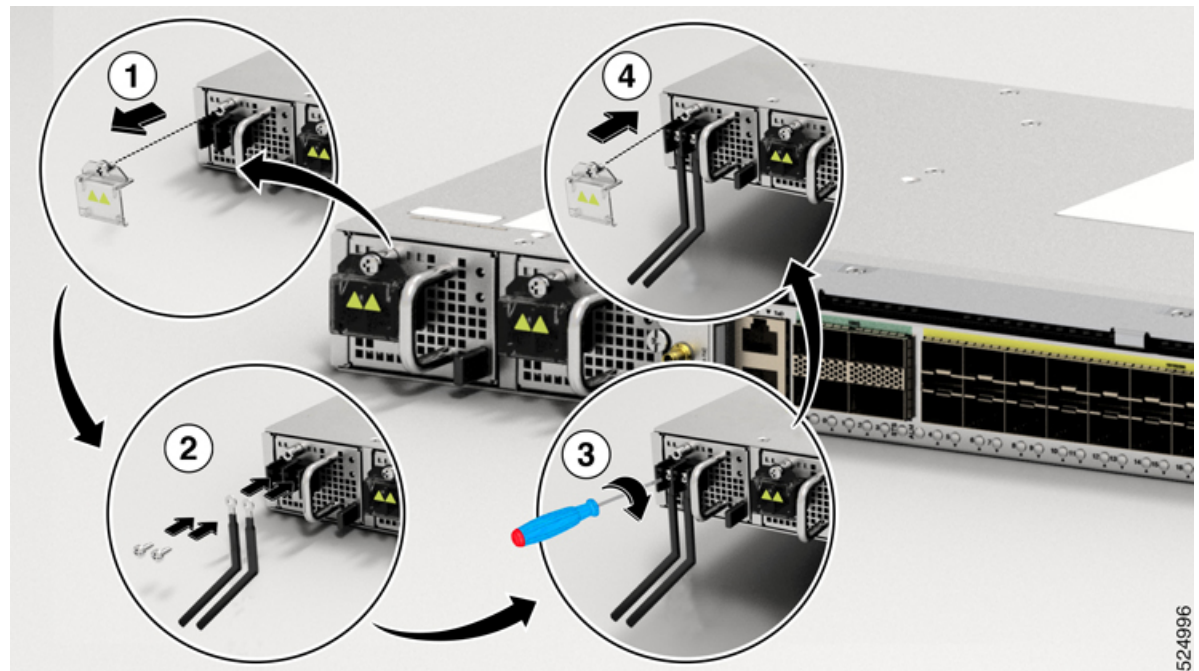


A	0.97 in. (2.4 cm)	C	0.27 in. (0.68 cm)
B	0.31 in. (0.78 cm)	D	0.17 in. (0.43 cm)
M	0.81 in. (2 cm)		

To connect to the DC power source:

1. Locate the terminal block plug.
2. Insert the DC-input power source wires into the terminal block plug.
3. Attach the DC supply wires using the designated screws.
4. Use a ratcheting torque screwdriver to torque the terminal block plug captive screw. (See the following figure.)

Figure 65: Attach the DC Power Supply Wires



Note These images are for representation purpose only. Certain Cisco 8010 Series Routers may not include a tie for the power supply cord.

Cisco 8011-32Y8L2H2FH Router

Prior to connecting the power supply to a power source, ensure that the chassis is properly grounded.



Note When installing DC power supply, use 14AWG for longer cables and 14-16AWG for shorter cables, 90°C temperature rated cable. The recommended cable length is three meters maximum from source.

**Note**

- Always ensure that the building's installation for short-circuit (overcurrent) protection does not exceed 30A.
- We recommend you to use a circuit breaker or a fast acting fuse with a maximum DC rating, based on the router variants for over current protection.

To connect the DC power supply directly to one or two DC power sources, follow these steps:

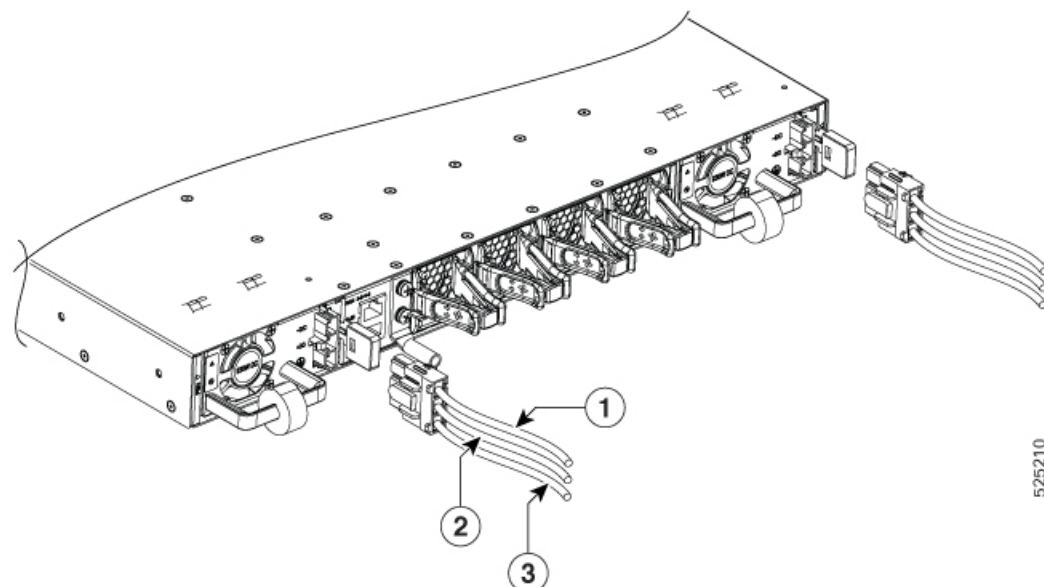
**Warning****Statement 1003—DC Power Disconnection**

To reduce risk of electric shock or personal injury, disconnect DC power before removing or replacing components or performing upgrades.

1. Plug the DC power cable (CAB-48DC-40A-8AWG) into the DC power supply.
2. Turn off the power at the circuit breakers for the portions of the DC grid power that you are connecting to and verify that all of the LEDs on the DC grid power supplies are off.
3. Depending on the power supply installed, install the two cables from the DC power supply to the DC power source as follows:
 - a. If the unconnected end of each power cable is not stripped off of its insulation for the last 0.75 inches (19 mm), use wire strippers to remove that amount of insulation.
 - b. Attach the negative cables to the negative terminals of a DC power source, and attach the positive cables to the positive terminals of the same power source.
4. Verify that the power supply is receiving power by checking that the LED is on and is amber or red. For more information about the power supply LEDs and the conditions that they indicate, see [Power Status LEDs](#).

When you first activate the power supply, you can verify the functionality of the LED by checking that LED turns on for a couple of seconds. If the LED is flashing amber or red, check the power connections on the power supply and the power source

Figure 66: Attach the DC Power Supply Wires



1	Negative terminal	3	Ground terminal
2	Positive terminal		



Note These images are for representation purpose only. Certain Cisco 8010 Series Routers may not include a tie for the power supply cord.

Cisco 8011-12G12X4Y Router

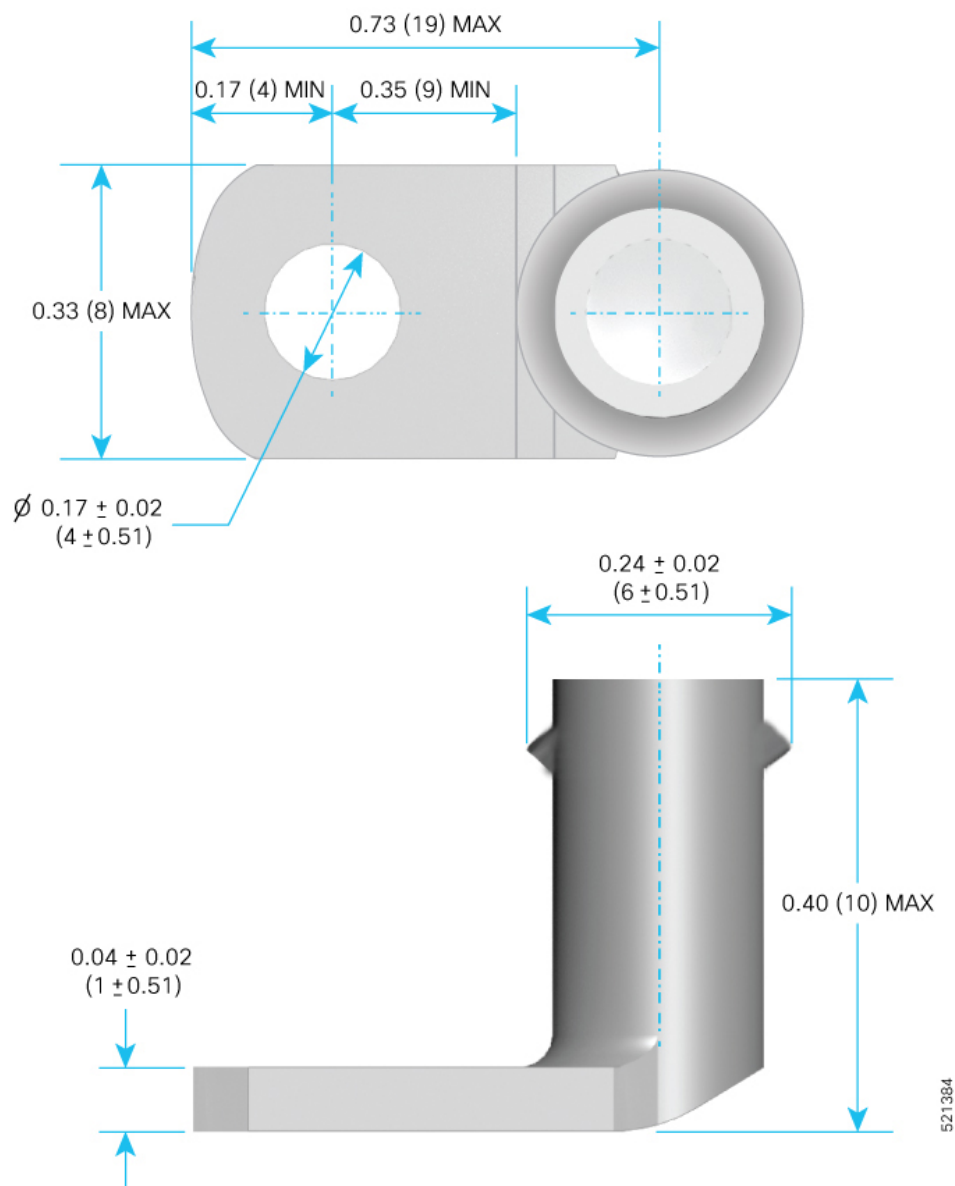


Note When installing DC power supply, use 14AWG for longer cables and 14-16AWG for shorter cables, 90°C temperature rated cable. The recommended cable length is three meters maximum from source.



- Note**
- Always ensure that the building's installation for short-circuit (overcurrent) protection does not exceed 6A.
 - We recommend you to use a circuit breaker or a fast acting fuse with a maximum DC rating, based on the router variants for over current protection.

Figure 67: DC Connector with In built Screw



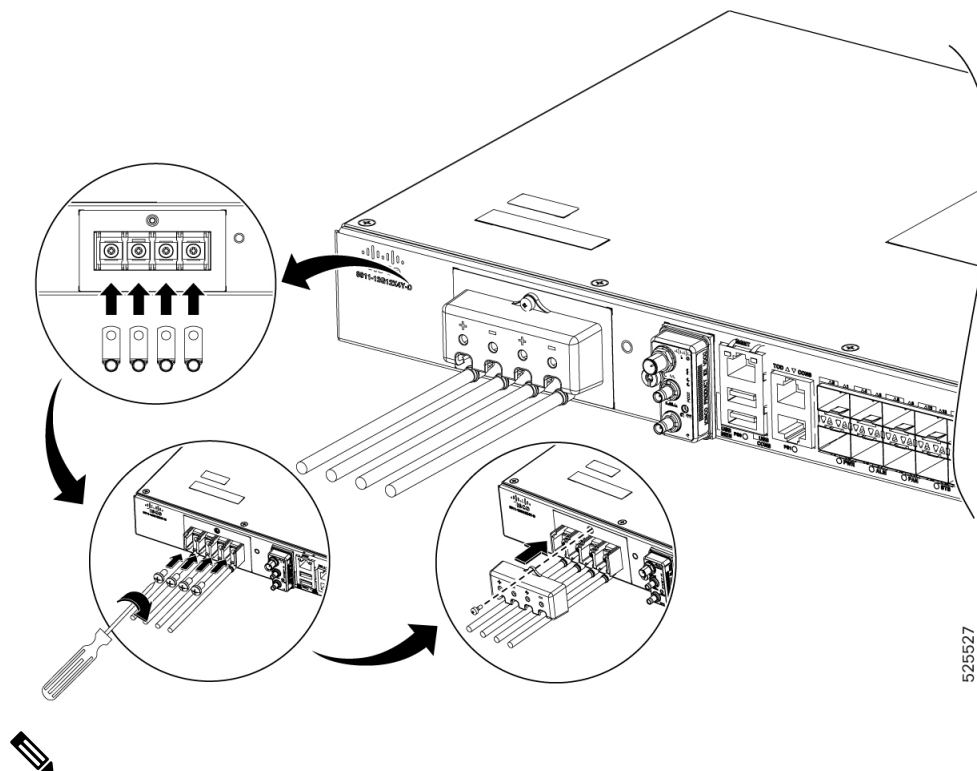
A	0.97 in. (2.4 cm)	C	0.27 in. (0.68 cm)
B	0.31 in. (0.78 cm)	D	0.17 in. (0.43 cm)
M	0.81 in. (2 cm)		

To connect to the DC power source:

1. Locate the terminal block plug.
2. Insert the DC-input power source wires into the terminal block plug.

3. Attach the DC supply wires using the designated screws.
4. Use a ratcheting torque screwdriver to torque the terminal block plug captive screw. (See the following figure.)

Figure 68: Attach the DC Power Supply Wires



Note These images are for representation purpose only. Certain Cisco 8010 Series Routers may not include a tie for the power supply cord.

Turn On a DC Power Supply Module

Perform this procedure to activate a DC power supply:

1. Verify the power supply operation by checking whether the respective power supply front panel LED (PS0 or PS1) is green.
2. If the LEDs indicate any issues with power problem, see *Appendix*.
3. If you are also connecting a redundant DC power supply, repeat these steps for the second power source.



Note If you are connecting a redundant mix mode (AC and DC) power supply, ensure that each power supply is connected to a separate power source to prevent power loss in the event of a power failure.

Operating Voltage Range

Table 18: Operating voltage range for mixed mode power supply

Port Connection Guidelines

These are the port connection guidelines:

- Depending on the chassis, you can use Quad Small Form-factor Pluggable Double Density (QSFP-DD), QSFP28, SFP, SFP+, and RJ45 connectors to connect the ports on the system to other network devices.
- To prevent damage to the fiber-optic cables, we recommend that you keep the transceivers disconnected from their fiber-optic cables when installing the transceiver in the system. Before removing a transceiver from the router, remove the cable from the transceiver.
- To maximize the effectiveness and life of your transceivers and optical cables, do the following:
 - Wear an ESD-preventative wrist strap that is connected to an earth ground whenever handling transceivers. The router is typically grounded during installation and provides an ESD port to which you can connect your wrist strap.
 - Do not remove and insert a transceiver more often than is necessary. Repeated removals and insertions can shorten its useful life.
 - Keep the transceivers and fiber-optic cables clean and dust free to maintain high signal accuracy and to prevent damage to the connectors. Attenuation (loss of light) is increased by contamination and should be kept below 0.35 dB.
 - Clean these parts before installation to prevent dust from scratching the fiber-optic cable ends.
 - Clean the connectors regularly; the required frequency for cleaning depends upon the environment. In addition, clean connectors when they are exposed to dust or accidentally touched. Both wet and dry cleaning techniques can be effective; refer to your site's fiber-optic connection cleaning procedures.
 - Do not touch the ends of connectors. Touching the ends can leave fingerprints and cause other contamination.
- Inspect routinely for dust and damage. If you suspect damage, clean and then inspect fiber ends under a microscope to determine if damage has occurred.

Console Port Connection

The system console port is an RJ45 receptacle for connecting a data terminal to perform the initial configuration of the router. After you create a local management connection, you can use the console to perform these functions through the management interface:

- Configure the router using the command-line interface (CLI).
- Monitor network statistics and errors.
- Configure Simple Network Management Protocol (SNMP) agent parameters.

- Download software updates.

The console cable is shipped with the hardware. Only RJ45 to DB-9 adapter cable is provided in the package.

Figure 69: Cisco 8011-4G24Y4H-I Router — Connecting the USB Type-A Console Cable to the Chassis

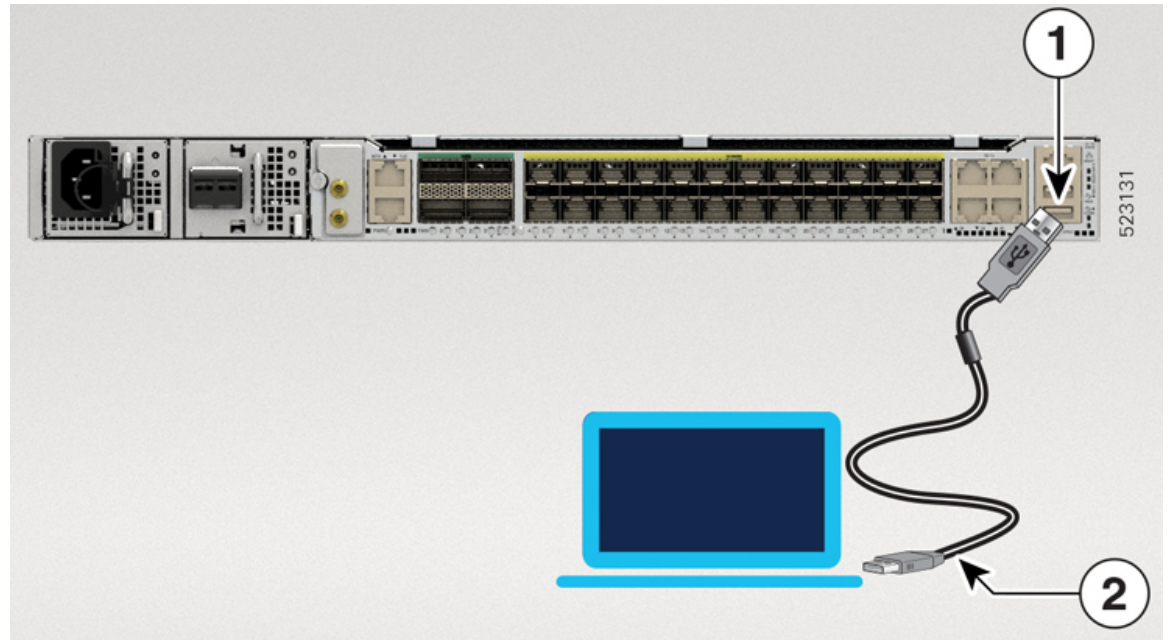


Figure 70: Cisco 8011-32Y8L2H2FH — Connecting the USB Type-A Console Cable to the Chassis

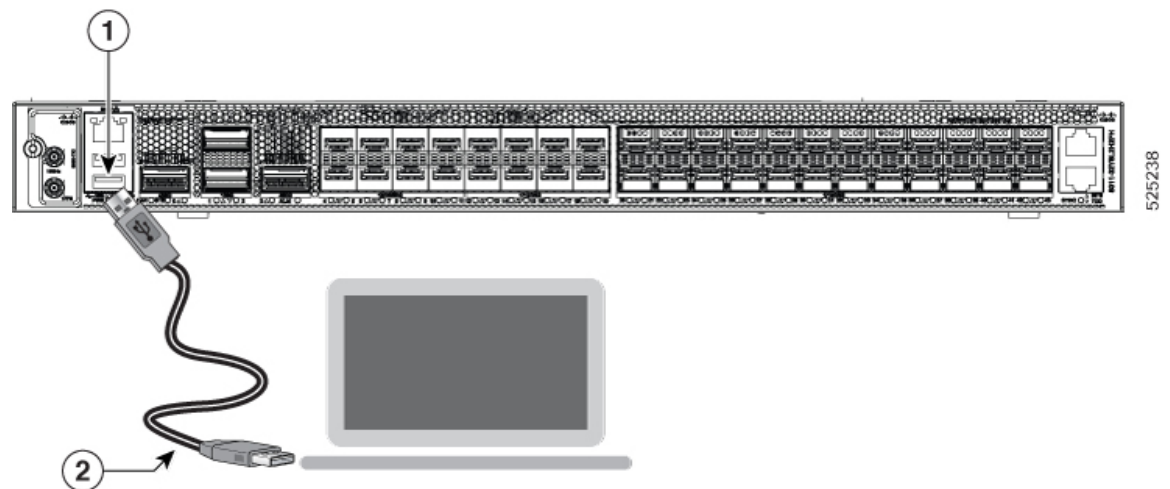
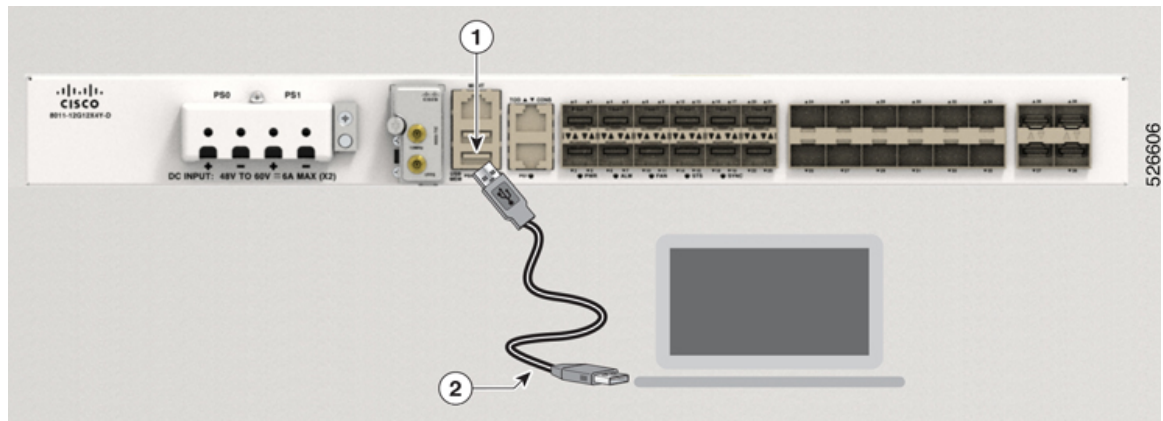


Figure 71: Cisco 8011-12G12X4Y — Connecting the USB Type-A Console Cable to the Chassis



1	USB Type-A console port	2	USB Type-A to USB Type-A cable
---	-------------------------	---	--------------------------------

Connect a Data Terminal to the Console Port

To connect a data terminal to the console port:

Before you begin

- Before you create a network management connection for the router or connect the router to the network, you must create a local management connection through a console terminal and configure an IP address for the router.

The router must be fully installed in its rack, connected to a power source, and grounded.

- The necessary cabling for the console, management, and network connections must be available.
 - An RJ45 rollover cable and DB9F/RJ45 adapter are provided in the router accessory kit.
 - Network cabling should already be routed to the location of the installed router.

Procedure

-
- Step 1** Set your terminal to these operational values: 115200 bps, 8 data bits, no parity, and two stop bits.
- Step 2** Attach the terminal end of the cable to the interface port on the data terminal.
- Step 3** Attach the other end of the cable to the console port.
-

Management Ethernet Port Connection

The management Ethernet port provides out-of-band management, which enables you to use the Command Line Interface (CLI) to manage the router by its IP address. This port uses a 10/100/1000 Ethernet connection with an RJ45 interface.

Figure 72: Cisco 8011-4G24Y4H-I — Management Ethernet Port

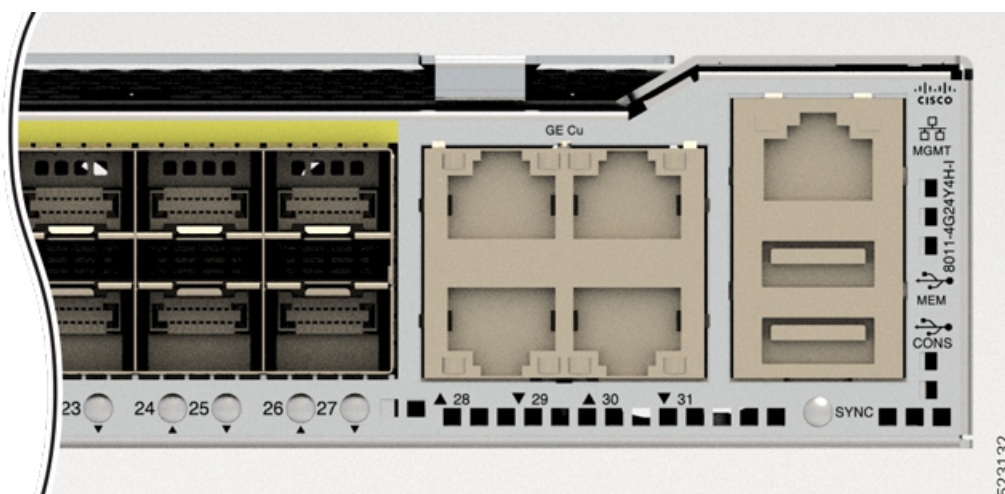


Figure 73: Cisco 8011-32Y8L2H2FH — Management Ethernet Port

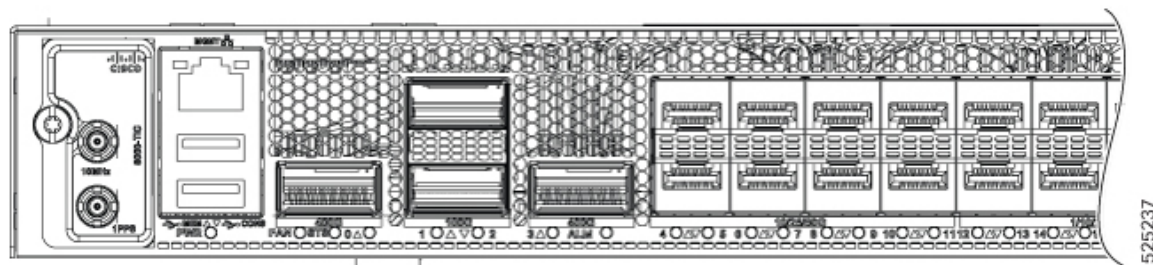
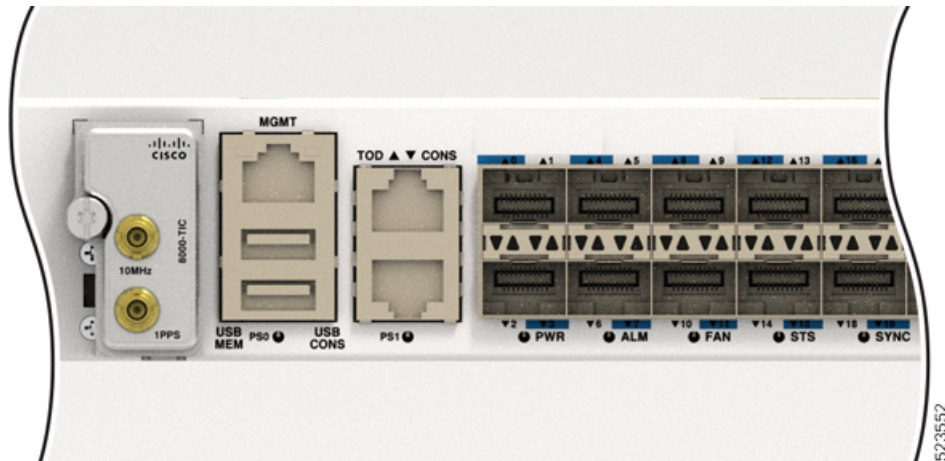


Figure 74: Cisco 8011-12G12X4Y — Management Ethernet Port



Note To comply with GR-1089-CORE, the intra-building port(s) of the equipment must use shielded intra-building cabling or wiring that is grounded at both ends.

Connect to the Management Ethernet Port

To connect cables to the system management port:

Before you begin

You must complete the initial router configuration.



Note To prevent an IP address conflict, do not connect the management Ethernet port until the initial configuration is complete.

Procedure

- Step 1** Attach category 5 cables directly to the RJ45 receptacle on the management Ethernet port.
- Step 2** Connect the network end of your RJ45 cable to a switch, hub, repeater, or other external equipment.

Connecting Cables to Timing Interfaces Card

Perform these steps on how to connect cables from the router to a Timing Interfaces Card unit for input 10Mhz or 1PPS interface.

1. Connect one end of a mini-coax cable to the Timing Interfaces Card unit.
2. Connect the other end of the mini-coax cable to the 10MHz or 1PPS port on the router.

Inserting and Securing Timing Interfaces Card with GNSS Module

Summary

This process involves physically integrating the module into a device and fastening it to ensure proper connection and stability.

The key components involved in the process are:

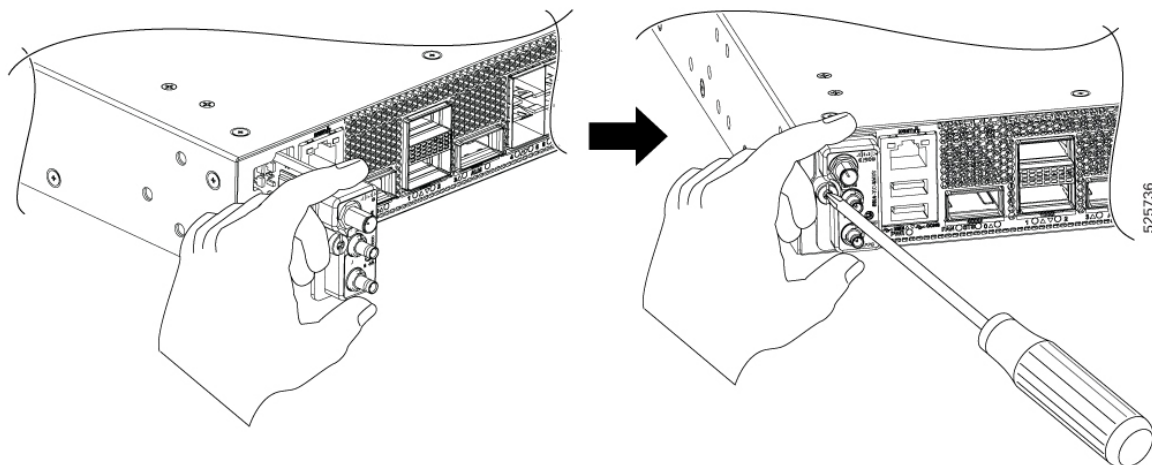
- Installer: The individual performing the physical installation and securing actions.
- GNSS module: The hardware component being installed, which provides Global Navigation Satellite System capabilities to the device.
- Device chassis: The main equipment or system where the GNSS module is inserted.
- Screwdriver: A tool used to tighten the captive screws on the module, ensuring it is firmly secured.

Workflow



Note You can insert and secure the Cisco 8000-TIC-GNSS module only on the Cisco 8011-32Y8L2H2FH Router.

Figure 75: Cisco 8011-32Y8L2H2FH Router: Inserting and securing the Cisco 8000-TIC-GNSS module



These are the stages of inserting and securing a Timing Interfaces Card with GNSS module:

1. **Module Insertion:** The installer carefully aligns the GNSS module with the designated slot on the device chassis and pushes it into place until it is fully seated within the device's internal connectors.

2. **Module Securing:** The installer uses a screwdriver to tighten the captive screws located on the faceplate of the GNSS module. This action secures the module firmly to the device chassis, preventing accidental dislodgement and ensuring a stable connection.



Note Ensure that the captive screws are tightened with a torque of 5lbs-in or 0.65 Nm.

Result

The GNSS module is successfully installed and secured within the device, establishing a stable physical connection and preparing it for operational use.

Install and Remove Transceiver Module



Caution Removing and installing a transceiver can shorten its useful life. Do not remove and insert transceivers more than it is absolutely necessary. We recommended that you disconnect cables before installing or removing transceivers to prevent damage to the cable or transceiver

This section shows how to install and remove transceiver module:

Safety Precautions for Module Installation and Removal

Be sure to observe the following safety precautions when you work on the chassis.



Warning **Statement 1006**—Chassis Warning for Rack-Mounting and Servicing

To prevent bodily injury when mounting or servicing this unit in a rack, you must take special precautions to ensure that the system remains stable. The following guidelines are provided to ensure your safety:

- This unit should be mounted at the bottom of the rack if it is the only unit in the rack.
 - When mounting this unit in a partially filled rack, load the rack from the bottom to the top with the heaviest component at the bottom of the rack.
 - If the rack is provided with stabilizing devices, install the stabilizers before mounting or servicing the unit in the rack.
-



Warning **Statement 1008**—Class 1 Laser Product

This product is a Class 1 laser product.

**Note** **Statement 1089**—Instructed and Skilled Person Definitions

An instructed person is someone who has been instructed and trained by a skilled person and takes the necessary precautions when working with equipment.

A skilled person or qualified personnel is someone who has training or experience in the equipment technology and understands potential hazards when working with equipment.

**Warning** **Statement 1090**—Installation by Skilled Person

Only a skilled person should be allowed to install, replace, or service this equipment. See statement 1089 for the definition of a skilled person.

Install and Remove SFP Modules

Before you remove or install an SFP module, read the installation information in this section.



Caution Protect all the unused ports by inserting clean dust covers or dust caps into them.

**Warning** **Statement 1055**—Class 1/1M Laser

Invisible laser radiation is present. Do not expose to users of telescopic optics. This applies to Class 1/1M laser products.

**Warning** **Statement 1056**—Unterminated Fiber Cable

Invisible laser radiation may be emitted from the end of the unterminated fiber cable or connector. Do not view directly with optical instruments. Viewing the laser output with certain optical instruments, for example, eye loupes, magnifiers, and microscopes, within a distance of 100 mm, may pose an eye hazard.



Caution Protect the system by inserting a clean SFP module cage cover (shown in the following figure) into the optical module cage when there is no SFP module installed. The SFP module cage cover is not a standard part of the accessories kit.

Figure 76: SFP Module Cage Cover

**Caution**

Protect the SFP or SFP+ modules by inserting clean dust covers into them after the cables are removed. Be sure to clean the optic surfaces of the fiber cables before you plug them back into the optical ports of another module. Avoid getting dust and other contaminants into the optical ports of your SFP modules, because the optics do not work correctly when obstructed by dust.

**Caution**

We strongly recommend that you do not install or remove the SFP module with fiber-optic cables attached to it because of the potential of damaging the cable, the cable connector, or the optical interfaces in the module. Disconnect all cables before removing or installing an SFP module. Removing and inserting a module can shorten its useful life; so you should not remove and insert modules more than it is absolutely necessary.

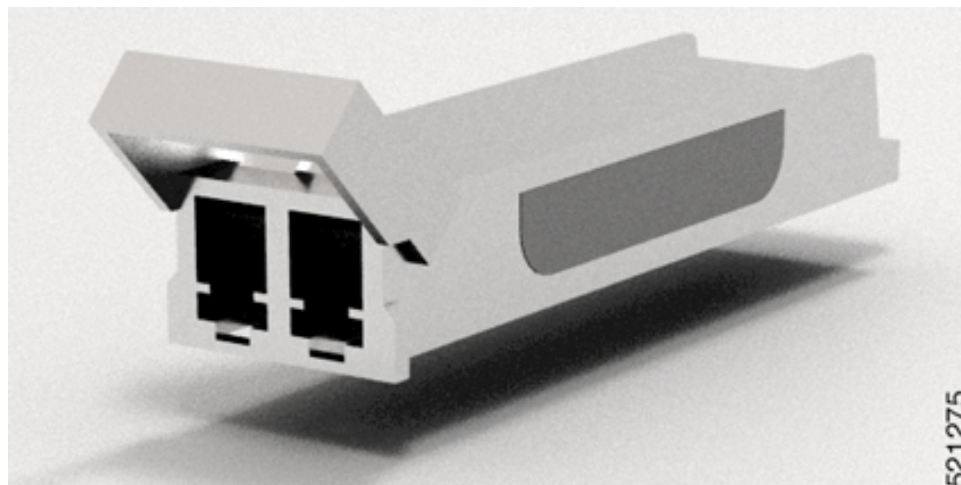
**Note**

When installing an SFP module, you would hear a click as the triangular pin on the bottom of the module snaps into position into the hole in the receptacle. The click indicates that the module is correctly seated and secured in the receptacle. Verify that the modules are completely seated and secured in their assigned receptacles on the system by firmly pushing on each SFP module.

Bale Clasp SFP Module

The bale clasp SFP module has a clasp that you use to remove or install the module. (See the figure below.)

Figure 77: Bale Clasp SFP Module

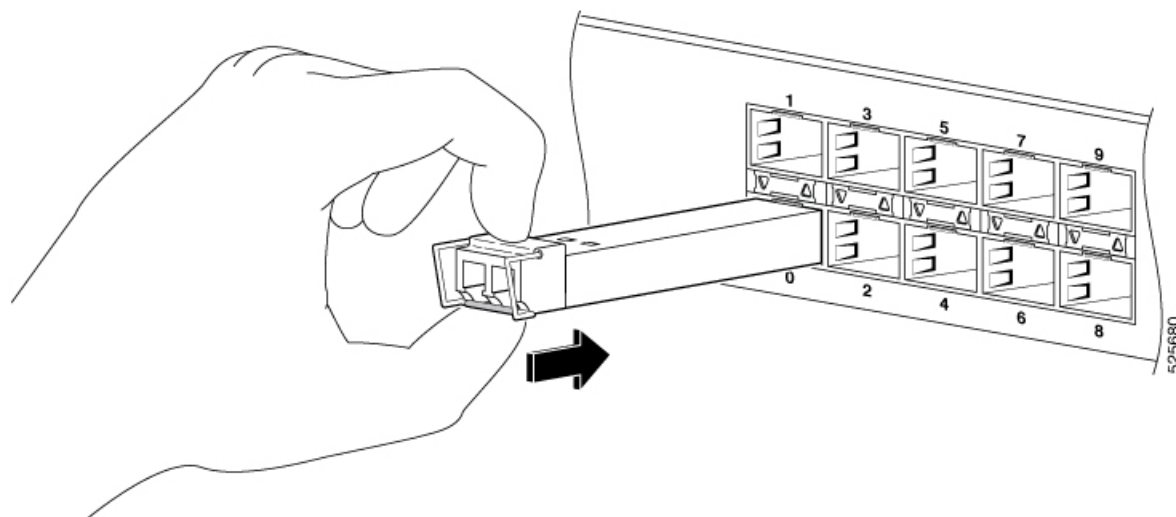


Install a Bale Clasp SFP Module

To install this type of SFP module:

1. Attach an ESD-preventive wrist or ankle strap and follow its instructions for use.
2. Close the bale clasp before inserting the SFP module.
3. Line up the SFP module with the port and slide it into the port. (See the figure below.)

Figure 78: Installing a Bale Clasp SFP Module into a Port



Note

When installing an SFP or SFP+ module, you should hear a click as the triangular pin on the bottom of the SFP module snaps into the hole in the receptacle. This click indicates that the module is correctly seated and secured in the receptacle. Verify that the SFP modules are completely seated and secured in their assigned receptacles on the system by firmly pushing on each SFP module.

Remove a Bale Clasp SFP Module

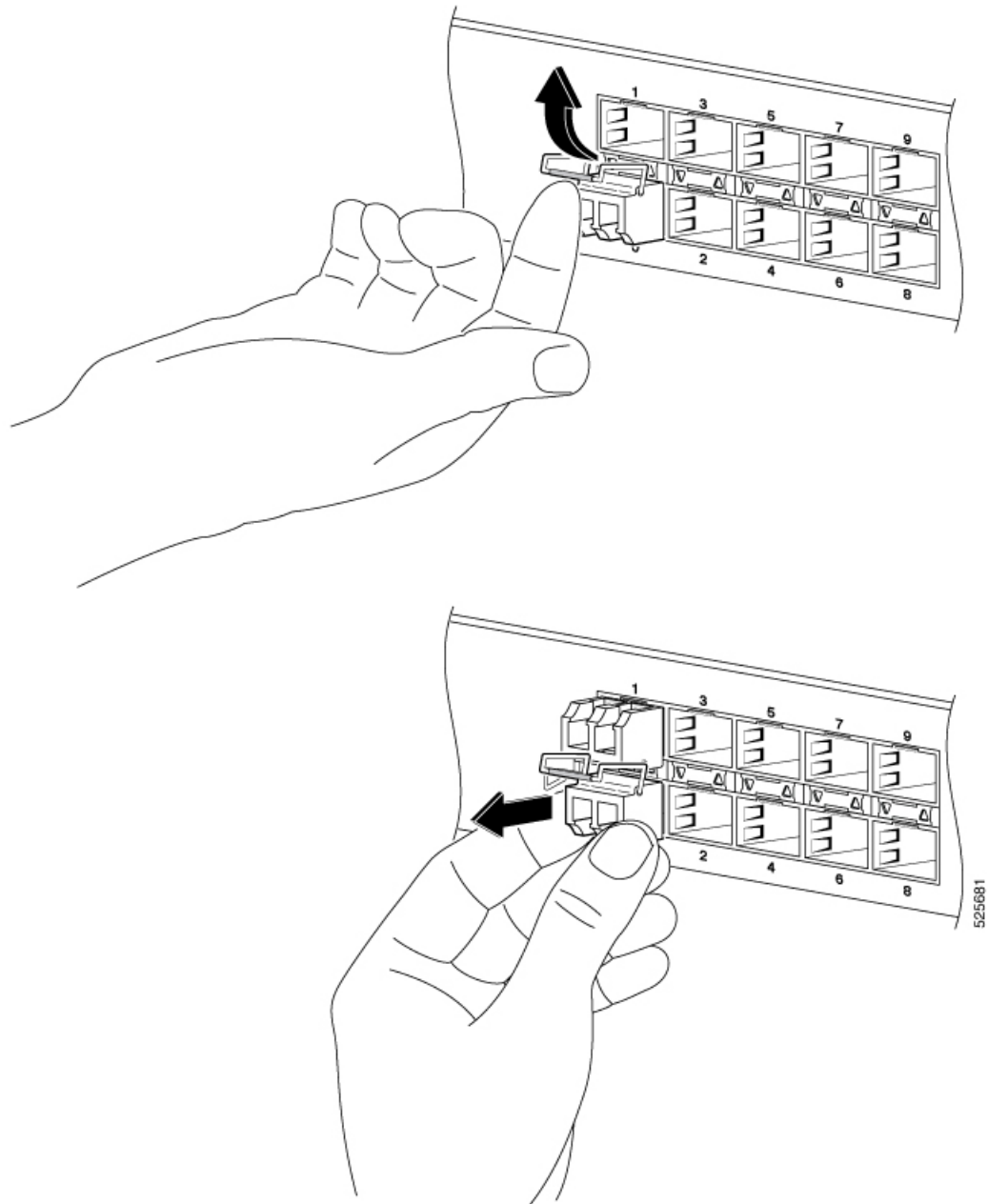
To remove this type of SFP module:

1. Attach an ESD-preventive wrist or ankle strap and follow its instructions for use.
2. Disconnect and remove all interface cables from the ports; note the current connections of the cables to the ports on the system.
3. Open the bale clasp on the SFP module with your index finger, as shown in the figure below. If the bale clasp is obstructed and if you cannot open it, use your index finger, use a small flat-blade screwdriver or other long, narrow instrument to open the bale clasp.
4. Grasp the SFP module between your thumb and index finger and carefully remove it from the port, as shown in the figure below.



Note This action must be performed during your first instance. After all ports are populated, this may not be possible.

Figure 79: Removing a bale clasp SFP module



5. Place the SFP module that you removed on an antistatic mat, or immediately place it in a static shielding bag if you plan to return it to the factory.
6. Protect your system by inserting a clean SFP module cage covers into the optical module cage when there is no SFP module installed.

Connect Interface Ports

You can connect optical interface ports on with other devices for network connectivity.

Connect a Fiber-Optic Port to the Network

Depending on the model that you are using, you can use either QSFP+ or QSFP28 transceivers. Some transceivers work with fiber-optic cables that you attach to the transceivers and other transceivers work with pre-attached copper cables. When installing fiber-optic cables for a port, you must install SFP transceivers for 1-Gigabit optical ports or install SFP+ transceivers for 10-Gigabit optical ports or QSFP+ transceivers for 100-Gigabit ports before installing the fiber-optic cable in the transceivers.

**Caution**

Removing and installing a transceiver can shorten its useful life. Do not remove and insert transceivers more than it is absolutely necessary. We recommended that you disconnect cables before installing or removing transceivers to prevent damage to the cable or transceiver.

Disconnect Optical Ports from the Network

When you need to remove fiber-optic transceivers, you must first remove the fiber-optic cables from the transceiver before you remove the transceiver from the port.

Maintain Transceivers and Optical Cables

To maintain high signal accuracy and to prevent damage to the connectors, transceivers and fiber-optic cables must be kept clean and free of dust. Attenuation (loss of light) is increased by contamination and should be below 0.35 dB.

Consider the following maintenance guidelines:

- Transceivers are static sensitive. To prevent ESD damage, wear an ESD-preventative wrist strap that is connected to the grounded chassis.
- Do not remove and insert a transceiver more than it is necessary. Repeated removals and insertions can shorten its useful life.
- Keep all optical connections covered when not in use. Clean them before use to prevent dust from scratching the fiber-optic cable ends.
- Do not touch the ends of connectors. Touching the ends would leave fingerprints and cause other contamination.
- Clean the connectors regularly; the required frequency of cleaning depends upon the environment. In addition, clean connectors if they are exposed to dust or have been accidentally touched. Both wet and dry cleaning techniques can be effective; refer to your site's fiber-optic connection cleaning procedures.
- Inspect routinely for dust and damage. Clean and then inspect fiber ends under a microscope to determine whether any damage has occurred.



CHAPTER 5

Configure the Router

Before you begin this task, ensure that you have read and understood the safety warnings in the *Compliance and Safety Information* section.

Configuring the Cisco router involves these tasks:

- [Create the Initial Router Configuration, on page 111](#)
- [Verify Router Installation, on page 112](#)

Create the Initial Router Configuration

You must assign an IP address to the router management interface so that you can then connect the router to the network.

When you initially power up the router, it boots up and asks a series of questions to help configure the router. To enable you to connect the router to the network, you can use the default choices for each configuration except for the IP address, which you must provide.

Before you begin

- A console device must be connected with the router.
- The router must be connected to a power source.

Procedure

Step 1

Power up the router.

The LEDs on each power supply light up (green) when the power supply units are sending power to the router, and the software asks you to specify a password to use with the router.

Step 2

When the system is booted up for the first time, a new username and a password is to be created. The following prompt appears:

```
!!!!!!!!!!!!!!!!!!!! NO root-system username is configured. Need to configure root-system username.
!!!!!!!!!!!!!!!!!!!!
```

```
--- Administrative User Dialog ---
```

```
Enter root-system username:  
% Entry must not be null.
```

```
Enter root-system username: root  
Enter secret:  
Use the 'configure' command to modify this configuration.  
User Access Verification
```

```
Username: root  
Password:
```

```
RP/0/RP0/CPU0:ios#
```

Step 3 Enter a new password to use for this router.

The software checks the security strength of it and rejects your password if it is not considered to be a strong password. To increase the security strength of your password, make sure that it adheres to the following guidelines:

- At least eight characters
- Minimizes or avoids the use of consecutive characters (such as "abcd")
- Minimizes or avoids repeating characters (such as "aaa")
- Does not contain recognizable words from the dictionary
- Does not contain proper names
- Contains both uppercase and lowercase characters
- Contains both numbers and letters

Note

Clear text passwords cannot include the dollar sign (\$) special character.

Tip

If a password is trivial (such as a short, easy-to-decipher password), the software rejects the password configuration. Be sure to configure a strong password as described by the guidelines in this step. Passwords are case sensitive.

If you enter a strong password, the software asks you to confirm the password.

Step 4 Re-enter the password.

When you enter the same password, the software accepts the password.

Verify Router Installation

After installing the Cisco router, you use the **show** commands to verify the installation and configuration. If any issue is detected, take corrective action before continuing with further configurations.

1. show inventory

Displays information about the field replaceable units (FRUs), including product IDs, serial numbers, and version IDs.

Example:

```
#show inventory
```

2. **show environment**

Displays all environment-related router information.

Example:

```
# show environment
```

3. **show environment temperature**

Displays temperature readings for on-board temperature sensors and for PSU. Each temperature sensor has three thresholds:

- Minor temperature threshold: When a minor threshold is exceeded, a minor alarm occurs and the following actions occur for all sensors:
 - System messages are displayed
 - SNMP notifications (if configured) are sent
 - Log environmental alarm event is triggered (Run the **show alarm** command to review this.)
- Major temperature threshold: When a major threshold is exceeded, a major alarm occurs and the following actions occur for all sensors:
 - System messages are displayed
 - SNMP notifications (if configured) are sent
 - Log environmental alarm event is triggered (Run the **show alarm** command to review this.)
- Critical temperature threshold: When a critical threshold is exceeded, a critical alarm occurs and the following actions occurs:
 - For all the main board sensors the system is shut down.
 - For the PSU sensor, the particular PSU is turned off .

4. **show environment power**

Displays the power usage information for the entire router.

Example:

```
#show environment power
```

5. **show environment voltage**

Displays the voltage for the entire router.

Example:

```
#show environment voltage
```

6. **show environment current**

Displays current for different voltage rails of the router.

Example:

```
#show environment current
```

7. **show environment fan**

Displays the speed of all the fans including the fan in PSU.

Example:

```
#show environment fan
```



CHAPTER 6

Replace Chassis Components

- [Replace Fan Modules, on page 115](#)
- [Replace Power Supply, on page 117](#)
- [Replace DC Power Supply Modules, on page 118](#)
- [Replace AC Power Supply Modules, on page 122](#)

Replace Fan Modules



Note Fan modules can be replaced only on the Cisco 8011-32Y8L2H2FH Router.



Caution If you cannot replace a fan module (FAN-1RU-PI-V3) within three minutes, we recommend that you leave it in the chassis until you are prepared to replace it within that specified time limit.

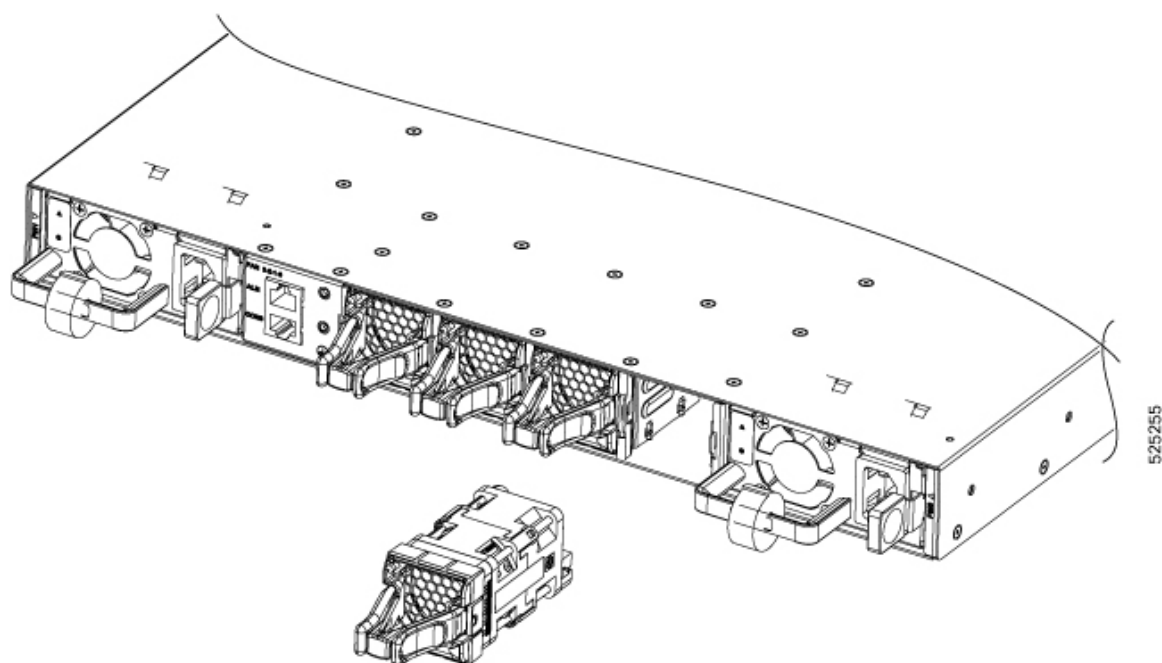


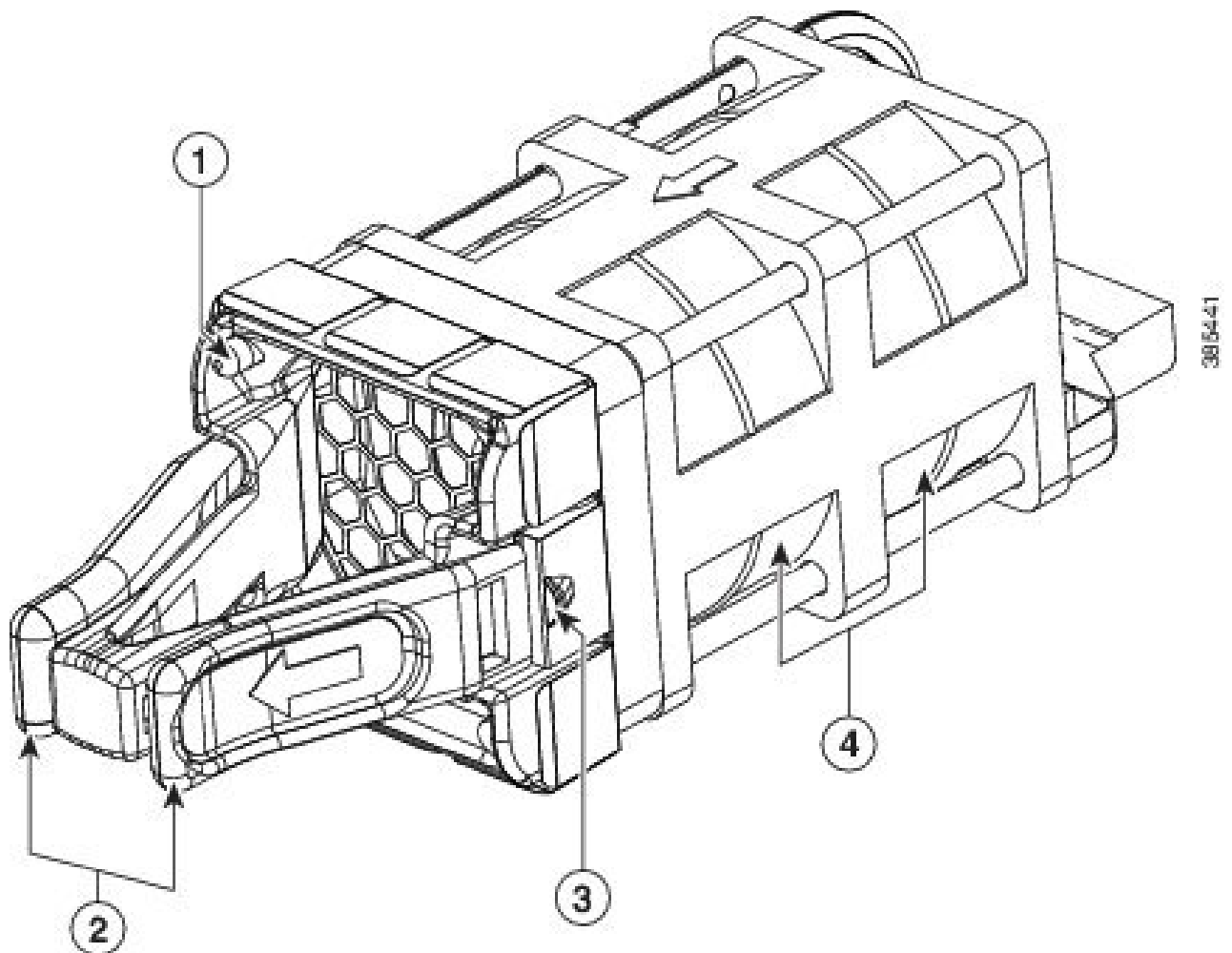
Note If you remove more than one fan module at a time during operations, the router allows up to 2 minutes of operations before shutting down, unless you replace extra missing fan modules within that time. If the router senses an over temperature condition when multiple fan modules are removed, the shutdown can occur in less than 2 minutes.

Procedure

- Step 1** Press both the latches on the fan module to disengage the fan module connection from the chassis.
- Step 2** Simultaneously press the latches, and pull the fan module fully out of the chassis.

Figure 80: Remove fan module from the Cisco 8011-32Y8L2H2FH chassis





1	Fan LED	3	Fan assembly retention latch
2	Fan assembly levers	4	Fan

Replace Power Supply

The Cisco 8010 Large Density Routers provide a choice of two different power supplies as AC and DC.

Cisco 8011-4G24Y4H-I Router

The router also supports a combination of AC (PWR-400-AC) and DC (PWR-400-DC) PSUs together:

- DC power (PWR-400-DC)—The DC power supply uses two-position terminal block-style connector with positive latching or securing, and labeled connections for + 48V and -48V. The terminal block connector is of suitable size to carry the appropriate AWG wire size to handle the input current of the power supply. No ON/OFF switch is provided.

- AC power (PWR-400-AC)—The AC power supply has an IEC 320-type power receptacle and a 16 Amp service connector. You can use standard right angle power cords with the AC power supply. The power supply includes a power cord retainer. No ON/OFF switch is provided.

Cisco 8011-32Y8L2H2FH Router

The router also supports a combination of AC (PWR-650-AC-R) and DC (PWR-930-DC-R) PSUs together:

- DC power (PWR-930-DC-R)—The DC power supply uses two-position terminal block-style connector with positive latching or securing, and labeled connections for + 48V and -48V. The terminal block connector is of suitable size to carry the appropriate AWG wire size to handle the input current of the power supply. No ON/OFF switch is provided.
- AC power (PWR-650-AC-R)—The AC power supply has an IEC 320-type power receptacle and a 16 Amp service connector. You can use standard right angle power cords with the AC power supply. The power supply includes a power cord retainer. No ON/OFF switch is provided.

You can install dual power supplies for redundancy.



Note Products that have an AC power connection are required to have an external SPD provided as part of the building installation to comply with the Telcordia GR-1089 NEBS standard for electromagnetic compatibility and safety.



Caution Don't use interface module and power supply ejector handles to lift the chassis; using the handles to lift the chassis can deform or damage the handles.

PSU Redundancy Lost Alarm:

PSU redundancy lost alarms are generated when there's no proper input feed applied on any one of Power Modules (PMs) (PM0 or PM1). The alarms are also generated when the output for PM0 or PM1 isn't proper.

The following alarms are raised for PSU redundancy lost event with a faulty PM0:

- Power Module Generic Fault
- Power Module Error
- Power Group Redundancy Lost

Replace DC Power Supply Modules

Remove the DC Power Supply Module

This section provides information about removing and replacing the DC power supply.

**Warning****Statement 1003—DC Power Disconnection**

To reduce risk of electric shock or personal injury, disconnect DC power before removing or replacing components or performing upgrades.

Procedure

-
- Step 1** Before servicing the power supply, switch off the circuit breaker in your equipment area. As an additional precaution, tape the circuit-breaker switch in the Off position.
 - Step 2** Slip on the ESD-preventive wrist strap that is included in the accessory kit.
 - Step 3** Switch the power supply circuit-breaker switch to the Off (O) position.
 - Step 4** Pull the terminal block plug cover out of the terminal block head in the power supply.
 - Step 5** Unscrew the terminal block screws and remove the cables.
 - Step 6** Grasp the power supply handle. Press the power supply lock towards the left and simultaneously pull the power supply out from the chassis while supporting it with the other hand.
-

Install the DC Power Supply Module

This equipment is suitable for installation in network telecommunications facilities and locations where the NEC applies.

This equipment is suitable for installations utilizing the Common Bonding Network (CBN).

The grounding architecture of this product is DC-Isolated (DC-I) for DC-powered products. DC-powered products have a nominal operating DC voltage of 48 VDC.

Perform the following procedure to install the power supply module:

Procedure

-
- Step 1** Ensure that the system (earth) ground connection is made. (See the following figure.)
 - Step 2** If necessary, remove the blank power supply filler plate from the chassis power supply bay opening by loosening the captive installation screws.
 - Step 3** Verify that power to the DC circuit connected to the power supply you are installing is turned off. To ensure that power has been removed from the DC circuits, locate the circuit breakers for the DC circuits, switch the circuit breakers to the OFF position, and tape the circuit-breaker switches in the OFF position.
 - Step 4** Grasp the power supply handle with one hand. Place your other hand underneath the power supply. Slide the power supply into the power supply bay. Make sure that the power supply is fully seated in the bay.

Figure 81: Cisco 8011-4G24Y4H-I — Install DC Power Supply Module

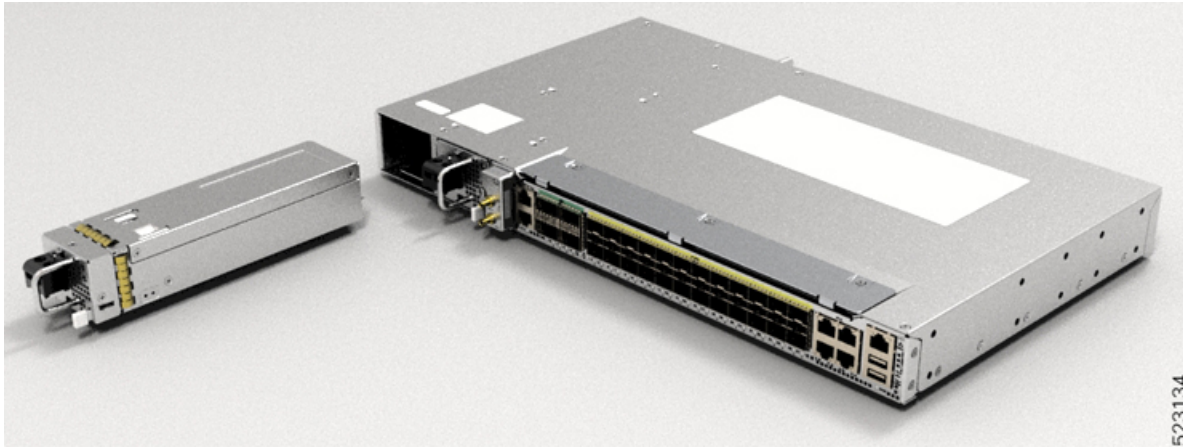
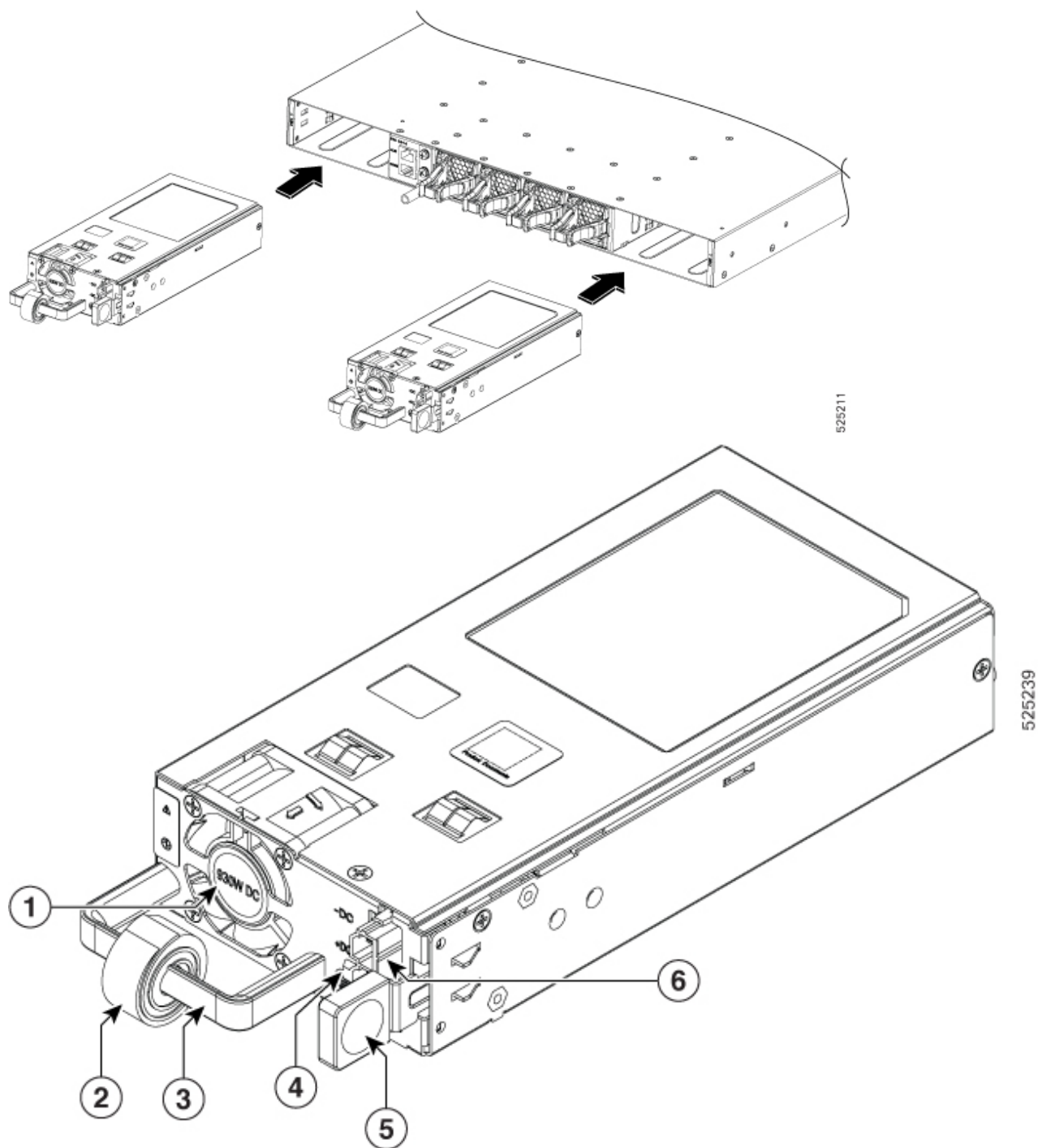


Figure 82: Cisco 8011-32Y8L2H2FH — Install DC Power Supply Module



1	PSU fan	2	Power cord retainer
3	Release handle	4	DC+ connector

5	Release latch	6	DC- connector
---	---------------	---	---------------

Replace AC Power Supply Modules

Remove the AC Power Supply Module

Before you begin

This section describes how to remove and replace the AC power supply.



Warning Statement 1046—Installing or Replacing the Unit

To reduce risk of electric shock, when installing or replacing the unit, the ground connection must always be made first and disconnected last.

If your unit has modules, secure them with the provided screws.



Warning Statement 1074—Comply with Local and National Electrical Codes

To reduce risk of electric shock or fire, installation of the equipment must comply with local and national electrical codes.

To remove and replace the AC power supply:

Procedure

- Step 1** Disconnect the power cord from the power source. Do not touch any metal on the power cord when it is still connected to the power supply.
- Step 2** Loosen the tie and remove the power cord from the tie-and holder.
- Step 3** Remove the power cord from the power connection on the power supply. Do not touch the metal prongs embedded in the power supply.
- Step 4** Grasp the power supply handle. Press the power supply lock towards the left and simultaneously pull the power supply out from the chassis while supporting it with the other hand.

Install the AC Power Supply Module

Follow these steps to install the AC power supply module:

Procedure

- Step 1** Ensure that the system (earth) ground connection is made.
- Step 2** If necessary, remove the blank power supply filler plate from the chassis power supply bay opening by loosening the captive installation screws.
- Step 3** Grasp the power supply handle with one hand. Place your other hand underneath the power supply. Slide the power supply into the power supply bay. Make sure that the power supply is fully seated in the bay. (See the following figure.)

Figure 83: Cisco 8011-4G24Y4H-I — Install AC Power Supply Module

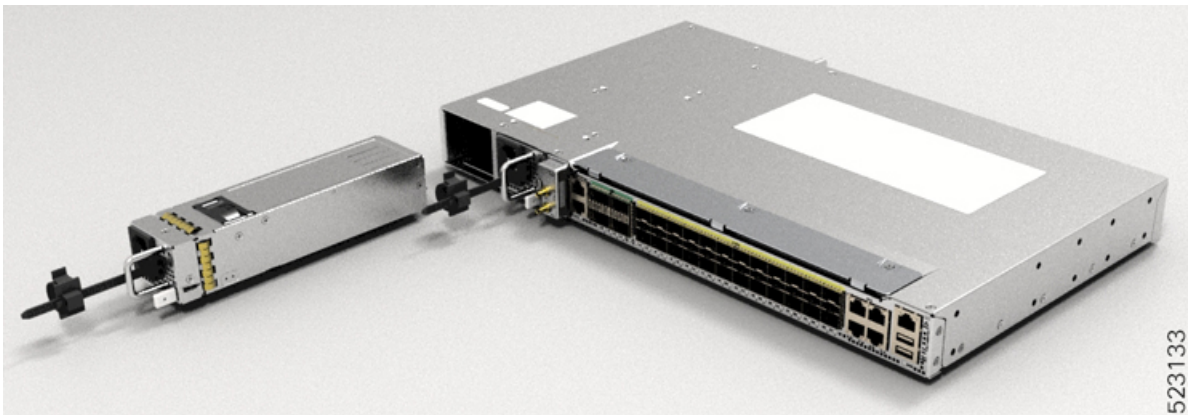
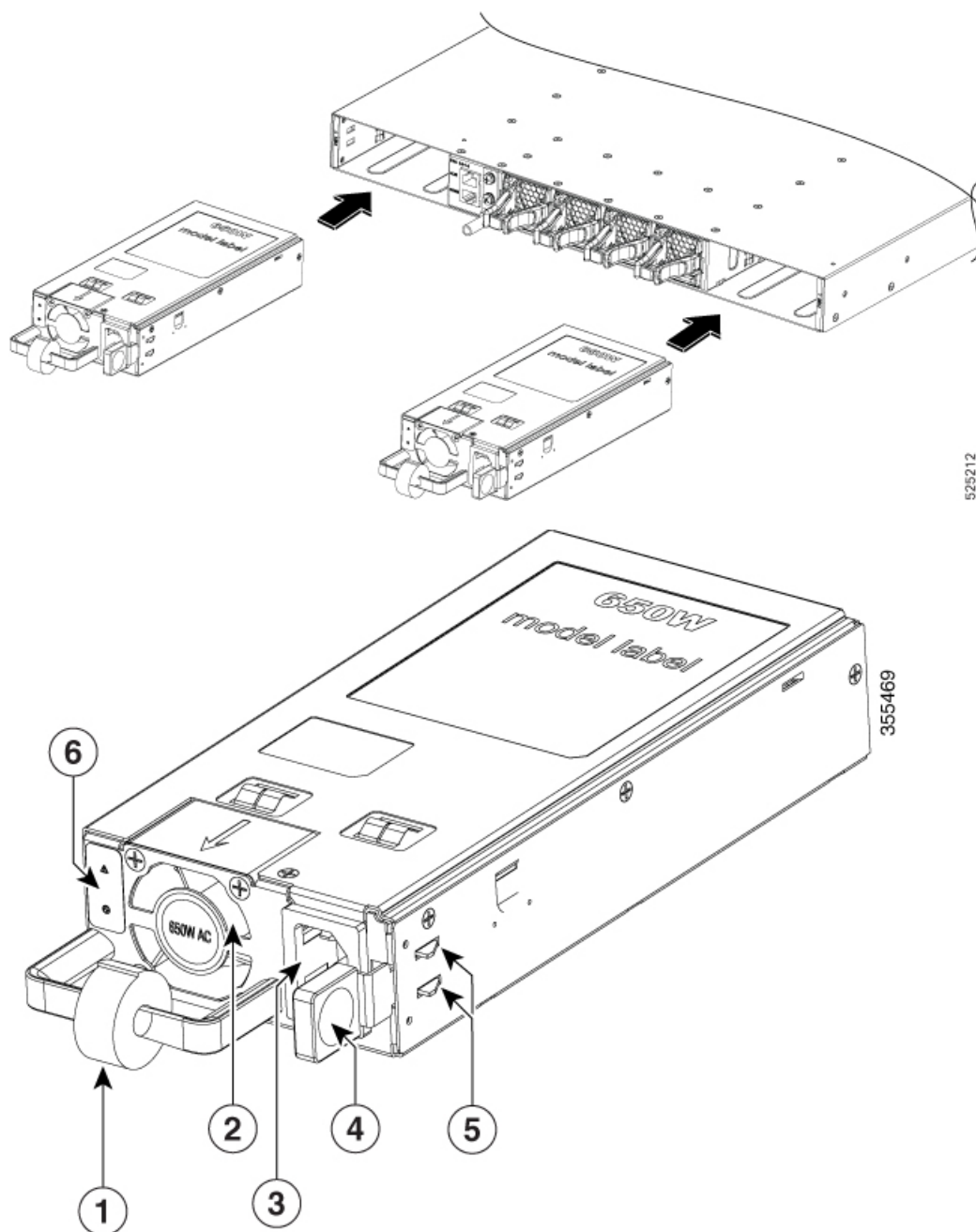


Figure 84: Cisco 8011-32Y8L2H2FH — Install AC Power Supply Module



1	Velco strap	4	Release latch
2	PSU fan	5	Retainer clips
3	AC power cord connector	6	Power status and power supply failure LEDs

Step 4 Slide the AC power supply cord inside the tie of the tie-and-holder and tighten the tie around the power supply cord.

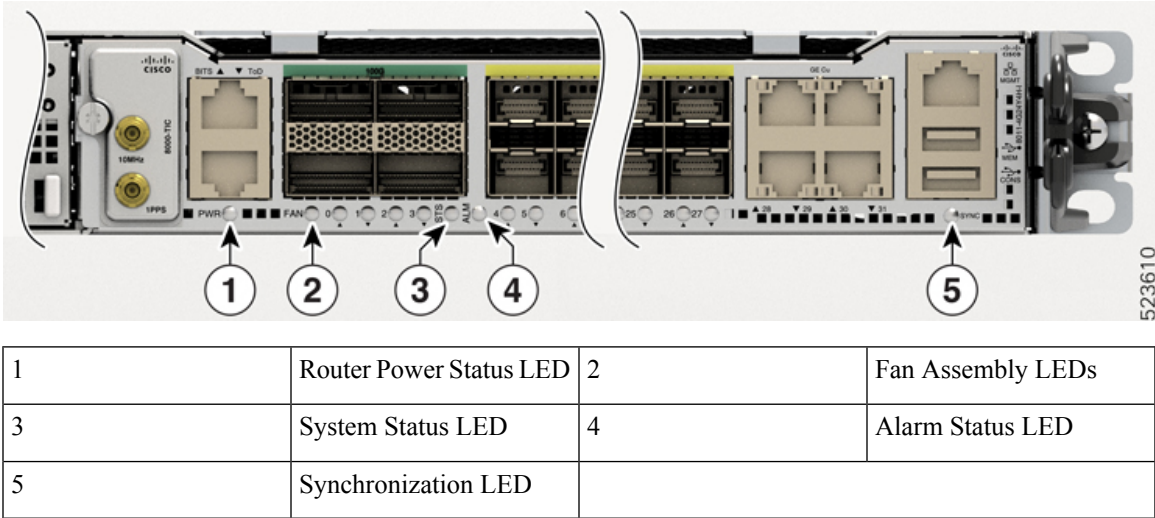
Step 5 Plug the power supply cord into the AC power supply.

Appendix

Certain troubleshooting aids of the Cisco 8010 Large Density Routers enable you to perform these tasks that assist the troubleshooting process:

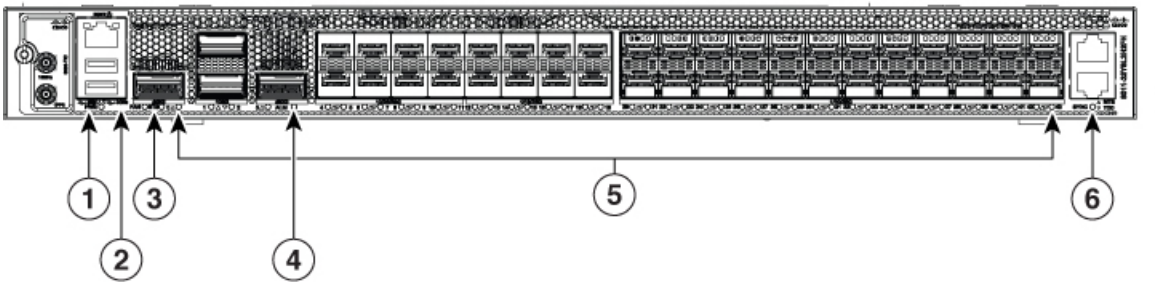
Cisco 8011-4G24Y4H-I Router

Figure 85: Cisco 8011-4G24Y4H-I Router LEDs



Cisco 8011-32Y8L2H2FH Router

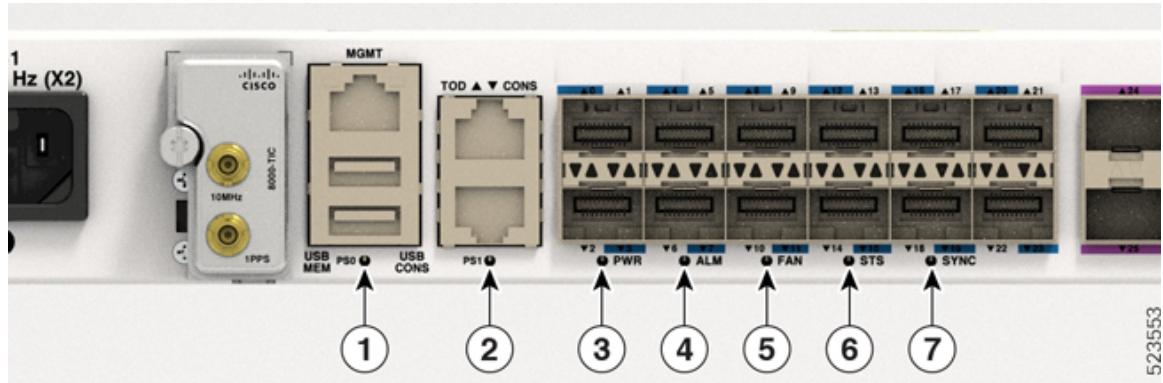
Figure 86: Cisco 8011-32Y8L2H2FH Router LEDs



1	Router power status LED	2	Fan assembly LEDs
3	System status LED	4	Port LED
5	Alarm LEDs	6	Synchronization LED

Cisco 8011-12G12X4Y Router

Figure 87: Cisco 8011-12G12X4Y Router LEDs



1, 2	Power Supply Module LEDs	3	Router Power Status LED
4	Alarm Status LED	5	Fan Assembly LEDs
6	System Status LED	7	Synchronization LED

- [LEDs](#), on page 128
- [System Specifications](#), on page 136

LEDs

The Cisco 8010 Large Density Router LEDs are similar for most of the variants, and any differences between the routers are specifically called out.

Router LEDs

All the data port LEDs in the Cisco router are at the front panel.

Figure 88: Cisco 8011-32Y8L2H2FH Router LEDs

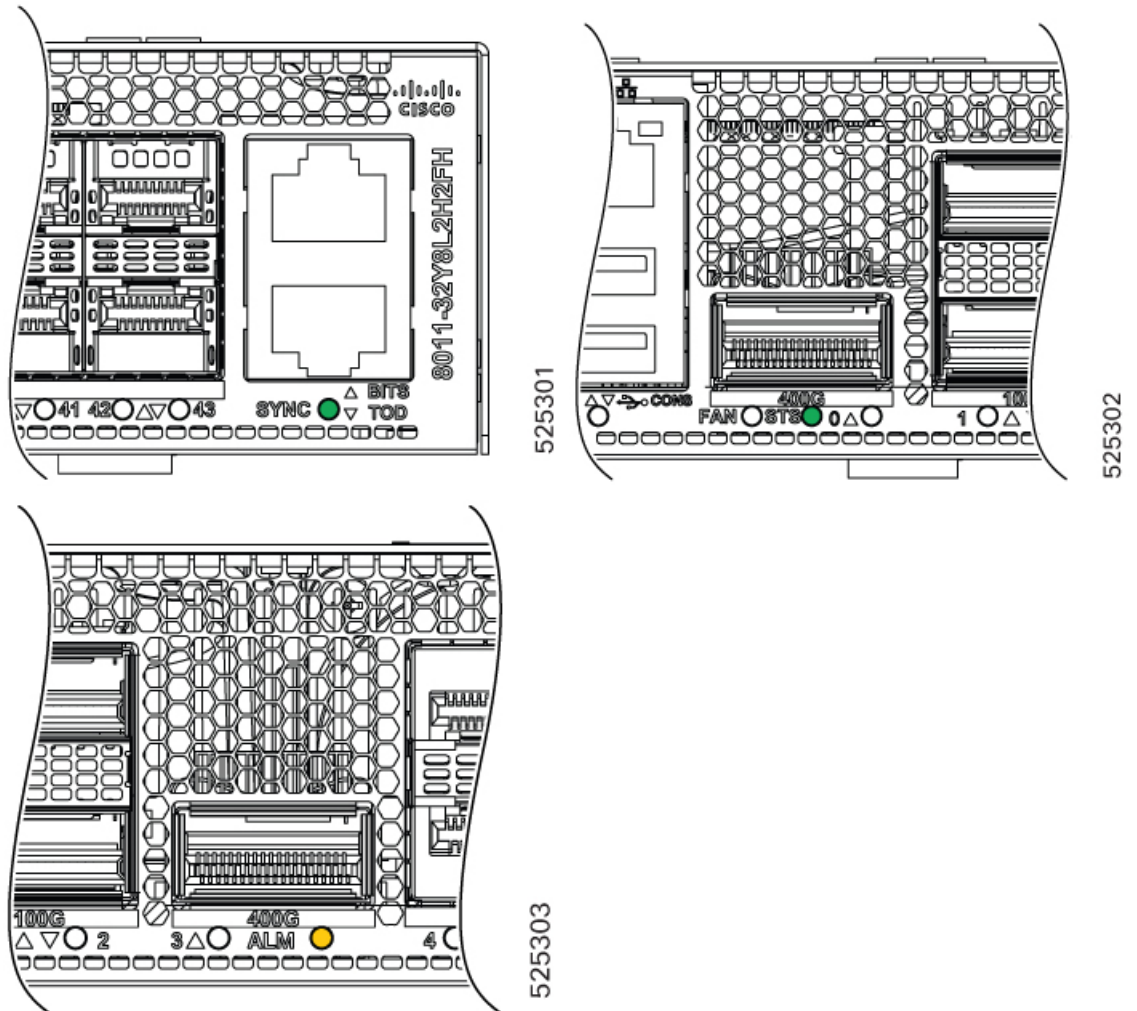


Table 19: Router LED Descriptions

LED Label	Color	Status
STS	Off	The module is powered-off (set by hardware); only standby power mode is available.
	Flashing Amber (Slow)	The module is booting up (set by IOFPGA).
	Flashing Amber (Fast)	The module is booting up (set by BIOS), shutting down, or is being reloaded.
	Amber	Host kernel is booted and is ready to start SysAdmin VM.
	Green	The module is operational and has no active major or critical alarms.
ALM	Off	No alarm
	Red	Critical alarm - system scope, critical temperature.
	Flashing Red	Critical alarm - Relating to voltage rail failures.
	Amber	Major alarm - system-scope.
	Flashing Amber	Minor alarm - system-scope
SYNC	Off	Time core clock synchronization is disabled or in free-running state.
	Green	Time core is synchronized to an external source including IEEE1588.
	Flashing Green	System is in Synchronous Ethernet mode.
	Amber	Acquiring state or Holdover: Time core is in acquiring state or holdover mode.

GNSS LEDs



Note GNSS LEDs are available only on the Cisco 8011-32Y8L2H2FH Router.

Figure 89: Cisco 8011-32Y8L2H2FH — GNSS LED

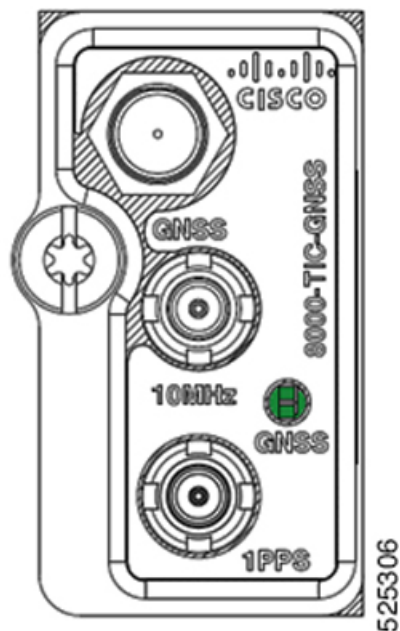


Table 20: Cisco 8011-32Y8L2H2FH — GNSS LEDs

LED	Color	Status
GNSS	Off	GNSS is not configured.
	Green	GNSS normal state. Self-survey is complete.
	Amber	Auto holdover.
	Red	Power up. GNSS is not tracking any satellite.
	Flashing green	Learning state is normal. Self-survey is not completed.

Fan Module LEDs

Figure 90: Cisco 8011-32Y8L2H2FH — Fan Module LED

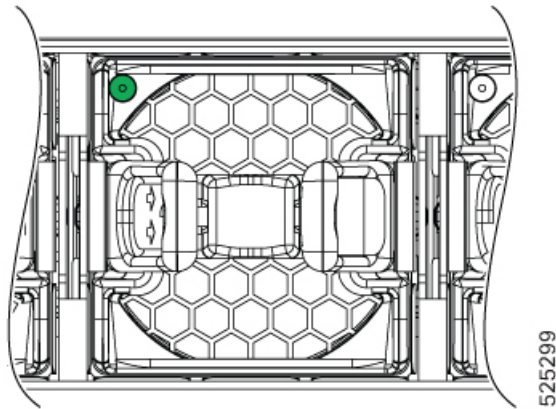


Table 21: Cisco 8011-32Y8L2H2FH — Fan Module LEDs

LED	Color	Status
STATUS	Green	Fan operating normally.
	Red	Fan failure.
	Off	Fan is not receiving power.

Fan Assembly LEDs

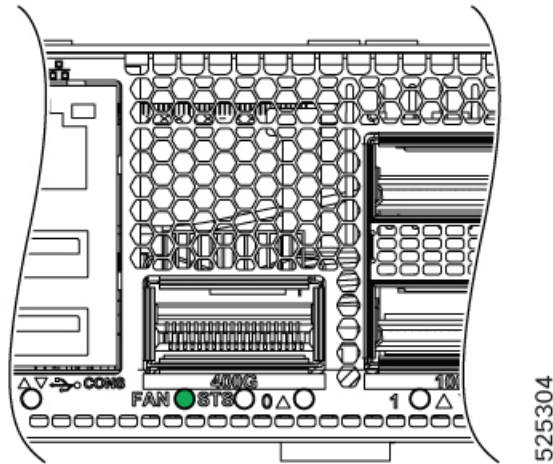
The **Cisco 8011-4G24Y4H-I Router** contains:

- 5 fixed fans at the back panel.
- Fan modules are numbered from left to right as Fan 4 to Fan 0.

The **Cisco 8011-32Y8L2H2FH Router** contains:

- 4 modular fans at the back panel.
- Fan modules are numbered from left to right as Fan 3 to Fan 0.

Figure 91: Cisco 8011-32Y8L2H2FH Router — Fan assembly LED



There is an LED on the front panel of the router that reflect the different status of the fans.

Table 22: Fan Assembly LEDs

LED Label	Color	Status
FAN	Off	Fan tray is not receiving power.
	Green	Fans are operating normally.
	Amber	Single fan failure.
	Red	More than one fan failure.

Power Supply LEDs

Table 23: Cisco 8011-32Y8L2H2FH — Power Supply LEDs

State	Color - Green (AC/DC power status)	Color - Amber (Power supply status)
Blinking	AC power OK, DC output not enabled, sleep mode, 12V standby is ON.	- Power supply warning (PS still operational; 12V main is ON) - Temperature warning - Current limit (OCP)

State	Color - Green (AC/DC power status)	Color - Amber (Power supply status)
Solid On	DC output OK regardless of warning.	• Critical power supply failure • 12V main is OFF • No AC input (Standby power comes from shared PSU): • High temperature (critical) • High voltage, low voltage • Fan failure
Off	• No AC power input • Has AC input; 12V main and 12V standby are off • PSKILL is true	No failures or warnings

Power Status LEDs

Table 24: Power Status LEDs

LED Label	Color	Status
PWR	Off	System is powered off.
	Green	All the power supplies are on and operating normally.
	Amber	Standby FPGA upgrade is in progress (this is expected to take about three to five minutes).
	Red	Power redundancy is lost due to a power feed failure or an internal power supply failure.

SFP Port LEDs

Table 25: SFP Port LEDs

LED Label	Color	Status
STATUS	Off	Admin is down
	Green	Link is up in 1/10/25G ports
	Yellow	Fault or Error or Link Down

Copper Port LEDs



Note Copper port LEDs are available only on the Cisco 8011-4G24Y4H-I Router.

Table 26: 1G Copper Port LEDs

LED Label	Color	Status
Left LED	Green	Link is up in 10/100/1000 Mbps
	Blinking Green	Activity in 1G/100/10 Mbps
	Amber or Orange	Fault/Error/Link is down
	Off	Admin is down
Right LED	Green	Link is up in full duplex

Management Port LEDs

Figure 92: Management port LED

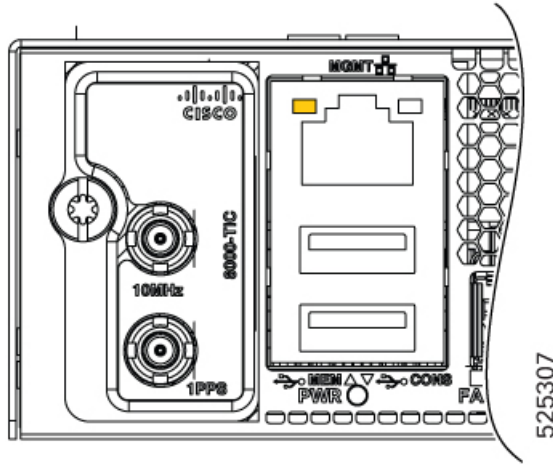


Table 27: Management port LEDs

LED Label	Color	Status
Left LED	Green	Link is up in 1000 Mbps
	Blinking Green	Activity in 1000 Mbps
	Amber or Orange	Link is up in 100/10Mbps
	Blinking Amber or Orange	Activity in 100/10Mbps
	Off	Link is down
Right LED	Off	Unused

System Specifications

For more information on the weight, power consumption, environmental specifications, and other details, see *Cisco 8010 Series Router Data Sheet*.

RJ45 Connectors

The RJ45 connector connects Category 3, Category 5, Category 5e, Category 6, or Category 6A foil twisted-pair or unshielded twisted-pair cable from the external network to the following module interface connectors:

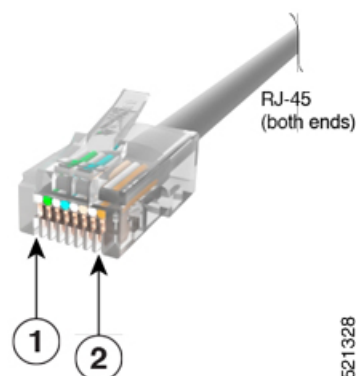
- Router chassis
 - CONSOLE port
 - MGMT ETH port

**Caution**

To comply with GR-1089 intrabuilding, lightning immunity requirements, you must use a foil twisted-pair (FTP) cable that is properly grounded at both ends.

The following figure shows the RJ45 connector.

Figure 93: RJ45 Connector



1	Transmit data (bidirectional)
2	NC (Not Connected)

Transceiver and Cable Specifications

To determine which transceivers and cables are supported by this router, see [Cisco Transceiver Modules Compatibility Information](#).

To see the transceiver specifications and installation information, see [Cisco Transceiver Modules Install and Upgrade Guides](#).

RJ45 ToD and 1PPS Port Pinouts

The following table summarizes the RJ45 ToD and 1PPS port pinouts:

Table 28: RJ45 ToD and 1PPS Port Pinouts

Pin	Signal Name	Direction	Description
1	NA	NA	NA
2	NA	NA	NA
3	1PPS_N	Output or Input	1PPS RS422 signal
4	GND	NA	NA
5	GND	NA	NA
6	1PPS_P	Output or Input	1PPS RS422 signal

Pin	Signal Name	Direction	Description
7	TOD_N	Output or Input	Time-of-Day character
8	TOD_P	Output or Input	Time-of-Day character

Console Port Pinouts

This summarizes the Console port pinouts:

Table 29: Console Port Pinouts

Pin	Signal Name	Direction	Description
1	ACONS-TX	Output	Aux Consoles transmit output, RS232
2	NC	NA	NA
3	CONS-TX	Output	Console RS232 transmit
4	GND	NA	Ground
5	GND	NA	Ground
6	CONS-RX	Input	Console RS232 receive
7	ACONS-RX	Input	Aux Consoles receive input, RS232
8	NC	NA	NA

Alarm Port Pinouts

This summarizes the alarm port pinouts:

Table 30: Alarm Port Pinouts

Pin	Signal Name	Description
1	ALARM1_IN	Alarm input 1
2	ALARM2_IN	Alarm input 2
3	NC	NA
4	ALARM3_IN	Alarm input 3
5	ALARM4_IN	Alarm input 4
6	NC	NA
7	NC	NA

Pin	Signal Name	Description
8	ALARM_I_COMMON	Alarm input COM

To set the description of the alarm:

```
RP/0/RP0/CPU0:ios(config)# environment alarm-contact contact-number description
description
```

To set the severity of the alarm:

```
RP/0/RP0/CPU0:ios(config)# environment alarm-contact contact-number severity
[critical | major | minor] [
```

To set the trigger for the alarm:

```
RP/0/RP0/CPU0:ios(config)# environment alarm-contact contact-number trigger [open
| closed]
```



Note You can configure up to four external alarms.

The *contact-number* is the pin number of the connected alarm port, that is Alarm input 1 to Alarm input 4.

The **description** string can be up to 80 alphanumeric characters in length and is included in any generated system messages.

For **severity**, enter any one of: **critical**, **major**, or **minor**.

Description and severity are both mandatory values.

Use the **show alarms** command in admin mode to view the alarm details. Use the **show logging** command to view the displays the state of syslog error and event logging.

An SNMP trap is sent for every external alarm that is raised or cleared on the system.

USB Port Console Pinouts

This table summarizes the USB port console pinouts:

Table 31: USB Port Console Pinouts

Pin	Signal Name	Description
A1	VCC	+5 VDC
A2	D–	Data–
A3	D+	Data+
A4	GND	Ground

USB Port Memory Pinouts

This table summarizes the USB port memory pinouts:

Table 32: USB Port Memory Pinouts

Pin	Signal Name	Description
A1	VCC	+5 VDC
A2	D–	Data–
A3	D+	Data+
A4	GND	Ground

Management Ethernet Port Pinouts

This table summarizes the management ethernet port pinouts:

Table 33: Management Ethernet Port Pinouts

Pin	Signal Name
1	TRP0+
2	TRP0–
3	TRP1+
4	TRP2+
5	TRP2–
6	TRP1–
7	TRP3+
8	TRP3–